

SCiBTM

**Battery System
for Social Infrastructure Applications**

BMU-2G

Model:

5P4E0124P001

5P4E0125P001

PUR-0000036

**System Integration Guidelines
for Host System**

Toshiba Corporation

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1 Outline

This engineering data intends to clarify the control method of the battery unit by clarifying the interface between BMU and the host system in the social infrastructure battery system.

Be sure to follow “Safety Precautions” provided in the related documents shown on the table below.

	related documents	Document ID
[1]	Battery System Component BMU-2G Instruction Manual	SPC-COM-E0224
[2]	Battery System Component BMU-2G F/W Instruction Manual	SPC-COM-E0085
[3]	Battery System Component Instruction Manual: Common Edition	6F4E0019
[4]	Battery Module Type 3-20 for Social Infrastructure Applications – Instruction Manual	SPC-COM-E0116
[5]	Instruction manual of Battery Module Type 3-23 for Social Infrastructure Applications	SPC-COM-E0113
[6]	Instruction manual of Battery Module Type 1-20	SPC-COM-E0118
[7]	Instruction manual of Battery Module Type 1-23	SPC-COM-E0117
[8]	Battery Module TypeS-20 (2P12S) Instruction Manual	SPC-COM-E0175
[9]	Battery Module TypeS-20 (2P9S) Instruction Manual	SPC-COM-E0173
[10]	Battery Module Type 1-23(2P11S) for Social Infrastructure Applications Instruction Manual	SPC-COM-E0222
[11]	Battery Module Type 3-23(2P11S) for Social Infrastructure Applications Instruction Manual	SPC-COM-E0220
[12]	Battery for Social Infrastructure Applications Type3-20HP Instruction Manual	SPC-COM-E0210
[13]	Battery for Social Infrastructure Applications Type3-26 Instruction Manual	SPC-COM-E0213

Refer to the latest edition of each document.

2 Definition of Terms and Abbreviations

Table 1 shows the definitions of terms and abbreviations used in this document.

Table 1 Term/abbreviation list

Term/ abbreviation	Explanation
BMU-1G	Battery Management Unit. BMU-1G refers to the model FMW-GAA0001P (housing color: white), and is the device that controls the whole battery unit. BMU-1G performs the protection operation, detects abnormalities or failures, and gives notification of the battery capacity in the battery unit.
BMU-2G	BMU-2G refers to BMU-2G-M12 (model 5P4E0125P001, PUR-0000036) and BMU-2G-RJ45 (model 5P4E0124P001) (housing color: black). BMU-2G is a high-performance device, with compatibility with Ethernet and Current Sensor-C Type added to BMU-1G.
It	Standard current value. The value is calculated by dividing the nominal capacity of the cell or the battery module by the time rate. The 1 hour rate is used as the time rate in this document. It is sometimes expressed as C. e.g. 1 It current of 20 Ah is 20 A. 1 It current of 40 Ah is 40 A.
CAN	Controller Area Network. CAN is the communication standard used for connected devices to transmit data to one another.

CMU	Cell Monitoring Unit. CMU is the device that is incorporated into each battery module to control the whole battery module, and to measure each cell voltage and temperature, etc.
Ethernet	LAN standard developed by Xerox Corporation and Digital Equipment Corporation. Ethernet has been standardized by IEEE 802.3 working group.
OCV	Open Circuit Voltage (or open voltage). OCV is the battery voltage with no load connected (with no current flow).
SNA	Signal Not Available. SNA is a status in which the transmitted data is either indeterminate, or invalid and unavailable because of hardware abnormality detection.
SOC	State Of Charge. SOC is the charge remaining in the battery shown in percentage points, where 0% indicates a fully discharged state and 100% indicates a fully charged state.
HSIMV	Host System Instructed Minimum Voltage. Minimum cell voltage value instructed to BMU for balance control by host system.
Self-discharge	Self-discharge is a phenomenon in chemical batteries in which the stored charge of the battery gradually decreases with the passage of time.
Battery unit	A battery unit is composed of a set of battery system components including BMU and one or more battery modules. A maximum of 22 battery units can be connected in parallel. (Also simply called "unit" for short.)
Battery module	Battery module is the generic name for 2P12S module, 2P11S module and 2P9S module. (Also simply called "module" for short.)
Data frame	Of the frames that make up CAN, data frame is the frame for the transmission unit to send messages to the reception unit. This document covers this data frame alone.
Arbitration field	Arbitration field is the field that contains the 11-bit ID that indicates the priority of the data in the standard format of CAN. In general, CAN ID refers to this arbitration field.
Alert	Alert is a failure where the battery module has a failure or abnormality that renders the battery unit inoperative.
Fault	Fault is a failure where the use of all battery units is prohibited.
Branch	Branch refers to one group of battery modules connected in series among the groups of battery modules connected in parallel under one BMU in the battery unit.
H/W failure	H/W failure is a temporary or continued failure of the connection line or of the part mounted on CMU.
2P12S module	2P12S module refers to the eight kinds of modules shown below. With Type3-20, Type3-23 and Type3-20HP, a maximum of 37 modules can be connected in series. With Type3-26, a maximum of 36 modules can be connected in series. With TypeS-20, a maximum of 28 modules can be connected in series. The maximum number of series connections for other modules is 22. • FM01202CCA01A: social infrastructure battery module Type1-20 • FM01202CCB02A: social infrastructure battery module Type1-23 • FM01202CCA04A: social infrastructure battery module Type3-20 • FM01202CCB01A: social infrastructure battery module Type3-23 • FM01202CCA05A: functional safety battery module TypeS-20 • AM01202CCA05A: 27.6V 40Ah battery module • FM01202CCE01A: social infrastructure battery module Type3-20HP • FM01202CCF01A: social infrastructure battery module Type3-26
2P11S module	2P11S module refers to the modules shown below. A maximum of 22 modules can be connected in series. • FM01102CCB02A: social infrastructure battery module Type1-23(2P11S) • FM01102CCB01A: social infrastructure battery module Type3-23(2P11S)
2P9S module	2P9S module refers to the module of the model shown below. With 20.7V 40Ah battery module, a maximum of 22 modules can be connected in series, whereas with TypeS-20(2P9S) battery module, in combination with TypeS-20, a maximum of 16 modules can be connected in series. • AM00902CCA04A: 20.7V 40Ah battery module • FM00902CCA03A TypeS-20(2P9S) battery module

3 Configuration

3.1 Component system

The figure below shows an example of the configuration of the battery panel made up of the social infrastructure battery system components. The host system can control a maximum of 22 battery units (hereinafter called "unit(s)"). (N=22)

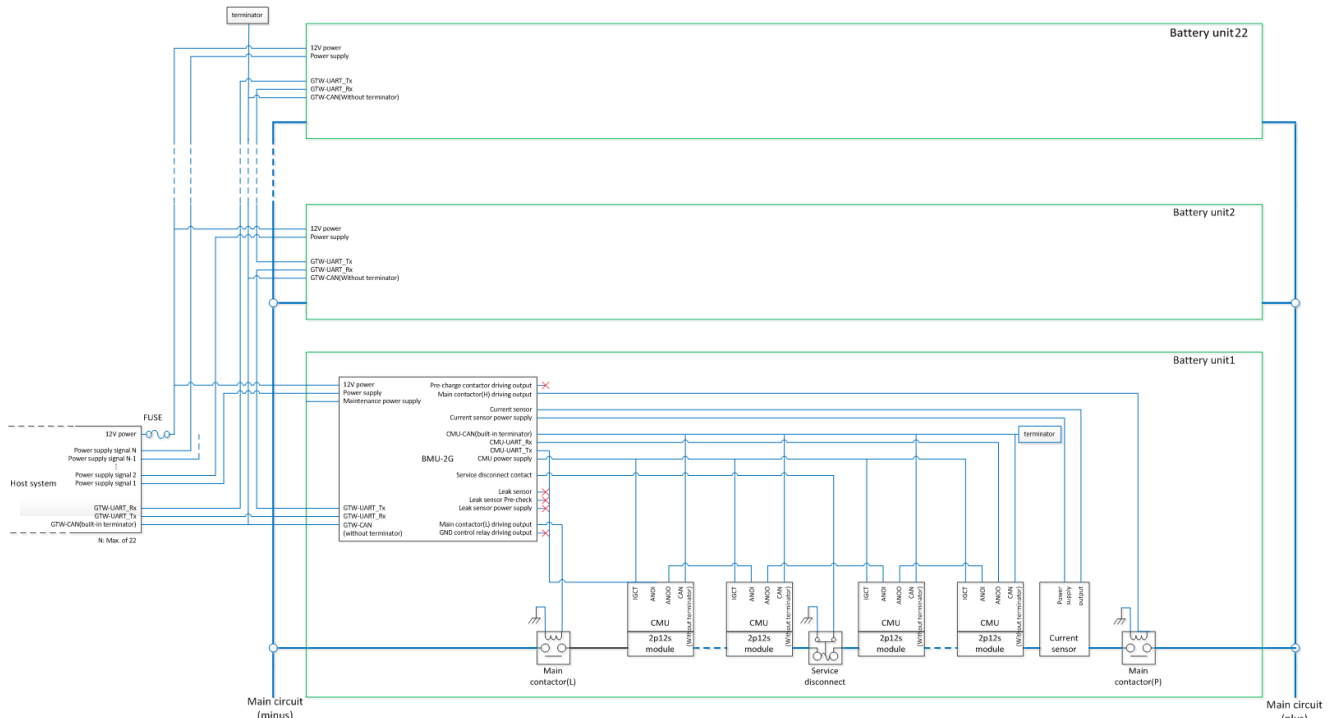


Figure 1 Component system configuration diagram

- * The ground wiring--omitted in the above diagram for the sake of simplicity--is necessary between BMU and the host system.
- * One BMU can control up to 56 battery modules. The maximum number of series connections for Type3-20, Type3-23 and Type3-20HP is 37 (for Type3-26 it is 36, for TypeS-20 it is 28, and for other modules it is 22), the maximum number of parallel connections is 56, and the maximum product of the number of series connections multiplied by that of parallel connections is 56. The 2P12S module(s), 2P11S module(s), and 2P9S module(s) need to be identical in number and arrangement in each branch. Note that, for 23 or more modules to be connected, the unit needs to be configured only with the social infrastructure battery module Type3-20 or later products (models FM01202CCA04A, FM01202CCB01A, FM01202CCA05A, FM01202CCE01A or FM01202CCF01A).
- * Change of BMU settings is required for the use of a pre-charge contactor or a pre-charge resistor, for the parallel connection of the modules, or for the configuration with the modules FM01202CCB02A, FM01202CCB01A, FM01202CCE01A or M01202CCF01A. For details on the BMU settings change, please contact our sales department.
- * On the CAN bus between BMU and CMU, do not connect devices other than the battery module and the current sensor, except for the monitoring device.

Instead of the contactor included in the battery system components, MCCB can be used for the configuration of the battery panel. The figure below shows an example of the battery panel configuration with MCCB. The battery panel is made up of six parallel connections of battery units, each unit composed of four battery modules connected in series, with each battery unit equipped with MCCB for separating the unit, and with the battery panel equipped with MCCB for separating the whole panel. The host system monitors the open/closed status of each MCCB, and each BMU can trip MCCB for separating the unit. The host system can control a maximum of 22 battery units. (N=22)

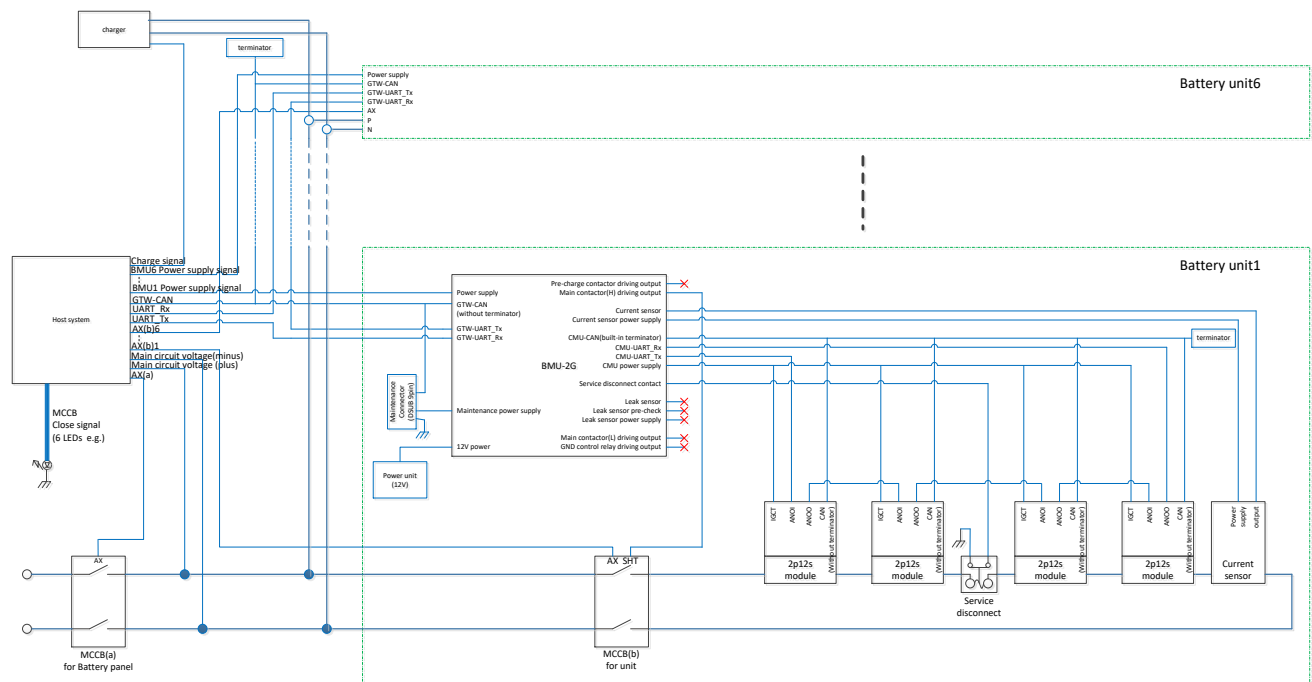


Figure 2 Component system configuration diagram (MCCB mode)

- * Change of BMU settings is required for the battery panel configuration with MCCB. For details on the BMU settings change, please contact our sales department.
- * If the battery panel is configured with MCCB, the pre-charge contactor cannot be used.
- * For MCCB(b) (for separating the unit), be sure to select one that is able to open MCCB on the trip instruction from BMU. The trip terminal (SHT) of MCCB(b) should ordinarily be connected to the drive output terminal of the main contactor(H) of BMU. The high output of BMU corresponds to the trip output.

3.2 Differences between contactor mode and MCCB mode

Shown below are the differences of the component system between the contactor mode and the MCCB mode. The standalone mode--where it is possible to start up without the host system--is available only in the case of a single unit with no parallel connections. See 5.5 for details on the behavior of the standalone mode.

Specification	Contactor mode	MCCB mode	Standalone mode ^{*1}
Maximum number of parallel unit connections	22 units	22 units	Parallel unit connection prohibited.
Component for main circuit breaker of unit	Contactor x2	MCCB x1 ^{*2} Monitor of the open/closed status, with external trip signal	Contactor x2
Pre-charge function	Available Pre-charge contactor and pre-charge resistor required	Unavailable	Unavailable
Current leak detection	Available Installation of designated current leak sensor required	Unavailable	Unavailable
BMU settings change	Unnecessary (Settings change required for use of pre-charge or current leak sensor)	Settings change required	Settings change required
Unit start up procedure	Host system controls the connecting of the main circuit in each unit. (Refer to 5.3.)	It is recommended to install the means to display the permission for connecting of the main circuit, and to close MCCB after confirmation of the permission. (Refer to 5.4.)	The system starts operation without a host system when BMU power supply signal is turned on. (Refer to 5.5.)

^{*1} To turn on the standalone mode, the last three digits of the Model No on the label attached to BMU needs to be equal to or greater than 102. "Model No: B2**102"

^{*2} In the parameter setting, set only one of the contactor/MCCB control ports as MCCB control.

4 Hardware Specification

4.1 Power supply specification

A 12 V power supply is required for BMU. In case the power is to be supplied by the host system, be sure to use a 12 V power supply with overcurrent protection, or to insert a fuse in the output of the power supply.

Signal name	Specification	Voltage range	Max. current
12 V power supply	12 V	10.2 to 13.2 V	5.3A*

* The values above are those with 56 modules connected and the contactors of the standard type in use.

For the influences of and measures against the power supply for BMU being interrupted for a duration of 1 ms or longer to 100 ms or shorter, refer to SPC-COM-E0224 "Battery System Component for Social Infrastructure Applications BMU-2G Instruction Manual."

4.2 Signal input/output

Signal name	Direction	Specification	Characteristics
BMU power supply signal*	Host system → BMU	Logic signal	ON state: 10.3 V or more OFF state: 0.8 V or less (Irregular: above 0.8 V to below 10.3 V)
GTW-CAN_H GTW-CAN_L	Host system ↔ BMU (bidirectional)	In accordance with ISO11898	CAN2.0B active 500kbps Comm. clock error: ±0.7% or less Terminator (120Ω) should be attached (terminator not built-in on BMU side)
Ethernet Tx Ethernet Rx	Host system ↔ BMU (bidirectional)	IEEE 802.3u	100BASE-Tx / 10BASE-T
GTW-UART_Tx	Host system ← BMU	In accordance with ISO9141 (V _{BATT} 12V) Max. possible drive capacity: 1000pF or more	Unidirectional, asynchronous system (UART) Data bit length: 8 bits Start bit length: 1 bit Stop bit length: 1 bit With parity (1 bit/even) Communication speed: 1200 bps (±1%) (Communication specification: refer to 7.3.2.)
GTW-UART_Rx	Host system → BMU	In accordance with ISO9141 (V _{BATT} 12 V) Input capacity: 220pF or less	

* When restarting BMU, please control the OFF period to be 200ms or more. (1 sec recommended)

In the standard configuration, the interface between the host system and BMU is CAN. Change of BMU settings is required for the use of Ethernet. For details on the BMU settings change, please contact our sales department.

5 State Transition

5.1 State transition of host system

For the parallel control of multiple units, the state of control required on the host system side transitions as shown in the example below.

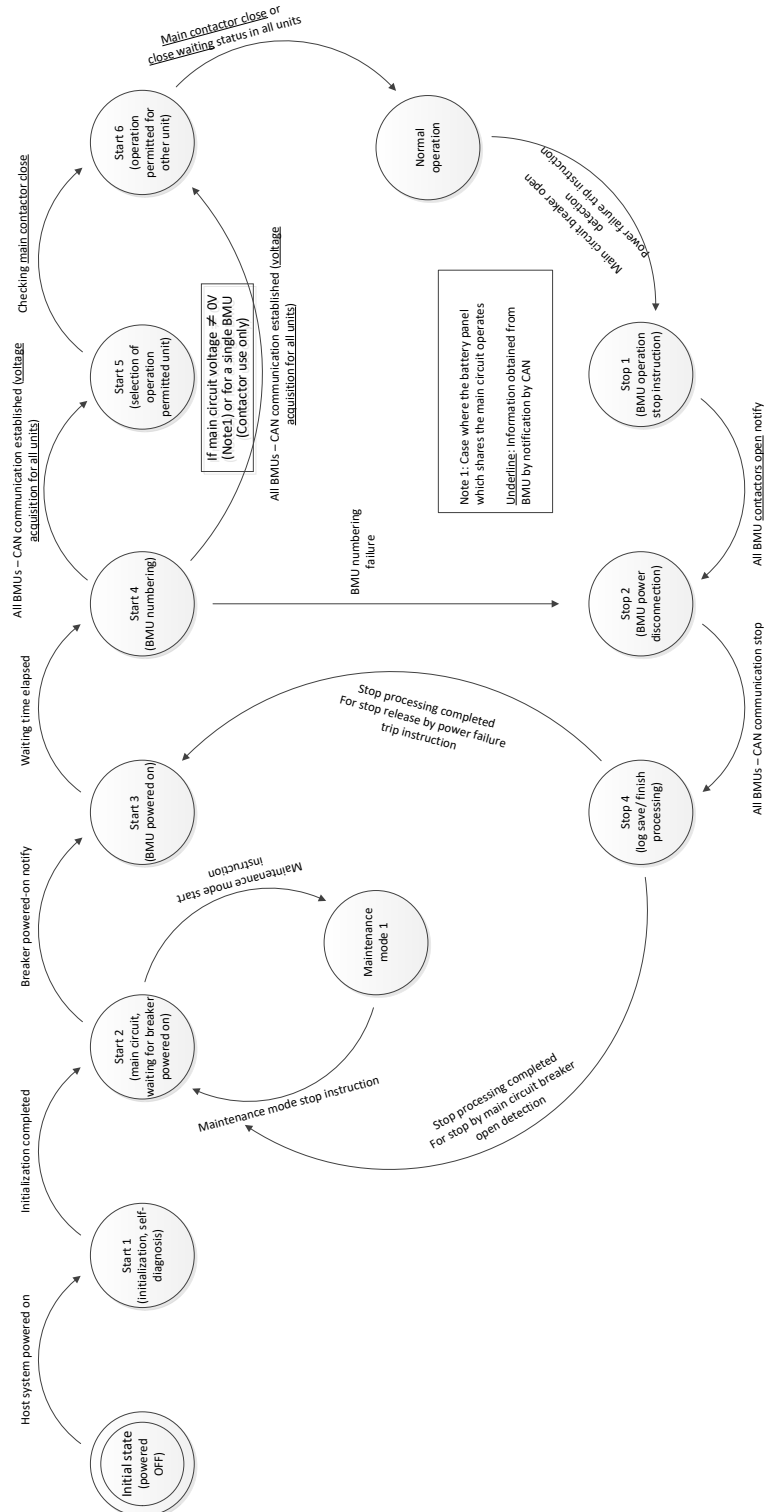


Figure 3 State transition of host system

5.2 State transition of BMU

Shown below is the basic control state diagram of BMU.

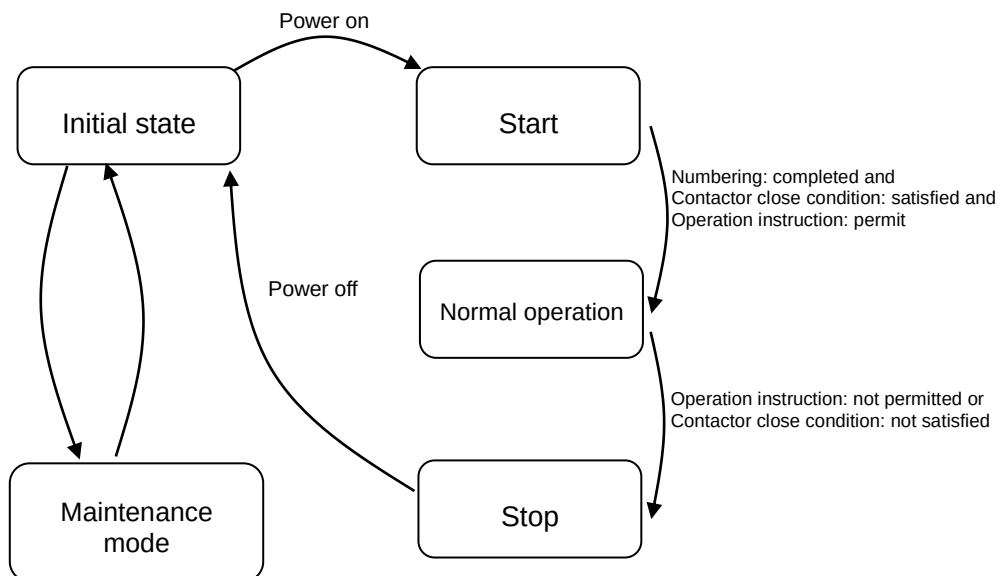


Figure 4 State transition of BMU

Each state and outline of process in the state

	State name	Outline of state/process
1	Initial state	State before powering on
2	Start	Check before start of normal operation; e.g. automatic numbering, contactor close condition
3	Normal operation	Normal operation state including the balance action
4	Stop	State reached by the instruction by the host system or abnormality detection. State of waiting for the return to the initial state by restart by the host system.
5	Maintenance mode	Used only for maintenance

5.3 Start process flow (with contactor in use)

Below is the flow chart showing the control of BMU powering on at the time of starting up, as shown in the status transition diagram, with the contactor included in the battery system components in use. The processes enclosed in the dotted line (S) in the flow chart are skipped in the standalone mode. For details, see 5.5.

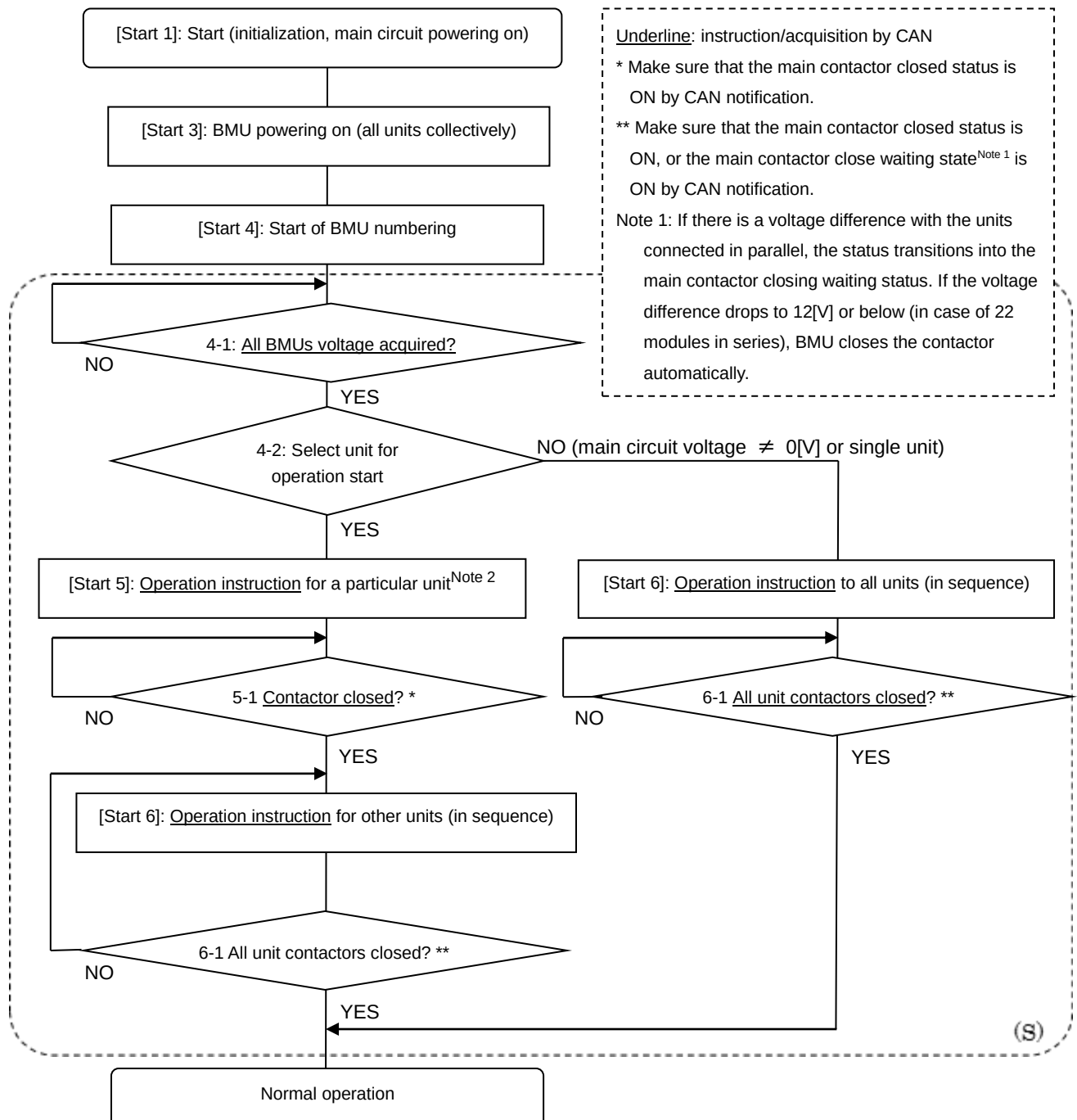


Figure 5 Unit start process flow of host system (contactor)

Note 2: [Start 5] About the procedure to send operation instruction to a particular BMU

In the start procedure to control multiple units, the host system needs to send the operation instruction to BMUs in sequence to close the contactor in each unit. The following are examples of the start procedure.

- To provide the operation permit in the ascending order from the unit with the lowest voltage
Send the main circuit voltage as 0x0000 to BMU of the unit with the lowest voltage alone, and then send the operation instruction (refer to 7.1.7.42). (Note that the measured main circuit voltage should be sent to other units.) All contactors in the unit with the lowest voltage are closed immediately, whereas other units wait for the main P contactor to be closed. After confirming that the contactors have been closed in the unit with the lowest voltage, send the measured main circuit voltage within 5 seconds. If the contactor in other units is to be closed, the charging should be carried out with the current equal to or lower than the current capacity for one unit (40 Ah). Contactors in other units are closed immediately when the voltage reaches the main circuit voltage. The charging should be continued until the contactors are closed in all units.

- To provide the operation permit in the descending order from the unit with the highest voltage
Send the main circuit voltage as 0x0000 to BMU of the unit with the highest voltage alone, and then send the operation instruction (refer to 7.1.7.42). (Note that the measured main circuit voltage should be sent to other units.) All contactors in the unit with the highest voltage are closed immediately, whereas other units wait for the main P contactor to be closed. After confirming that the contactors have been closed in the unit with the highest voltage, send the measured main circuit voltage within 5 seconds. If the contactor in other units is to be closed, the charging should be carried out with the current equal to or lower than the current capacity for one unit (40 Ah). Contactors in other units are closed immediately when the voltage reaches the main circuit voltage. The charging should be continued until the contactors are closed in all units.

It is recommended that, after confirming that the contactors are closed in all units, an additional charging should be carried out until it reaches SOC 70% to correct the SOC error or the imbalance in the cell voltage.

- In case of a single unit
If only one unit is to be controlled, no unit selection is necessary. In this case, before sending the operation instruction in [Start 6], either send the main circuit voltage as 0x0000 to BMU or send the unit voltage value provided by BMU to BMU as the main circuit voltage (refer to 7.1.7.43).

Note that, in the case of sending the main circuit voltage as 0x0000, the measured main circuit voltage should be sent within 5 seconds after confirming that the contactors have been closed in the unit. The main circuit voltage may be the unit voltage value provided by BMU.

In case the correct main circuit voltage fails to be recognized within 5 seconds after closing the contactors, BMU judges it to be the main circuit disconnection and opens the contactor.

- In case the pre-charge circuit is in place
In case the pre-charge circuit is in place, the operation instruction can be sent from any unit.

5.4 Start process flow (with MCCB in use)

Below is the flow chart showing the control of BMU powering on at the time of starting up, as shown in the status transition diagram (Figure 3) based on the configuration in Figure 2, with MCCB used instead of the contactor in the battery panel. As the MCCB mode requires the manual connection of the unit to the main circuit, MCCB needs to be closed individually for each unit while making sure that the unit voltage matches with the main circuit voltage.

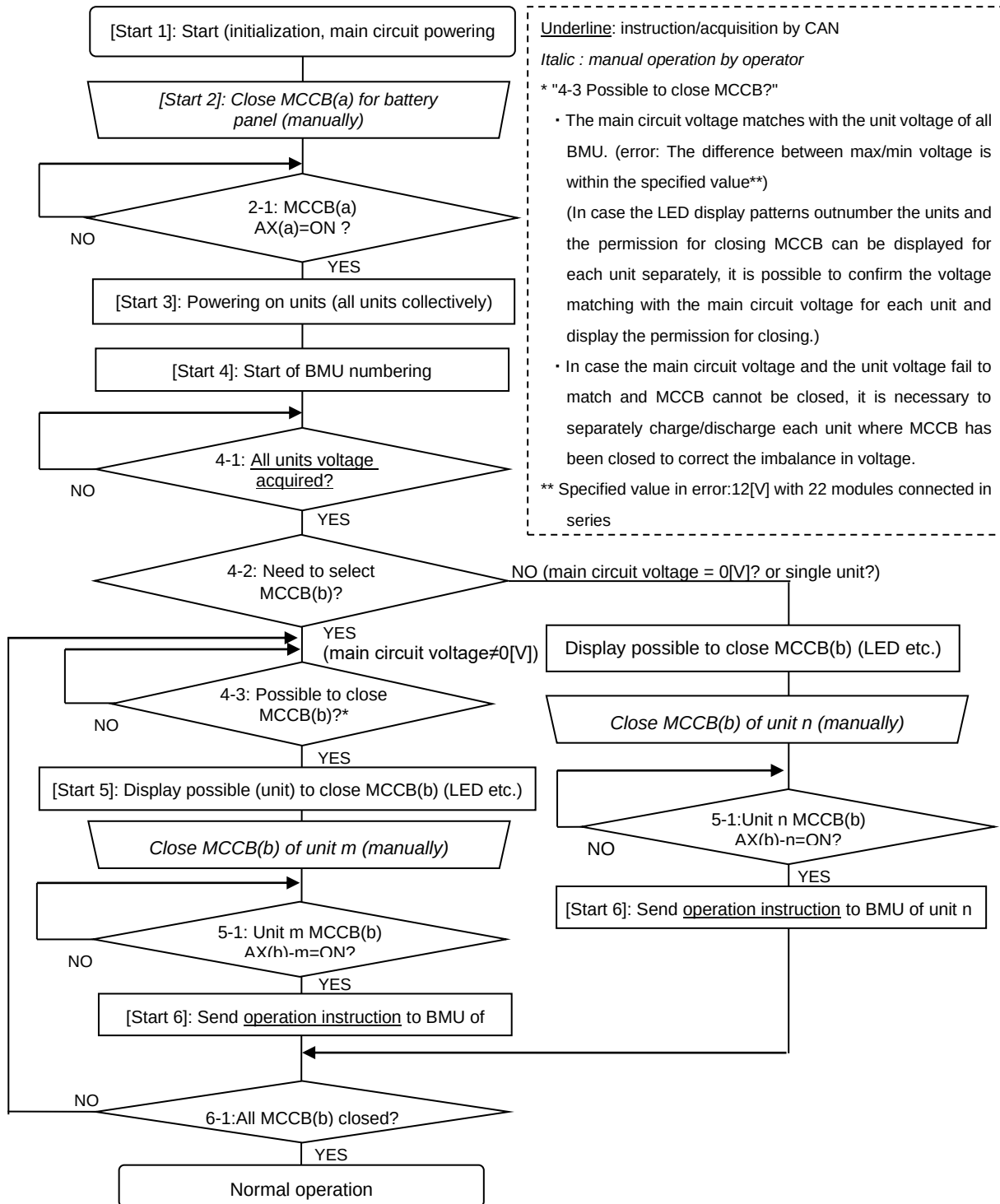


Figure 6 Unit start process flow of host system (MCCB)

About the procedure to close MCCB of each unit in the MCCB mode in Figure 6

To control multiple BMUs, it is necessary to confirm that the main circuit voltage matches with each unit voltage, close MCCB of the unit manually, and then send the operation instruction to BMU. The following are examples of the start procedure.

- In case a display (LED etc.) can be installed for the instruction to close MCCB(b)

To prevent fuse blowing or MCCB welding caused by the large current at the time of closing MCCB, it is necessary to confirm that the main circuit voltage matches with each unit voltage before closing MCCB(b). Note that, if the measured main circuit voltage--including the battery panel connected in parallel that shares the main circuit connection--is 0[V], the closing of MCCB(b) of any one unit can be permitted, in which case caution is advised against the inrush current.

The host system should measure the main circuit voltage and, after the voltage difference with the unit voltage is confirmed to be within the specified range ($12[V] \times [\text{module number in series} / 22]$), indicate the permission for closing MCCB(b) on the display such as LED. MCCB(b) should be closed manually after the permission is confirmed on the display.

In case the permission for each unit cannot be displayed individually (as when only one display is available, for example), the permission for closing MCCB(b) should be indicated only if all unit voltage and the main circuit voltage are within the specified range. In case the main circuit voltage and the unit voltage lie outside the specified range, the charging/discharging is required on the unit where MCCB(b) is already closed so that the voltage should be within the specified range with other units before MCCB(b) of other units are closed. (It is recommended that the displays should be installed in the same number as the units.)

When MCCB(b) of each unit is confirmed to be closed (by way of AX(b) signal), send the operation instruction to the BMU of the unit, after which the unit runs in the normal operation.

- In case no display can be installed for the instruction to close MCCB(b)

Measure the main circuit voltage and the battery voltage between each unit with a device such as a circuit tester, confirm that the measured voltage is within the specified range ($12[V] \times [\text{module number in series} / 22]$), and close MCCB(b) of the unit before starting up the unit. In case the main circuit voltage and the voltage of each unit lie outside the specified range, it is necessary to start up one unit at a time and correct the voltage, with reference to the procedure for a single unit provided below.

When MCCB(b) of all units are closed, start up all units, and send the main circuit voltage/operation instruction to each unit. Each unit runs in the normal operation after receiving the operation instruction.

- In case of a single unit

If only one unit is to be controlled, MCCB(b) may be closed any time with no need of the unit selection, in which case caution is advised against the inrush current. In this case, before sending the operation instruction in [Start 6], either send the main circuit voltage as 0x0000 to BMU or send the unit voltage value provided by BMU to BMU as the main circuit voltage (refer to 7.1.7.43).

Note that, in the case of sending the main circuit voltage as 0x0000, the correct main circuit voltage should be sent within 5 seconds after confirming that MCCB(b) has been closed in the unit (by way of AX signal). The main circuit voltage may be the unit voltage value provided by BMU.

* In the case of MCCB control, no pre-charge circuit can be used.

5.5 Start process flow (standalone mode)¹

In the standalone mode, BMU can transition to the normal operation only by powering on the 12 V power supply of BMU and the power supply permit signal, without the control by the host system. In the standalone mode, only "[Start 3]: BMU powering on" and "[Start 4]: Start of BMU numbering" in Figure 5 apply, with no need for [Start 5] and [Start 6]. The processes enclosed in the dotted line (S) in Figure 5 are skipped.

For details on the start procedure in the standalone mode, refer to 6.3.1. The processes enclosed in the dotted line (S) are skipped as in Figure 5.

[Restrictions on standalone mode]

- No pre-charge contactor or current leak sensor can be used.
- Notifications to the host system (CAN/Ethernet) operate the same as in non-standalone mode.
- Part of the failure diagnosis does not work. Refer to Battery System Component BMU-2G F/W Instruction Manual ^[2] for details of the failure diagnosis.
 - SDC failure / Fusing / Cable error between modules (Open) / Contactor open failure
 - Host system communication error (communication timeout, life signal error, synchronous counter error)
 - Ethernet error
 - Pre-charge contactor failure
- Instructions from the host system via CAN/Ethernet are not accepted, with some exceptions (refer to 7.1.7.42).
 - Operation instruction: BMU always treats it as ON.
 - Balancer drive instruction: BMU always treats it as ON.
 - Current leak detection instruction: BMU always treats it as disabled.
 - Power failure trip instruction: BMU accepts it from the host system.
- Signals from the host system are treated by BMU as shown below.
 - Main circuit voltage: BMU discards it. (See 7.1.7.43)
 - Time data: BMU accepts it. (See 7.1.7.45 and 7.1.7.46)
 - Data acquisition location module No.: BMU accepts it. (See 7.1.7.47)

¹ To turn on the standalone mode, the last three digits of the Model No on the label attached to BMU need to be equal to or greater than 102. "Model No: B2***102"

6 System Configuration

This chapter provides specific examples to explain how “Battery system” should be configured with the battery unit that contains the host system and multiple BMUs.

6.1 System configuration example

Figure 7 shows an example of the battery system where 22 battery modules are connected in series to make up one battery unit, 16 battery units are connected in parallel² to make up one battery panel, and 10 battery panels are systematically connected by way of the power conditioner (PCS) to make up the battery system. BMU can control a maximum of 56 module connections.

Note that the configuration below is only one example of the possible system configurations with any number of series connections within the limits of each component.

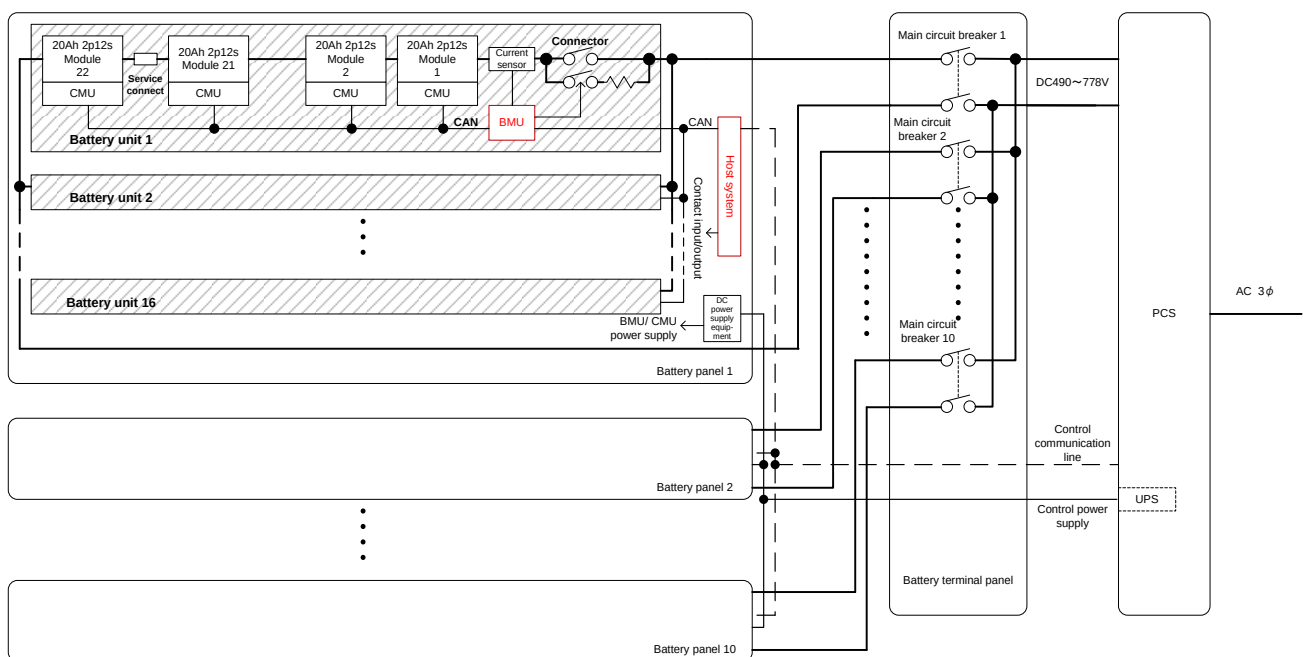


Figure 7 Configuration example of battery system

² Change of BMU settings is required for the parallel connection of the modules. For details on the BMU settings change, please contact our sales department.

6.2 Cautions for system configuration

The restrictions listed below should be considered for the system configuration.

- The system connection requires consideration for surge protection. See each component specifications for details on the surge resistance.
- Short-circuit of the DC bus bar between the battery panel and PCS may lead to the fuse blowout in each unit. The wiring for the DC bus bar should be able to endure the current until the fuse blowout.
- The ground fault of the DC bus bar cannot be detected because BMU opens the contactor and activates the current leak sensor after receiving the current leak detection instruction. Thus, consideration should be given as to whether the DS ground fault detection relay needs to be installed. Send the current leak detection instruction when the ground fault is detected. Also, make sure that the DC bus bar is wired correctly.
- Periodic repetition of abnormal and normal communication may cause BMU to be restored and stopped repeatedly, resulting in a shortened life span for the contactor or MCCB. Periodic occurrence of restoration and stop should be able to be stopped manually.
- If UPS is to be used for the control power supply, the system should be configured so that the UPS abnormality can be detected (e.g. voltage abnormality, or built-in battery abnormality).
- Operations on the host system side (e.g. the host system or PCS) should be able to stop BMU (by opening the contactor by the operation instruction turned off) in the case of the main circuit breaker abnormality. Also, make sure that the host system and the auxiliary contact of the main circuit breaker are wired correctly.
- The system should be designed in consideration of the detection of, and measures against, battery panel failures, such as an abnormal temperature in the battery panel, the power supply abnormality of the host system, and the abnormality in the DC power supply.
- The system should be designed in consideration of the possibility of the power supply abnormality of the host system or PCS leading to the battery panels or the battery units under them collectively stopping or losing control.
- If multiple battery panels are connected, the system should be configured so that the host system can detect the open circuit failure of the DC bus bar in the battery panel.
- The fuse in the service disconnecter can cut off a maximum of 16 battery units connected in parallel in the case of a short-circuit in the battery panel. For 17 or more battery units connected in parallel, the use of the fuse 1000GH of HINODE ELECTRIC CO.,LTD. is recommended. Note that the protection coordination for the battery system should be considered on the side of the battery panel and the host system.
- The mutual monitoring means such as the heart beat signal should be installed on the host system in the battery panel and its host system (PSC) to detect the communication disruption between them.
- The current in the main circuit should be measured and controlled on the host system side so as not to exceed the maximum allowed current.

6.3 Sequence diagram

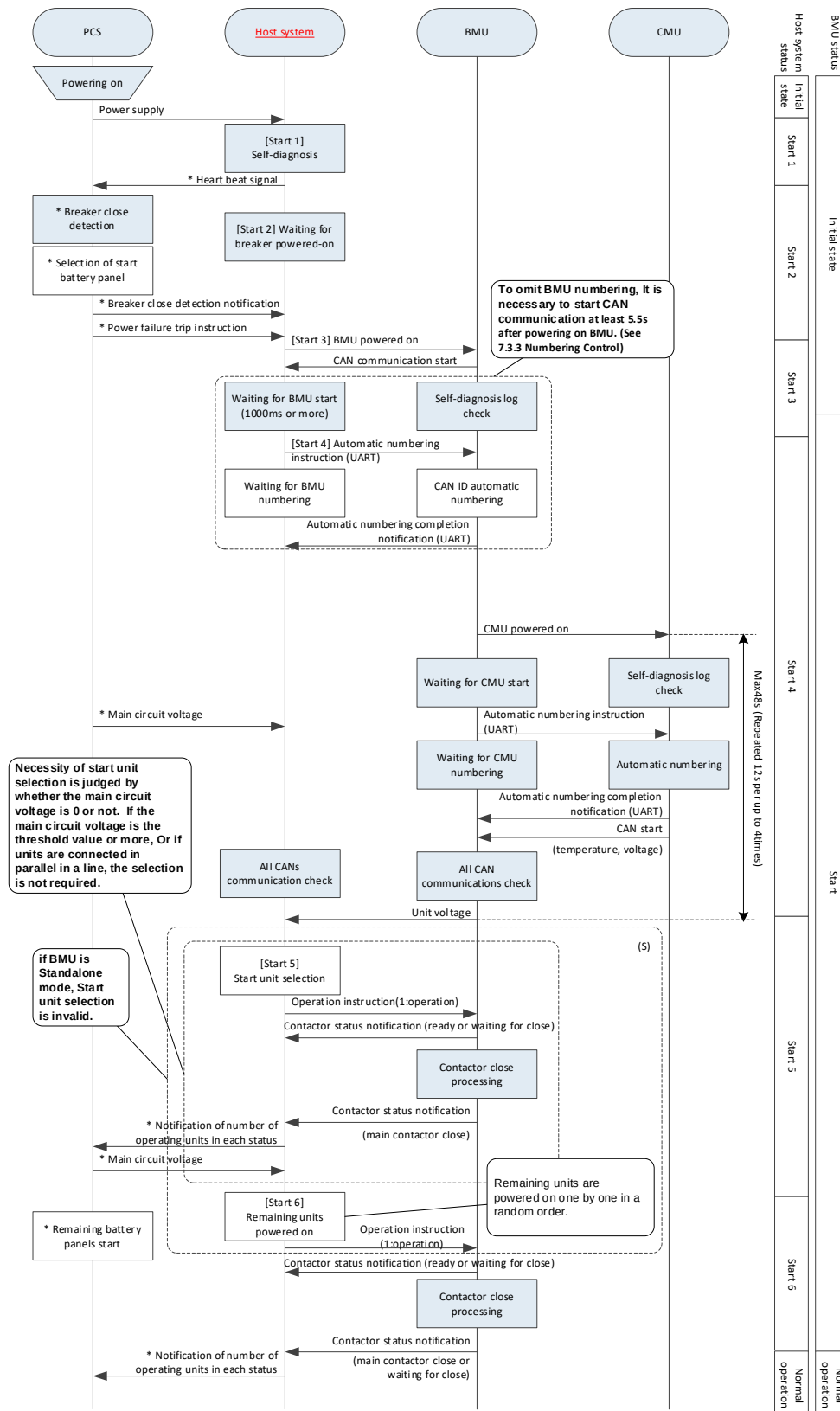
This section provides sequence diagrams to explain the control specifications of BMU, the host system, and its host system PCS shown in the configuration example of the battery system in Figure 7. The table below lists the procedures explained in this section.

Note that the asterisks (*) in the sequence diagrams indicate examples on the host system in the system configuration, and do not mean that the implementation of the control specification is mandatory.

Section	Procedures	Description
6.3.1	Start procedure (with contactor in use)	Start procedure with multiple battery panels in place (with contactor in use)
6.3.2	Start procedure (with MCCB in use)	Start procedure with multiple battery panels in place (with MCCB in use)
6.3.3	Normal stop procedure	Stop procedure when the main circuit breaker is opened
6.3.4	Power failure procedure	Stop procedure when power failure/UPS failure is detected
6.3.5	Restoration procedure	Restart procedure in the case of restoring from power failure/UPS failure
6.3.6	Cell balance procedure	Cell balance instruction procedure
6.3.7	Overcharge/overdischarge (alert) detection procedure	Procedure when overcharge/overdischarge is detected
6.3.8	Overcharge/overdischarge (fault) detection procedure	Procedure when overcharge/overdischarge (e.g. fault or contactor welding) is detected
6.3.9	Overdischarge warning detection procedure	Procedure when overdischarge warning is detected (only warning)
6.3.10	Final charge/final discharge detection procedure	Procedure when final charge/final discharge is detected (only warning)
6.3.11	Over-temperature (alert) detection procedure	Procedure when over-temperature (alert) is detected
6.3.12	High temperature protection detection procedure	Procedure when high temperature protection is detected (only warning)
6.3.13	Time adjustment procedure	Procedure to update the time information of BMU
6.3.14	Current leak detection procedure	Procedure to make BMU execute current leak detection

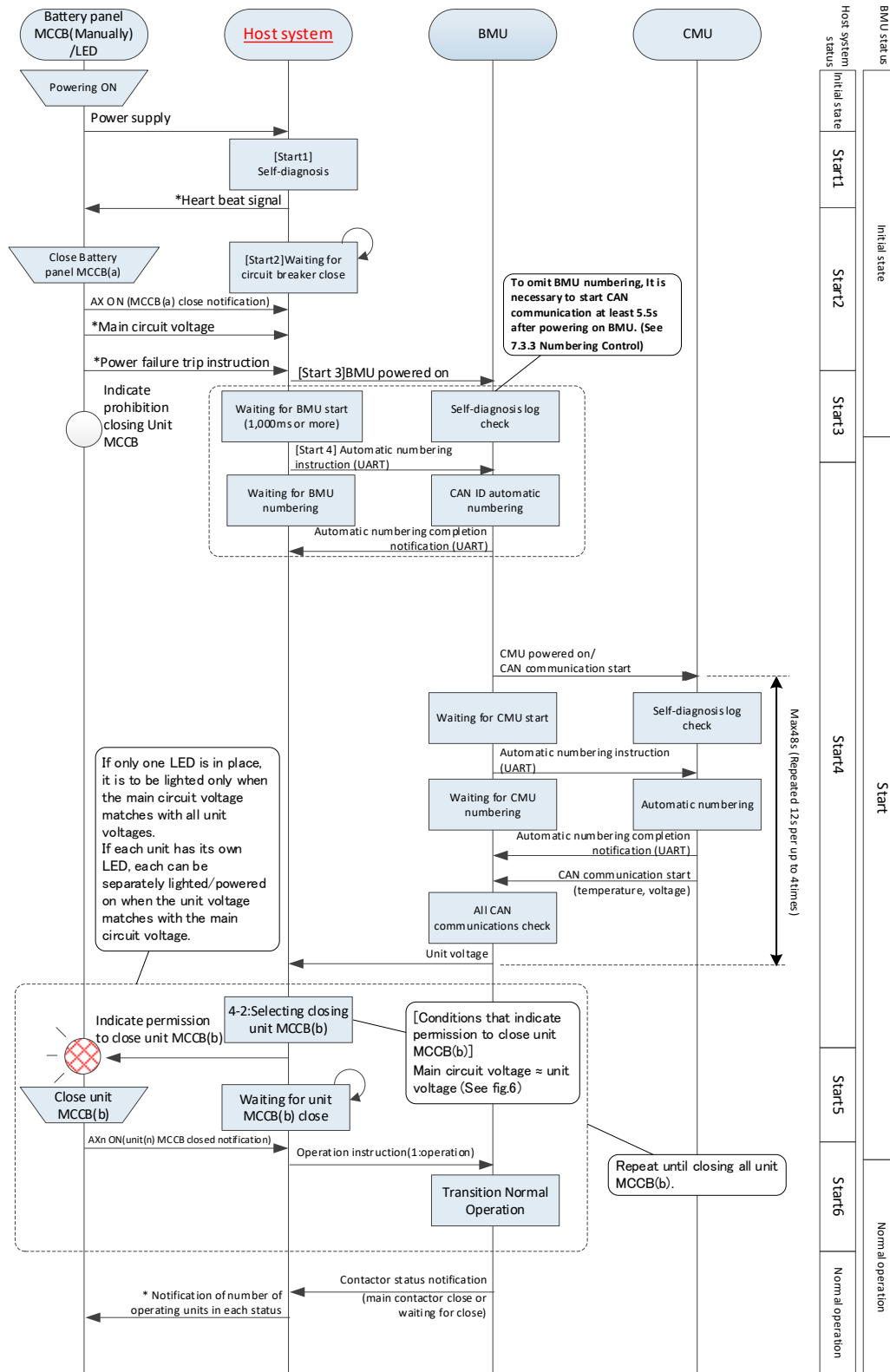
6.3.1 Start procedure (with contactor in use)

The diagram below shows the start procedure when the contactor included in the battery system components is used. The processes enclosed in the dotted line (S) in the diagram are skipped in the standalone mode. Refer to 5.5 for details.



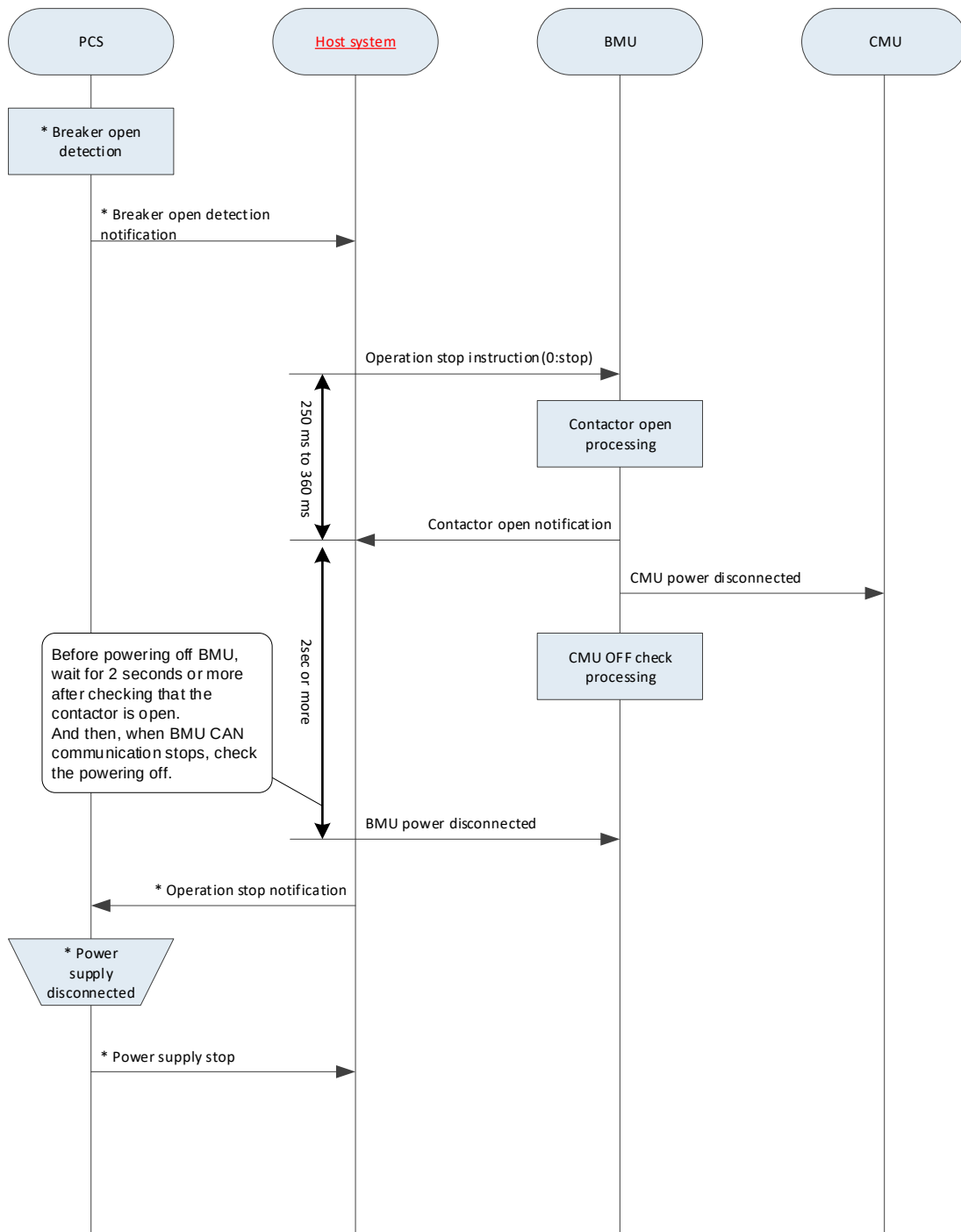
6.3.2 Start procedure (with MCCB in use)

The diagram below shows the start procedure for the system configuration in Figure 2, where the battery panel MCCB(a) and the unit MCCB(b) are installed, the host system can monitor the closed status (AX(a)/AX(b)-n), and the LED display is in place to indicate the permission for closing MCCB(b).



6.3.3 Normal stop procedure

Send the operation stop instruction and, after receiving the contactor open notification (= contactor closed status (0)), be sure to wait 2 seconds or more before powering off BMU to save the log data of CMU.

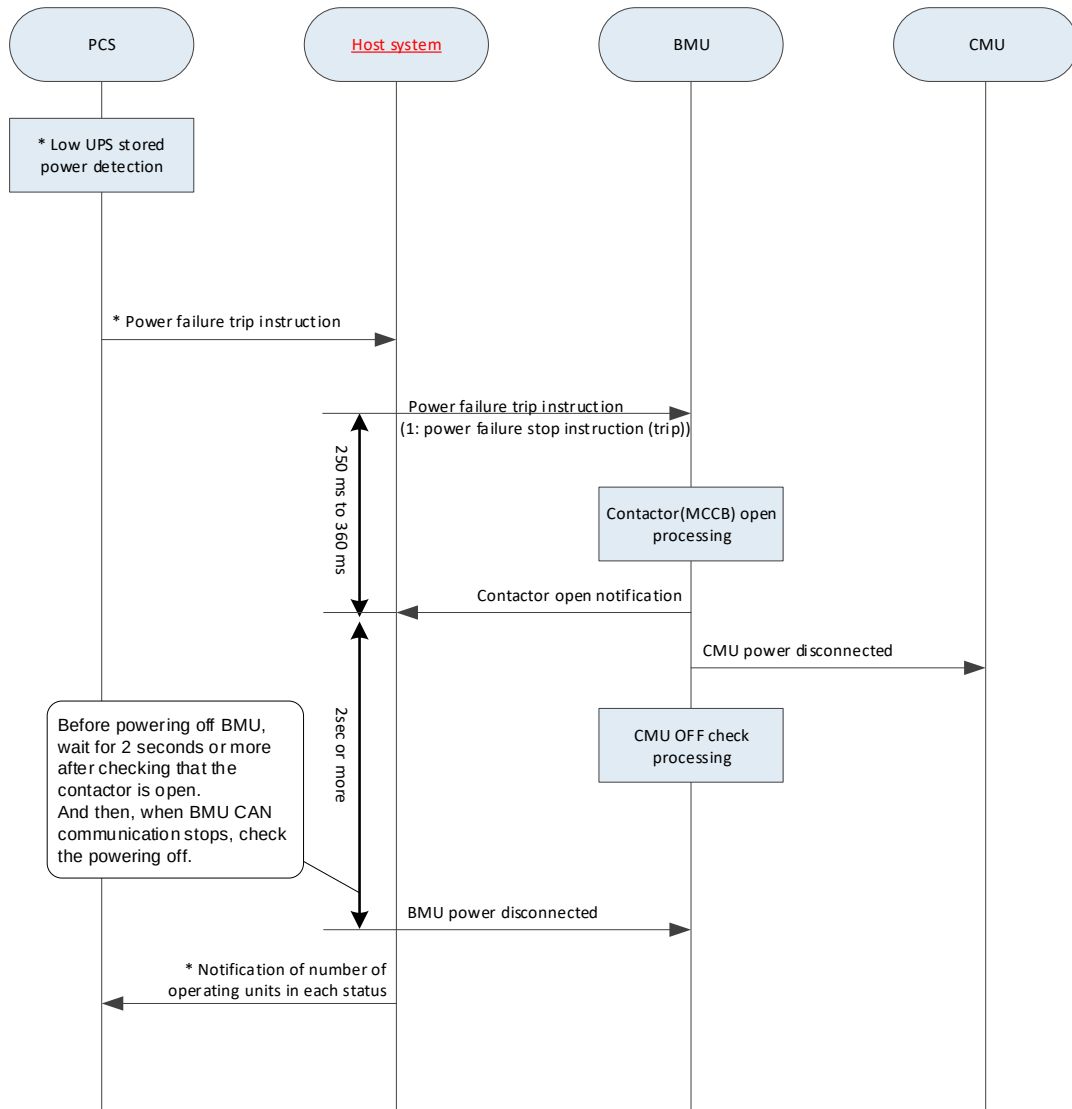


【Note】 Note that, in the case of using MCCB, MCCB remains closed and BMU continues the normal operation even if the host system sends the operation stop instruction to BMU.

*1: By default, BMU stops operation after *three* consecutive detections of the operation stop instruction, but can be made to stop with a *single* detection with a parameter change. (For the parameter change, the last three digits of the Model No on the label attached to BMU need to be equal to or greater than 109, e.g. "Model No: B2***109.")

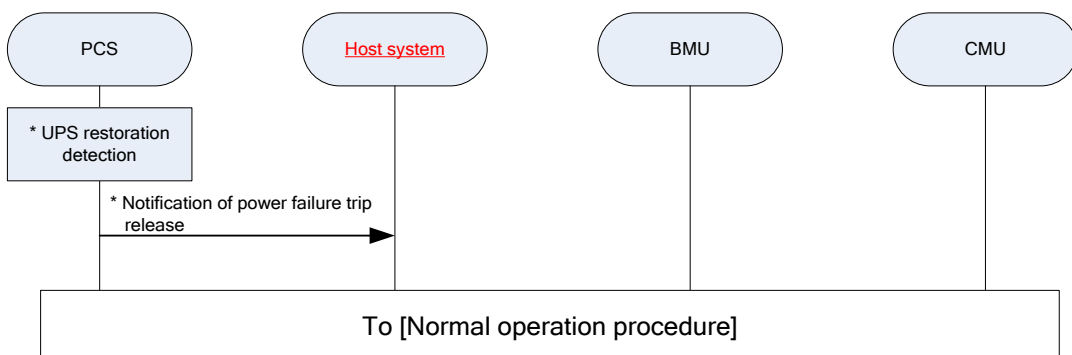
6.3.4 Power failure procedure

Send the power failure trip instruction and, after receiving the contactor open notification (= contactor closed status (0)), be sure to wait 2 seconds or more before powering off BMU to prevent a risk not be able to save or read CMU log data.



*1: As with the normal stop procedure, BMU can be made to stop with a *single* detection with a parameter change.

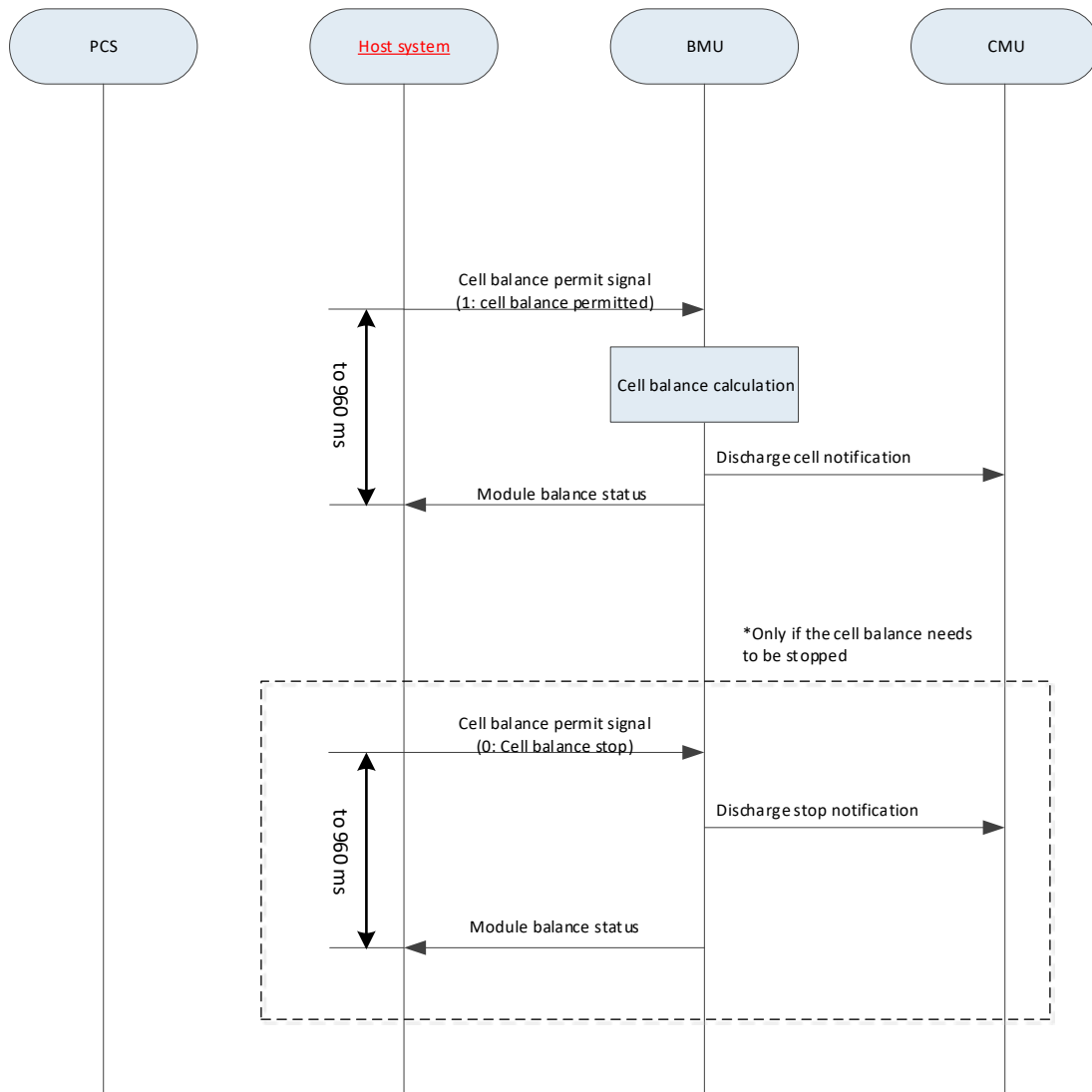
6.3.5 Restoration procedure



6.3.6 Cell balance procedure

Shown below is the cell balance procedure. Cell balance is available at the start and during the normal operation.

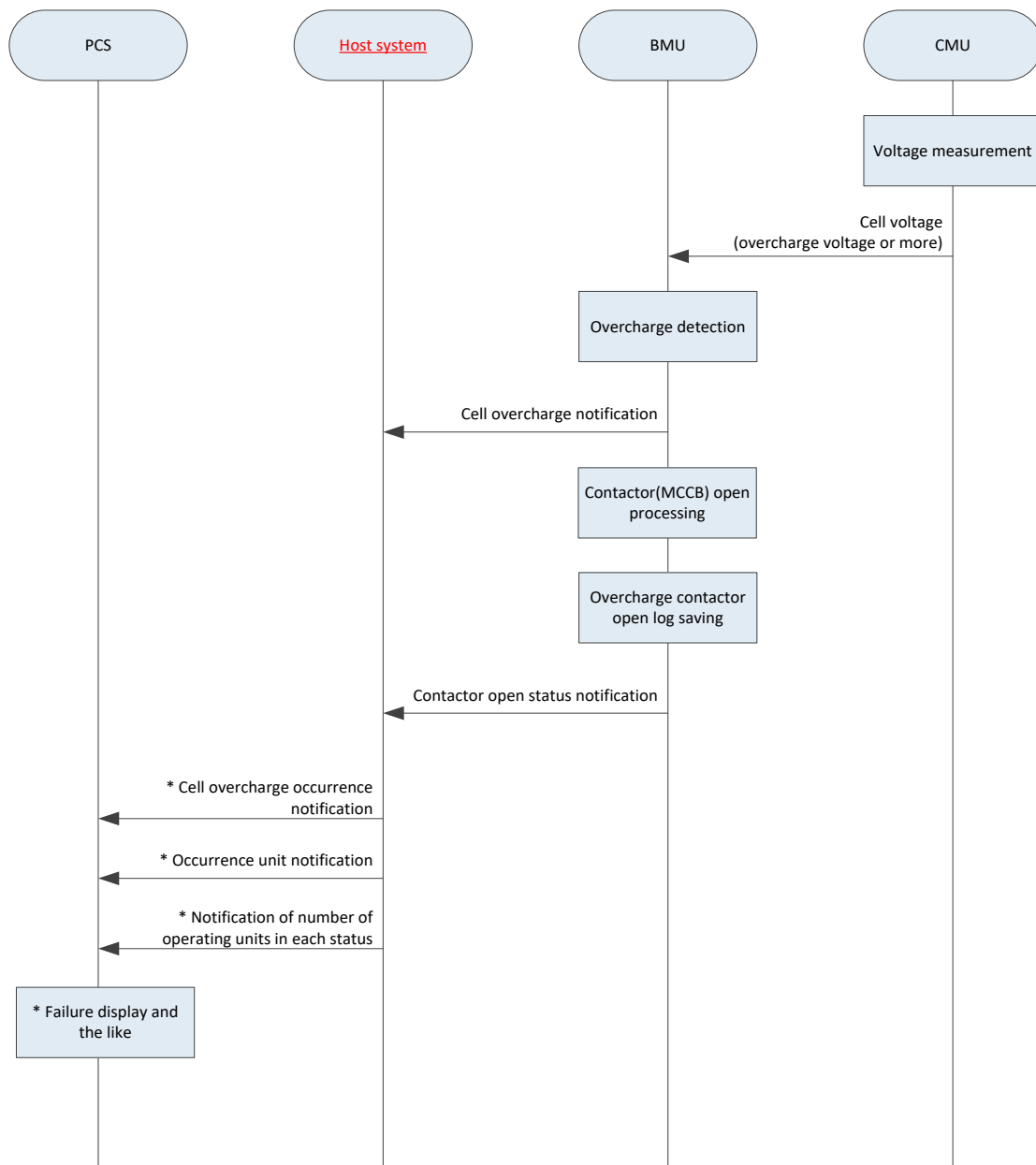
If the cell balance needs to be stopped, send the cell balance stop notification from the host system. Because the cell balance process itself does not affect the normal battery operation, the permission status may be left as it is, unless there is a special purpose.



Cell balance can be executed even before the contactor closes.

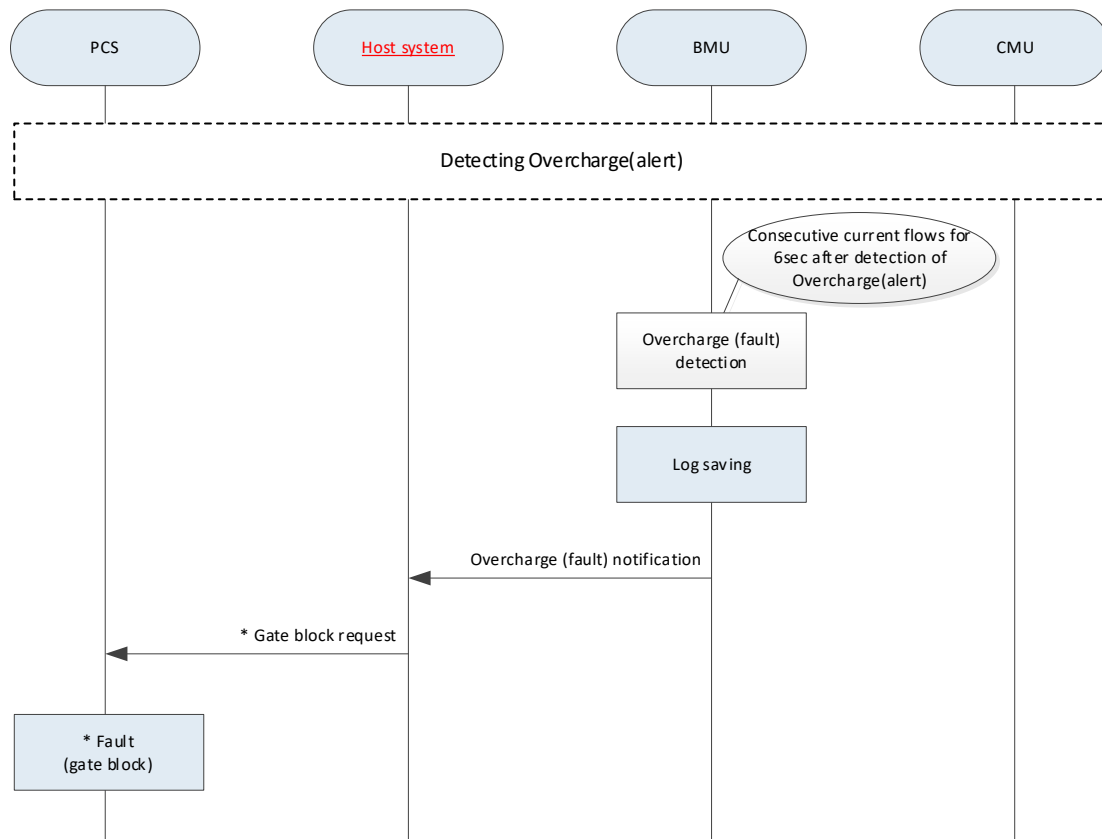
6.3.7 Overcharge/overdischarge (alert) detection procedure

The diagram below is one of overcharge, and the same diagram applies to overdischarge.



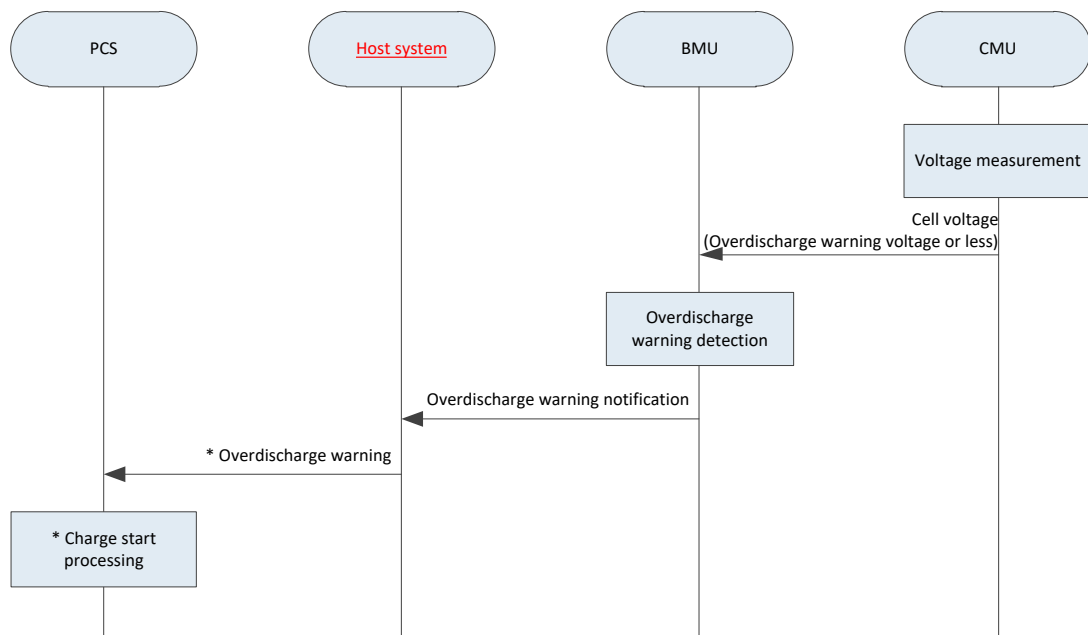
6.3.8 Overcharge/overdischarge (fault) detection procedure

The diagram below is one of overcharge. The same diagram applies to overdischarge.



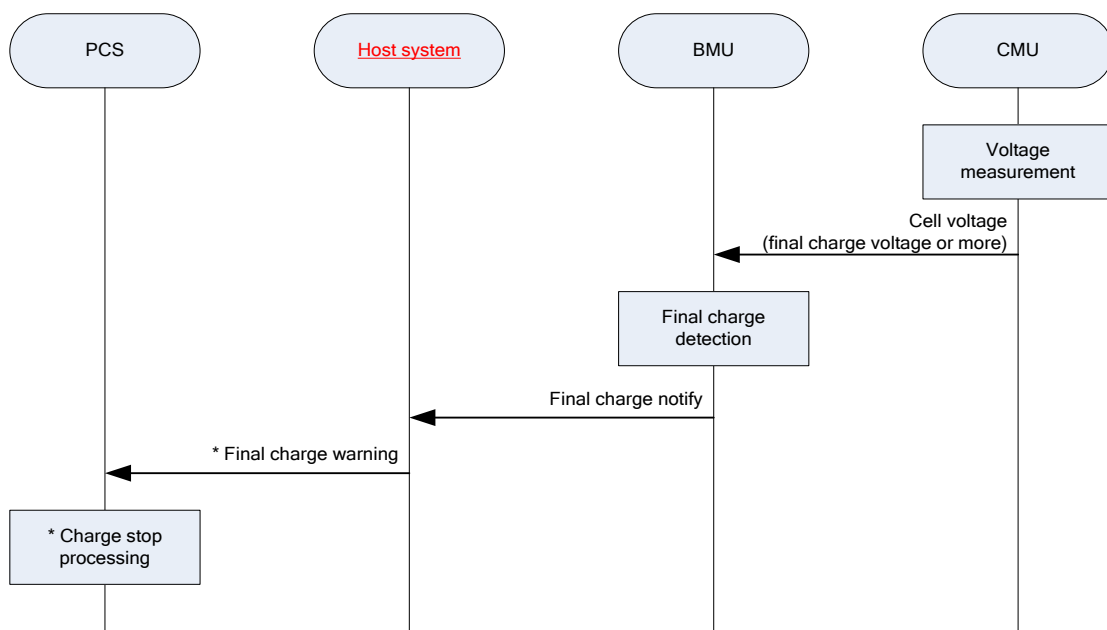
When a fault is detected, the circuit cannot be disconnected from the battery unit side. To prevent serious accidents such as fire, disconnect the battery unit from the circuit (gate block) by opening the circuit breaker of the host system.

6.3.9 Overdischarge warning detection procedure

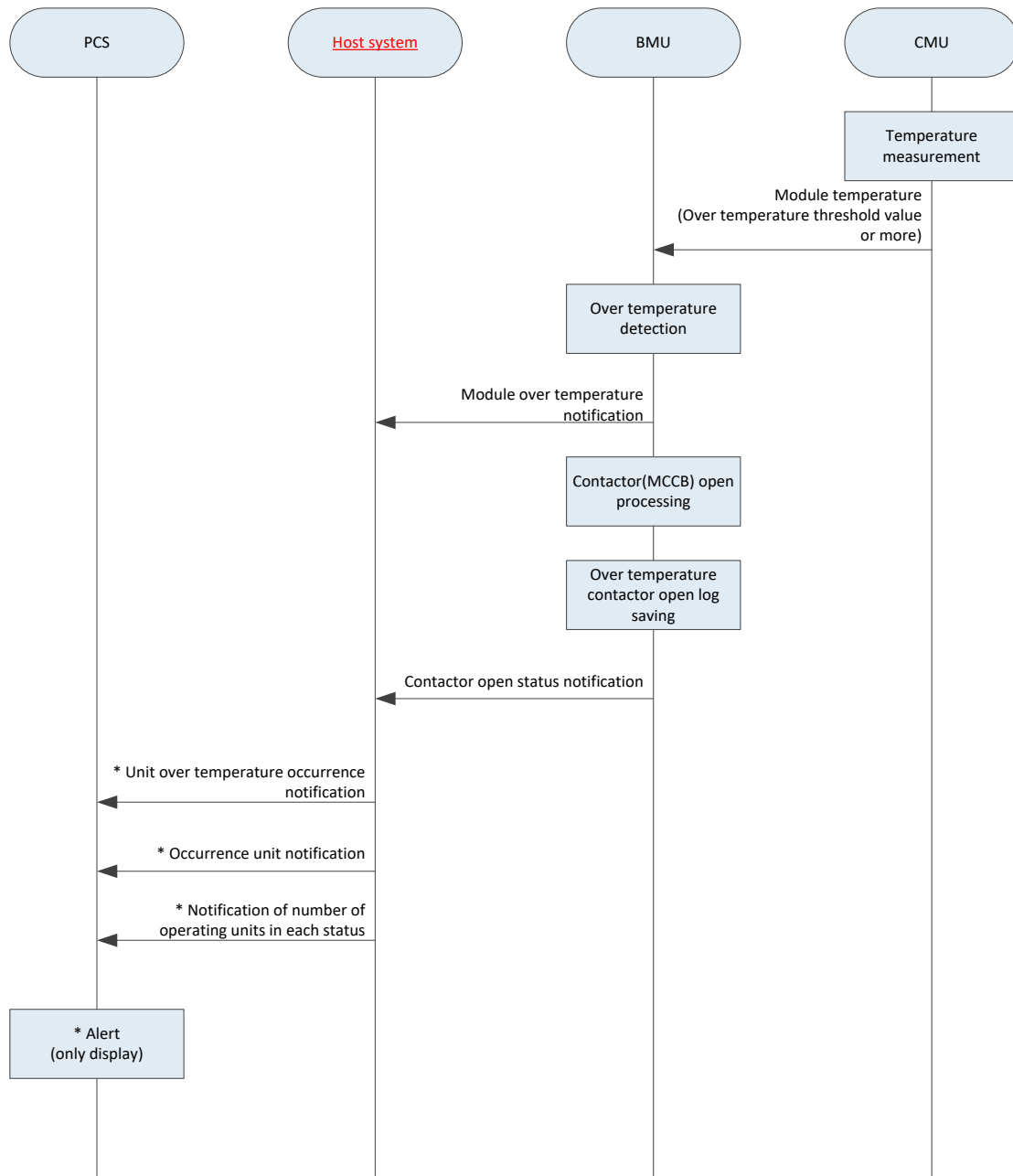


6.3.10 Final charge/final discharge detection procedure

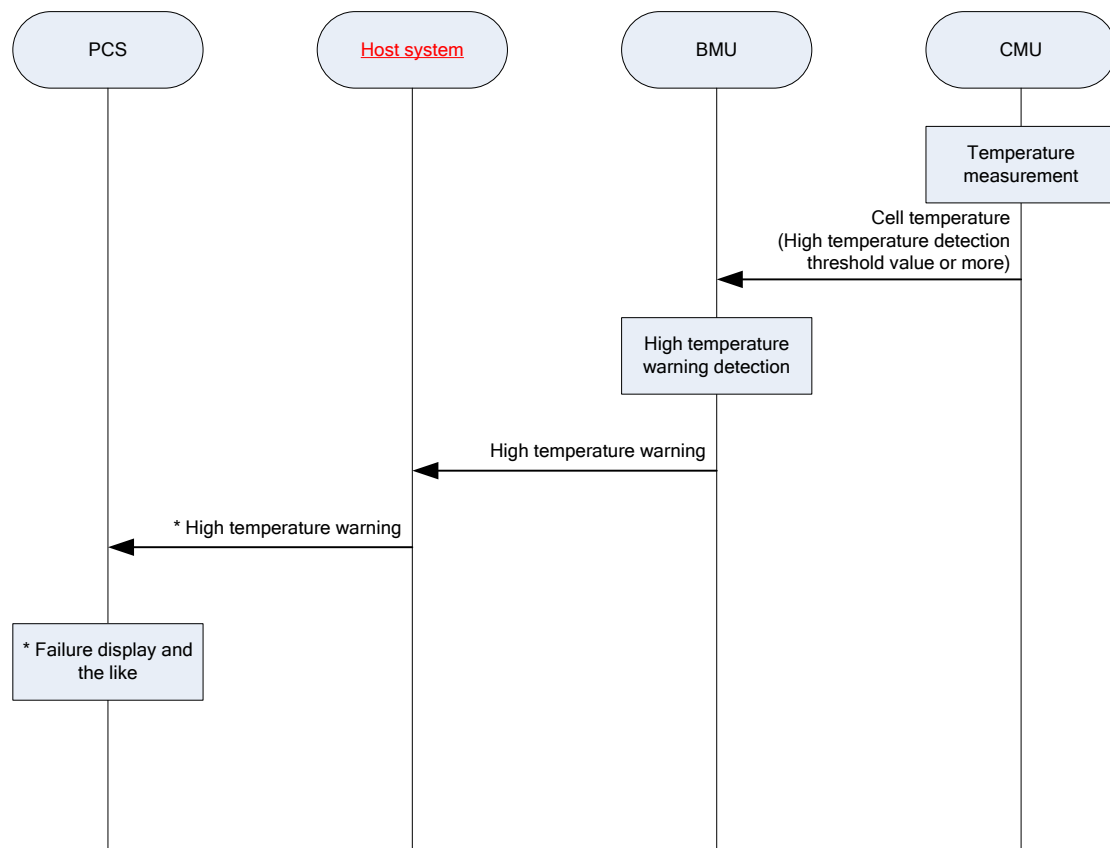
The diagram below is one of final charge, and the same diagram applies to final discharge.



6.3.11 Over-temperature (alert) detection procedure

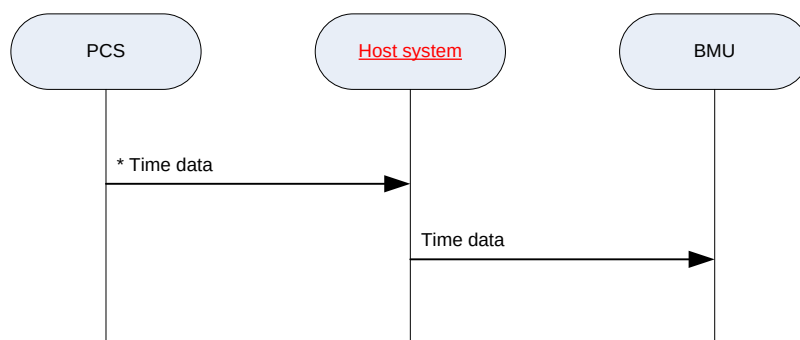


6.3.12 High temperature protection detection procedure

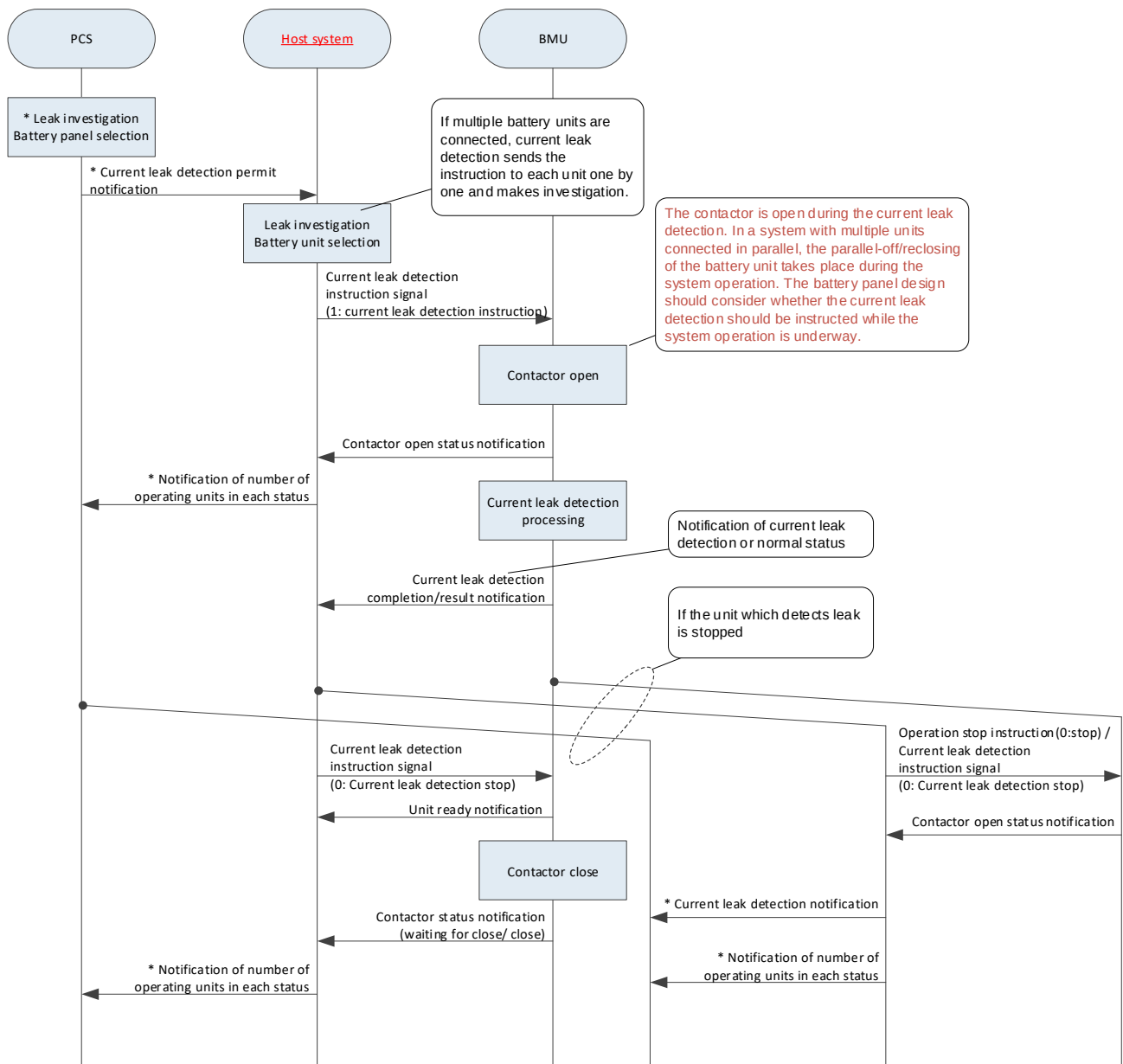


6.3.13 Time adjustment procedure

The time data can be sent at any time. Note that, to initialize the internal time of BMU at the time of start or restart, the time data should be sent immediately after checking the unit voltage of BMU. After the initialization is completed, the time of BMU should be adjusted regularly, with one-minute intervals as a standard.



6.3.14 Current leak detection procedure



The designated current leak sensor is required. Change of BMU settings is required for the installation of the current leak sensor. For details on the BMU settings change, please contact our sales department. The diagram above shows the case where the contactor included in the battery system components is used. Note that, in case MCCB is used instead of the contactor, BMU discards the current leak detection instruction.

*1: As with the normal stop procedure, BMU can be made to stop with a *single* detection with a parameter change.

7 Communication Specification between Host System and BMU

7.1 CAN communication control

7.1.1 General specification

This section explains the communication specification between BMU and the host system. The information is sent/received between BMU and the host system via CAN communication shown in Table 2. For the general specification of CAN communication, refer to the engineering data provided by the CAN promotion group CiA (CAN in Automation: <http://www.can-cia.org/>).

Table 2 CAN basic communication specification between host system and BMU

Item	Description
CAN protocol specification	CAN2.0B (standard format adopted) ISO11898
Data format	Standard format (CAN ID:11Bit) DLC : 8 fixed
Communication speed	500 kbps (High Speed CAN)
Message management	Full-CAN mode
Sample point	81%
BTR0, BTR1	DBT (data bit time): 16Tq SS (synchronization segment): 1Tq TSEG1 (PTS+PBS1): 12Tq TSEG2 (PBS2): 3Tq SJW: 3Tq

7.1.2 Assignment of arbitration field

The 11-bit arbitration field in the data frame is the ID (CAN-ID) for the node identification which defines the type of the transmission/reception and the destination ID of the data frame. Figure 8 shows the assignment of the arbitration field. The arbitration field consists of two fields; i.e. the node ID indicating the destination or source BMU, and the frame ID indicating the data type. Note that, if the node ID is not defined (i.e. BMU ID is not numbered), BMU sends "0x01" as the node ID and receives the frame with the node ID "0x30."

- CAN-ID assignment (frame format for BMU transmission/host system reception)

Node ID						Frame ID				
b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0x01 (without ID), 0x02(BMU 1) to 0x17(BMU 22)						0x00 to 0x1F (max. 32 frames)				

- CAN-ID assignment (frame format for BMU reception /host system transmission)

Node ID							Frame ID			
b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
0x30 (without ID), 0x31(BMU 1) to 0x40(BMU 16)							0x00 to 0x0F (max. 16 frames)			

Figure 8 Assignment of arbitration field (CAN-ID)

7.1.3 CAN ID assignment range

Based on the assignment of the arbitration field described in the previous section, the CAN-ID assignment range of the frame sent by the host system and BMU is as shown in Table 3. Note that the frames in Table 3 (b) primarily contain the information of the modules 29 to 56 and, in case they are not needed, the transmission can be controlled by the parameters.

Table 3 CAN-ID assignment range(a) BMU → host system 1

Source	Destination	Node ID						Frame ID					CAN-ID range (HEX)
		b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0	
BMU 0	Host system	0	0	0	0	0	1	(without BMU ID numbering)					0x20 to 3F
BMU 1		0	0	0	0	1	0	(Based on 7.1.6 Data format)					0x40 to 5F
BMU 2		0	0	0	0	1	1						0x60 to 7F
BMU 3		0	0	0	1	0	0						0x80 to 9F
BMU 4		0	0	0	1	0	1						0xA0 to BF
BMU 5		0	0	0	1	1	0						0xC0 to DF
BMU 6		0	0	0	1	1	1						0xE0 to FF
BMU 7		0	0	1	0	0	0						0x100 to 11F
BMU 8		0	0	1	0	0	1						0x120 to 13F
BMU 9		0	0	1	0	1	0						0x140 to 15F
BMU 10		0	0	1	0	1	1						0x160 to 17F
BMU 11		0	0	1	1	0	0						0x180 to 19F
BMU 12		0	0	1	1	0	1						0x1A0 to 1BF
BMU 13		0	0	1	1	1	0						0x1C0 to 1DF
BMU 14		0	0	1	1	1	1						0x1E0 to 1FF
BMU 15		0	1	0	0	0	0						0x200 to 21F
BMU 16		0	1	0	0	0	1						0x220 to 23F
BMU 17		0	1	0	0	1	0						0x240 to 25F
BMU 18		0	1	0	0	1	1						0x260 to 27F
BMU 19		0	1	0	1	0	0						0x280 to 29F
BMU 20		0	1	0	1	0	1						0x2A0 to 2BF
BMU 21		0	1	0	1	1	0						0x2C0 to 2DF
BMU 22		0	1	0	1	1	1						0x2E0 to 2FF

Table 3 CAN-ID assignment range(b) BMU → host system 2

Source	Destination	Node ID						Frame ID					CAN-ID range (HEX)
		b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0	
BMU 0	Host system	1	0	1	0	0	0	0	(without BMU ID numbering)				0x500 to 50F
BMU 1		1	0	1	0	0	0	1	(Based on 7.1.6 Data format)				0x510 to 51F
BMU 2		1	0	1	0	0	1	0					0x520 to 52F
BMU 3		1	0	1	0	0	1	1					0x530 to 53F
BMU 4		1	0	1	0	1	0	0					0x540 to 54F
BMU 5		1	0	1	0	1	0	1					0x550 to 55F
BMU 6		1	0	1	0	1	1	0					0x560 to 56F

Source	Destination	Node ID							Frame ID	CAN-ID range
BMU 7		1	0	1	0	1	1	1		0x570 to 57F
BMU 8		1	0	1	1	0	0	0		0x580 to 58F
BMU 9		1	0	1	1	0	0	1		0x590 to 59F
BMU 10		1	0	1	1	0	1	0		0x5A0 to 5AF
BMU 11		1	0	1	1	0	1	1		0x5B0 to 5BF
BMU 12		1	0	1	1	1	0	0		0x5C0 to 5CF
BMU 13		1	0	1	1	1	0	1		0x5D0 to 5DF
BMU 14		1	0	1	1	1	1	0		0x5E0 to 5EF
BMU 15		1	0	1	1	1	1	1		0x5F0 to 5FF
BMU 16		1	1	0	0	0	0	0		0x600 to 60F
BMU 17		1	1	0	0	0	0	1		0x610 to 61F
BMU 18		1	1	0	0	0	1	0		0x620 to 62F
BMU 19		1	1	0	0	0	1	1		0x630 to 63F
BMU 20		1	1	0	0	1	0	0		0x640 to 64F
BMU 21		1	1	0	0	1	0	1		0x650 to 65F
BMU 22		1	1	0	0	1	1	0		0x660 to 66F

Table 3 CAN-ID assignment range(c) Host system → BMU

Source	Destination	Node ID							Frame ID				CAN-ID range (HEX)
		b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0	
Host system	BMU 0	0	1	1	0	0	0	0	(without BMU ID numbering)				0x300 to 30F
	BMU 1	0	1	1	0	0	0	1	(Based on 7.1.6 Data format)				0x310 to 31F
	BMU 2	0	1	1	0	0	1	0					0x320 to 32F
	BMU 3	0	1	1	0	0	1	1					0x330 to 33F
	BMU 4	0	1	1	0	1	0	0					0x340 to 34F
	BMU 5	0	1	1	0	1	0	1					0x350 to 35F
	BMU 6	0	1	1	0	1	1	0					0x360 to 36F
	BMU 7	0	1	1	0	1	1	1					0x370 to 37F
	BMU 8	0	1	1	1	0	0	0					0x380 to 38F
	BMU 9	0	1	1	1	0	0	1					0x390 to 39F
	BMU 10	0	1	1	1	0	1	0					0x3A0 to 3AF
	BMU 11	0	1	1	1	0	1	1					0x3B0 to 3BF
	BMU 12	0	1	1	1	1	0	0					0x3C0 to 3CF
	BMU 13	0	1	1	1	1	0	1					0x3D0 to 3DF
	BMU 14	0	1	1	1	1	1	0					0x3E0 to 3EF
	BMU 15	0	1	1	1	1	1	1					0x3F0 to 3FF
	BMU 16	1	0	0	0	0	0	0					0x400 to 40F
	BMU 17	1	0	0	0	0	0	1					0x410 to 41F
	BMU 18	1	0	0	0	0	1	0					0x420 to 42F
	BMU 19	1	0	0	0	0	1	1					0x430 to 43F
	BMU 20	1	0	0	0	1	0	0					0x440 to 44F
	BMU 21	1	0	0	0	1	0	1					0x450 to 45F
	BMU 22	1	0	0	0	1	1	0					0x460 to 46F

7.1.4 Frame ID and data transmission interval

The frame ID needs to be managed by classifying the transmission intervals into three types depending on the importance of data to be sent. Table 4 shows the relationship between each frame ID and the data transmission interval. For the time data sent from the host system to BMU, refer to 6.3.13 Time adjustment procedure.

Table 4 Frame ID and data transmission interval

Data direction	Frame ID range	Data transmission accuracy	Main data contents
BMU → host system	0x00 to 0x03	Transmission at 120-ms interval	Max./min. cell voltage, temperature, current, protection info., SOC
	0x04 to 0x0F	Transmission at 960-ms interval	Failure info., balance info.
	0x10 to 0x1F	Irregular (only when requested)	All cell voltage/all cell temperature
Host system → BMU	0x00	Transmission at 120-ms interval	Start/stop instruction, system voltage
	0x01 to 0x07	Transmission at 1-minute interval (example)	Time
	0x08 to 0x0F	Irregular (only when necessary)	Cell voltage/temperature acquisition instruction

* All data of 2 bytes or more are arranged in the big-endian order.

The ensured data transmission accuracy range of BMU is within $\pm 10\%$ of each interval.

7.1.5 Frame ID and data item

Table 5 shows the relationship between each frame ID and the corresponding data item. Each data item includes the data size (the number of bytes), the presence/absence of a plus or minus sign, and the unit (engineering value per count). All data of 2 bytes or more are arranged in the big-endian order.

Table 5 Frame ID and data item

○ Host system → BMU

Frame ID	Data type	Item	Byte	Effective bits	Contents	1 count	Unit
0x00	Control signal	Operation instruction	1	5	Operation instruction signal for BMU	-	-
	System info.	Main circuit voltage	2	16	System voltage	0.1	[V]
0x01	Battery panel control info.	Time data	4	32 ^{Note 1}	Time info. of battery panel (lower 32 bits)	1	[s]
0x02	Battery panel control info.	Time data	4	32	Time info. of battery panel (upper 32 bits)	2 ³²	[s]
0x08	Control signal	Data acquisition location	1	8	Cell voltage/temperature acquisition location module No.	(No.1 to 56)	-

○ BMU → Host system

Frame ID	Data type	Item	Byte	Effective bits	Contents	1 count	Unit
0x00	Battery unit info.	Status	1	8	Unit contactor/abnormality info.	-	-
		Final charge/discharge info.	1	4	Status display of final charge/ discharge	-	-
		Overcharge/overdischarge/overtemp.	1	6	Status display of overcharge/overdischarge/overtemp.	-	-
		Unit SOC	2	16	State of charge of unit	1	[mAh]
		Unit SOH	2	16	Battery capacity deterioration status of unit	1	[mAh]
0x01	Battery unit info.	Max. cell voltage	2	16	Max. voltage value of all cells in unit	1	[mV]
		Min. cell voltage	2	16	Min. voltage value of all cells in unit	1	[mV]
		Max. cell temp.	1	7+S ^{Note 2}	Max. temp. value of all cells in unit ^{Note 6}	±1	[°C]
		Average cell temp.	1	7+S ^{Note 2}	Average temp. value of all cells in unit ^{Note 6}	±1	[°C]
		Min. cell temp.	1	7+S ^{Note 2}	Min. temp. value of all cells in unit ^{Note 6}	±1	[°C]
0x02	Battery unit info.	Max. cell module	1	8	Module No. containing max. cell voltage	(No.1 to 56)	-
		Min. cell module	1	8	Module No. containing min. cell voltage	(No.1 to 56)	-
		Max. temp. module	1	8	Module No. containing max. cell temp.	(No.1 to 56)	-
		Min. temp. module	1	8	Module No. containing min. cell temp.	(No.1 to 56)	-
0x03	Battery unit info.	Unit charge current	2	16	Charge current value of unit	10	[mA]
		Unit discharge current	2	16	Discharge current value of unit	10	[mA]
		Unit voltage	2	16	Unit voltage (total voltage value of all cells)	0.1	[V]
		BMU status supplemental information	1	7	Low power mode /overcurrent / BMU shutdown permission / pre-charge contactor state / current leak detection / current detection / SDC state	-	-
0x04	Battery unit info.	Max. cell temp. 2	2	15+S ^{Note 2}	Max. temp. value of all cells in unit ^{Note 7}	±0.1	[°C]
		Average cell temp. 2	2	15+S ^{Note 2}	Average temp. value of all cells in unit ^{Note 7}	±0.1	[°C]
		Min. cell temp. 2	2	15+S ^{Note 2}	Min. temp. value of all cells in unit ^{Note 7}	±0.1	[°C]
		BMU abnormality detail	1	8	BMU abnormality component and detailed information	-	-
0x05	Battery unit info.	BMU monitoring info.	1	3	Functional safety life signal error ^{Note 8} / Cell voltage deviation detection	-	-
		Unit SOC[%]	1	8	Percentage of state of charge of unit	1	[%]
0x06	Battery unit info.	CMU failure status	3	22	Bit used to display failed CMU	(No.1 to 22) (No.29 to 50)	-
			1	6	Bit used to display failed CMU	(No.23 to 28) (No.51 to 56)	-
		Failure/balance Status	2	13 ^{Note 4}	Bit used to display failed cell in module and presence of balance discharge	(No.28) (No.56)	-
0x07 to 0x0F	Battery unit info.	Failure/balance Status	2	13 ^{Note 4}	Bit used to display failed cell in module and presence of balance discharge	(No.1 to 27) (No.29 to 55)	-
0x07	Battery unit info.	Parallel no. of branch	1	8	Parallel no. of modules connected in parallel	0 to 56	-

0x10	Battery unit info.	Data acquisition location	1	8	Module No. of cell voltage/temp. acquisition location	(No.1 to 56)	-
0x10 to 0x13	Battery unit info.	Cell voltage η ^{Note 3}	2	16	Voltage of cell "n"	1	[mV]
0x16	Battery unit info.	Cell temp. m ^{Note 3}	1	7+S ^{Note 2}	Temp. value (6 points) of thermistor "m"	±1	[°C]
0x17, 0x18	Battery unit info.	Cell temp. p ^{Note 5}	1	7+S ^{Note 2}	Temp. value (13 points) of thermistor "p"	±1	[°C]

Note 1: Counter value in the unit of second with 1970/1/1 00:00:00'00 (UTC) as the base (64 bits in total)

Note 2: Uppermost bit "S" indicates the plus or minus sign, with a negative value indicated if S is "1." (2's complement expression)

Note 3: n: cell No. in the module (1 to 12), m: thermistor No. in the module (1 to 6)

Note 4: The uppermost bit indicates that some cell in the module is in the balance discharge status.

Note 5: p: Thermistor No. (1 to 13) in the module

Note 6: Calculated value based on the 6-point temperature sent by each module

Note 7: Calculated value based on the 13-point temperature (Max./Avg./Min.in 0.1°C) sent by each module

Note 8: Functional safety life signal error can only be used with functional safety battery modules.

7.1.6 Data format

Table 6 shows the data arrangement on the data frame of each data item shown in Table 5. CAN-ID frame not listed in the table is not sent. Note that the data is sent by BMU in a random order, so that the data is not necessarily sent in the display order of CAN-ID in Table 6.

For the reception process on the host system side, the implementation should not depend on the order. The received data should be judged after checking the CAN-ID.

(Note) Precaution common to CAN / Ethernet communication

Be sure to set 0 (0x00) for every "reserved" in the data fields for (Host system → BMU).

Table 6 Arrangement of data fields

• Data format (BMU → Host system) [BMU No.: 1 to 22(0x01 to 0x16)]

CAN ID	1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte
0x20+BMU No.×0x20	Sync. counter	Status	Final charge/discharge info.	Overcharge/overdischarge /overtemp. info.	Unit SOC		Unit SOH	
0x21+BMU No.×0x20	Sync. counter	Max. voltage value of cells in unit		Min. voltage value of cells in unit		Max. temp. value of cells in unit	Average temp. value of cells in unit	Min. temp. value of cells in unit
0x22+BMU No.×0x20	Sync. counter	Max. cell voltage value module	Min. cell voltage value module	Max. cell temp. value module	Min. cell temp. value module	(don't care) *1		
0x23+BMU No.×0x20	Sync. counter	Unit charge current		Unit discharge current		Unit voltage		BMU status supplemental information
0x24+BMU No.×0x20	Sync. counter	Max. temp. value of cells in unit 2		Average temp. value of cells in unit 2		Min. temp. value of cells in unit 2		BMU abnormality detail

CAN ID	1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte
0x25+BMU No.×0x20	Sync. counter	BMU monitoring info. *	reserved	reserved	reserved	reserved	reserved	Unit SOC [%]
0x26+BMU No.×0x20	Sync. counter	CMU failure status (Module 1 to 22)			CMU failure status (Module 23 to 28)	Module 28 Cell failure/balance status		reserved
0x27+BMU No.×0x20	Sync. counter	Module 1 Cell failure/balance status		Module 2 Cell failure/balance status		Module 3 Cell failure/balance status		Parallel no. of branch
0x28+BMU No.×0x20	Sync. counter	Module 4 Cell failure/balance status		Module 5 Cell failure/balance status		Module 6 Cell failure/balance status		reserved
0x29+BMU No.×0x20	Sync. counter	Module 7 Cell failure/balance status		Module 8 Cell failure/balance status		Module 9 Cell failure/balance status		reserved
0x2A+BMU No.×0x20	Sync. counter	Module 10 Cell failure/balance status		Module 11 Cell failure/balance status		Module 12 Cell failure/balance status		reserved
0x2B+BMU No.×0x20	Sync. counter	Module 13 Cell failure/balance status		Module 14 Cell failure/balance status		Module 15 Cell failure/balance status		reserved
0x2C+BMU No.×0x20	Sync. counter	Module 16 Cell failure/balance status		Module 17 Cell failure/balance status		Module 18 Cell failure/balance status		reserved
0x2D+BMU No.×0x20	Sync. counter	Module 19 Cell failure/balance status		Module 20 Cell failure/balance status		Module 21 Cell failure/balance status		reserved
0x2E+BMU No.×0x20	Sync. counter	Module 22 Cell failure/balance status		Module 23 Cell failure/balance status		Module 24 Cell failure/balance status		reserved
0x2F+BMU No.×0x20	Sync. counter	Module 25 Cell failure/balance status		Module 26 Cell failure/balance status		Module 27 Cell failure/balance status		reserved
0x30+BMU No.×0x20	Sync. counter	Cell voltage 1		Cell voltage 2		Cell voltage 3		Data acquisition location (Module No.) ^{*3}
0x31+BMU No.×0x20	Sync. counter	Cell voltage 4		Cell voltage 5		Cell voltage 6		reserved
0x32+BMU No.×0x20	Sync. counter	Cell voltage 7		Cell voltage 8		Cell voltage 9		reserved
0x33+BMU No.×0x20	Sync. counter	Cell voltage 10		Cell voltage 11		Cell voltage 12		reserved
0x36+BMU No.×0x20 ^{*3}	Sync. counter	Cell temp. 1 (Temp. 6 points)	Cell temp. 2 (Temp. 6 points)	Cell temp. 3 (Temp. 6 points)	Cell temp. 4 (Temp. 6 points)	Cell temp. 5 (Temp. 6 points)	Cell temp. 6 (Temp. 6 points)	reserved
0x37+BMU No.×0x20 ^{*3,4}	Sync. counter	Cell temp. 1 (Temp. 13 points)	Cell temp. 2 (Temp. 13 points)	Cell temp. 3 (Temp. 13 points)	Cell temp. 4 (Temp. 13 points)	Cell temp. 5 (Temp. 13 points)	Cell temp. 6 (Temp. 13 points)	Cell temp. 7 (Temp. 13 points)
0x38+BMU No.×0x20 ^{*3,4}	Sync. counter	Cell temp. 8 (Temp. 13 points)	Cell temp. 9 (Temp. 13 points)	Cell temp. 10 (Temp. 13 points)	Cell temp. 11 (Temp. 13 points)	Cell temp. 12 (Temp. 13 points)	Cell temp. 13 (Temp. 13 points)	reserved
0x500+BMU No.×0x10 ^{*2}	Sync. counter	CMU failure status (Module 29 to 50)			CMU failure status (Module 51 to 56)	Module 56 Cell failure/balance status		reserved
0x501+BMU No.×0x10 ^{*2}	Sync. counter	Module 29 Cell failure/balance status		Module 30 Cell failure/balance status		Module 31 Cell failure/balance status		reserved

CAN ID	1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte
0x502+BMU No.×0x10 ^{*2}	Sync. counter	Module 32 Cell failure/balance status		Module 33 Cell failure/balance status		Module 34 Cell failure/balance status		reserved
0x503+BMU No.×0x10 ^{*2}	Sync. counter	Module 35 Cell failure/balance status		Module 36 Cell failure/balance status		Module 37 Cell failure/balance status		reserved
0x504+BMU No.×0x10 ^{*2}	Sync. counter	Module 38 Cell failure/balance status		Module 39 Cell failure/balance status		Module 40 Cell failure/balance status		reserved
0x505+BMU No.×0x10 ^{*2}	Sync. counter	Module 41 Cell failure/balance status		Module 42 Cell failure/balance status		Module 43 Cell failure/balance status		reserved
0x506+BMU No.×0x10 ^{*2}	Sync. counter	Module 44 Cell failure/balance status		Module 45 Cell failure/balance status		Module 46 Cell failure/balance status		reserved
0x507+BMU No.×0x10 ^{*2}	Sync. counter	Module 47 Cell failure/balance status		Module 48 Cell failure/balance status		Module 49 Cell failure/balance status		reserved
0x508+BMU No.×0x10 ^{*2}	Sync. counter	Module 50 Cell failure/balance status		Module 51 Cell failure/balance status		Module 52 Cell failure/balance status		reserved
0x509+BMU No.×0x10 ^{*2}	Sync. counter	Module 53 Cell failure/balance status		Module 54 Cell failure/balance status		Module 55 Cell failure/balance status		reserved
0x50B+ BMU No.×0x10 ^{*4}	Sync. counter	BMU abnormality detail 2						
0x50C+ BMU No.×0x10 ^{*5}	Sync. counter	CMU-ID	CMU Serial No.			CMU Lot No.		

*1: An undefined value is entered in the 6th, 7th, and 8th bytes of 0x22+BMU No.×0x20. The data should be ignored.

*2: Transmission of this CAN-ID can be controlled by BMU parameter settings if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 101, e.g. "Model No.: B2***101."

*3: If the data acquisition location (Module No.) is 1 to 56, the temperature of 6 points is sent with 0x36+BMU No.×0x20. If the data acquisition location (Module No.) is 101 to 156, the temperature of 13 points (of the module 1 to 56) is sent with 0x37+BMU No.×0x20 and 0x38+BMU No.×0x20. If the data acquisition request is received while the power supply for the module (CMU) is not turned on, SNA is sent for the cell voltage and temperature of 0x30 to 0x33, 0x36, 0x37, and 0x38.

*4: The transmission can be permitted with the parameter setting if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 106, e.g. "Model No.: B2***106."

*5: The transmission can be permitted with the parameter setting if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 110, e.g. "Model No.: B2***110."

*6: BMU monitoring info. can be sent if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 201, e.g. "Model No.: B2***201."

• Data format (Host system → BMU) [BMU No.:1 to 22(0x01 to 0x16)]

CAN ID	1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte
0x300+ BMU No.×0x10	Sync. counter	Status	Main circuit voltage		Host system instructed minimum cell voltage ^{*3}		reserved	reserved
0x301+ BMU No.×0x10	Sync. counter	Time (lower 32 bits)				reserved	reserved	reserved
0x302+ BMU No.×0x10	Sync. counter	Time (upper 32 bits)				reserved	reserved	reserved
0x308+ BMU No.×0x10	Sync. counter	Data acquisition location (Module No.) ^{*1}	reserved	reserved	reserved	reserved	reserved	reserved

* In the range from 0x300+BMU No. X 10 to 0x302+BMU No. X 0x10, transmission to all BMUs is possible by sending the CAN-ID from 0x7E1 to 0x7E3 (broadcasting).

* Be sure to set 0x00 for "reserved" above.

*1: To acquire the temperature of 6 points, designate 1 to 56. To acquire the temperature of 13 points^{*2}, designate 101 to 156 (100 + Module 1 to 56). Note that, if the host system communication is CAN, the data is ignored even if 100 is designated.

*2: The last three digits of the Model No. on the label attached to BMU need to be equal to or greater than 106, e.g. "Model No.: B2***106."

*3: The last three digits of the Model No. on the label attached to BMU need to be equal to or greater than 110, e.g. "Model No.: B2***110."

7.1.7 Details of data

Details on the format of each data item are provided below.

<Common to all frames>

7.1.7.1 Synchronization counter

All IDs (1st byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

• 0 to 255

The counter that is incremented for each BMU/host system for each frame transmission interval. When it is reset, it starts from 0. In the normal condition, BMU/host system receives the CAN data with the continuous counter values for each interval.

<BMU → Host system>

7.1.7.2 Status

CAN ID 0x20+BMU No.×0x20 (2nd byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0

- <b0> Main contactor closed status (normal operation) 0: Open/1: close
- <b1> Main contactor closing waiting status 0: Not wait/1: wait for close
- <b2> Current leak detection completed 0: Not finished detection/1: waiting to return from leak detection finished
- <b3> Unit preparation 0: Not prepared/1: prepared
- <b4> BMU abnormality 0: Normal/1: Abnormal
- <b5> Pre-charge circuit abnormality 0: Normal/1: Abnormal
- <b6> Current leak detection 0: Normal/1: Abnormal
- <b7> BMU life signal (toggled at interval of 960 ms) 0/1 (inverse value is set at interval of 960 ms)

- *1: <b0> With MCCB in use, it does not synchronize with the actual MCCB closing status at the start.
- *2: <b1> Becomes 1 when waiting for the main circuit voltage and unit voltage to match after diagnosing contactor welding.
- *3: <b2> To be turned ON when the current leak check is completed, whether the current leak is detected or not. Status of waiting for the current leak detection instruction to clear.
When the current leak detection of <b6> is 0, the contactor can be closed again by turning the current leak detection instruction OFF.
- *4: <b3> To be turned ON when powered on, and turned OFF after the operation instruction is turned ON.
- *5: <b4> BMU abnormality³ is the status where the contactor cannot be closed due to BMU failure detection.
- *6: <b5> Available only if pre-charge contactor is in use.
- *7: <b6> Leak status. Turn the operation permit instruction OFF and stop BMU.
- *8: Notify <b2> = 0, <b6> = 0 in case of no current leak sensor.

³ The host system is notified via 7.1.7.23 BMU abnormality detail and 7.1.7.24 BMU abnormality detail 2. For details on BMU abnormality, refer to “5.1. Battery unit protection & error notification” of “Battery System Component BMU-2G F/W Instruction Manual [2]”.

7.1.7.3 Final charge/discharge information

CAN ID 0x20+BMU No.×0x20 (3rd byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0

- <b0> Nearly final charge 0: Cell voltage is smaller than the nearly final charge/1: Cell voltage is equal to or greater than the nearly final charge.
- <b1> Final charge 0: Cell voltage is smaller than the final charge/1: Cell voltage is equal to or greater than the final charge.
- <b2> to <b3> reserved
- <b4> Final discharge 0: Cell voltage is greater than the final discharge/1: Cell voltage is equal to or smaller than the final discharge.
- <b5> Overdischarge warning 0: Cell voltage is greater than the overdischarge recovery/1: Cell voltage is equal to or smaller than the overdischarge warning.
- <b6> Nearly final discharge 0: Cell voltage is greater than the nearly final discharge/1: Cell voltage is equal to or smaller than the nearly final discharge.
- <b7> reserved

7.1.7.4 Overcharge/overdischarge/over temperature information

CAN ID 0x20+BMU No.×0x20 (4th byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0

- <b0> Overcharge 1=overcharge (alert) 0: Normal/1: Abnormal
- <b1> Overcharge 2=overcharge (fault) 0: Normal/1: Abnormal
- <b2> Overdischarge 1=overdischarge (alert) 0: Normal/1: Abnormal
- <b3> Overdischarge 2=overdischarge (fault) 0: Normal/1: Abnormal
- <b4> Temp. high 1=high temp. protection 0: Normal/1: Abnormal
- <b5> Temp. high 2= overtemp. (alert) 0: Normal/1: Abnormal
- <b6> to <b7> reserved

7.1.7.5 Unit SOC

CAN ID 0x20+BMU No.×0x20 (5th and 6th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0xFFFF: SOC not determined or unable to be calculated (SNA)]

SOC is indicated by 1mAh/LSB (65535mAh-0mAh). In case of a parallel connection, the SOC of one branch is indicated.

7.1.7.6 Unit SOH

CAN ID 0x20+BMU No.×0x20 (7th and 8th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF

SOH (battery capacity with deterioration taken into consideration) is indicated by 1mAh/LSB (65535mAh-0mAh). In case of a parallel connection, the SOH of one branch is indicated.

This feature is optional. The SOH value to be set is a fixed value (battery module rated capacity).

7.1.7.7 Max. voltage value of cells in unit

CAN ID 0x21+BMU No.×0x20 (2nd and 3rd bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0xFFFF: SNA (undetermined or H/W failure)]

The cell voltage is indicated by 1mV/LSB (65534mV-0mV).

7.1.7.8 Min. voltage value of cells in unit

CAN ID 0x21+BMU No.×0x20 (4th and 5th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0xFFFF: SNA (undetermined or H/W failure)]

The cell voltage is indicated by 1mV/LSB (65534mV-0mV).

7.1.7.9 Max. temperature value of cells in unit

CAN ID 0x21+BMU No.×0x20 (6th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: SNA (undetermined or H/W failure)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C. Rounded off to the nearest whole number.

7.1.7.10 Average temperature value of cells in unit

CAN ID 0x21+BMU No.×0x20 (7th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: SNA (undetermined or H/W failure)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C.
Rounded off to the nearest whole number.

7.1.7.11 Min. temperature value of cells in unit

CAN ID 0x21+BMU No.×0x20 (8th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: SNA (undetermined or H/W failure)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C. Rounded off to the nearest whole number.

7.1.7.12 Max. cell voltage value module

CAN ID 0x22+BMU No.×0x20 (2nd byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 [0x00: Voltage not determined status]

Module No. containing the max. cell voltage value in unit. Usage range: module No. 1 to 56.

If there are multiple modules, the smallest module No. is sent to the host system.

7.1.7.13 Min. cell voltage value module

CAN ID 0x22+BMU No.×0x20 (3rd byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 [0x00: Voltage not determined status]

Module No. containing the min. cell voltage value in unit. Usage range: module No. 1 to 56

If there are multiple modules, the smallest module No. is sent to the host system.

7.1.7.14 Max. cell temperature value module

CAN ID 0x22+BMU No.×0x20 (4th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 [0x00: Temperature not determined status]

Module No. containing the max. cell temperature value in unit. Usage range: module No. 1 to 56.

If there are multiple modules, the smallest module No. is sent to the host system.

7.1.7.15 Min. cell temperature value module

CAN ID 0x22+BMU No.×0x20 (5th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 [0x00: Temperature not determined status]

Module No. containing the min. cell temperature value in unit. Usage range: module No. 1 to 56.

If there are multiple modules, the smallest module No. is sent to the host system.

7.1.7.16 Unit charge current

CAN ID 0x23+BMU No.×0x20 (2nd and 3rd bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF

The unit current in the charge direction is indicated by 0.01A/LSB (655.35A to 0A).

0A when discharging.

The measured value starts to be sent at the BMU startup, and if the BMU detects a current sensor error, the previous value is retained.

7.1.7.17 Unit discharge current

CAN ID 0x23+BMU No.×0x20 (4th and 5th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF

The unit current in the discharge direction is indicated by 0.01A/LSB (655.35A to 0A).

0A when charging.

The measured value starts to be sent at the BMU startup, and if the BMU detects a current sensor error, the previous value is retained.

7.1.7.18 Unit voltage

CAN ID 0x23+BMU No.×0x20 (6th and 7th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF

The unit voltage is indicated by 0.1V/LSB (6553.5V to 0V). In case of a parallel connection, the smallest branch voltage in the unit is indicated by calculating the voltage of each branch connected in parallel by adding up the cell voltages.

The total value of cell voltages starts to be sent at the BMU startup, and if the BMU detects a CMU abnormality, the previous value is retained.

7.1.7.19 BMU status supplemental information

CAN ID 0x23+BMU No.×0x20 (8th byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0
<b0> Low power mode	0: Not low power mode/1:Low power mode (CMU power off)							
<b1> Overcurrent	0: Normal/1:Overcurrent detected							
<b2> BMU shutdown permission	0: Not shutdown/1: Waiting for shutdown							
<b3> Pre-charge contactor state	0: Open/1: Close							
<b4> Leak detection	0: Not detecting/1: Detecting							
<b5> reserved								
<b6> SDC (service disconnecter) state	0:Close/1:Open							
<b7> Non-volatile storage failure	0:Normal/1:Failure (initialized)							

7.1.7.20 Max. temp. value of cells in unit 2

CAN ID 0x24+BMU No.×0x20 (2nd and 3rd bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0x7FFF: SNA (undetermined or H/W failure)]

Maximum cell temperature in unit is indicated by 0.1°C/LSB (2's complement multiplied by 10).

The value is effective if the parameter "CMU type" is either Type3(0) or TypeS(2). Otherwise, the value is SNA.

7.1.7.21 Average temp. value of cells in unit 2

CAN ID 0x24+BMU No.×0x20 (4th and 5th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0x7FFF: SNA (undetermined or H/W failure)]

Average of cell temperature in unit is indicated by 0.1°C/LSB (2's complement multiplied by 10).

The value is effective if the parameter "CMU type" is either Type3(0) or TypeS(2). Otherwise, the value is SNA.

7.1.7.22 Min. temp. value of cells in unit 2

CAN ID 0x24+BMU No.×0x20 (6th and 7th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0x7FFF: SNA (undetermined or H/W failure)]

Minimum cell temperature in unit is indicated by 0.1°C/LSB (2's complement multiplied by 10).

The value is effective if the parameter "CMU type" is either Type3(0) or TypeS(2). Otherwise, the value is SNA.

7.1.7.23 BMU abnormality detail

CAN ID 0x24+BMU No.×0x20 (8th byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0

<b7> to <b4>: Abnormal component, range: 0x0 to 0xF

<b3> to <b0>: Detail information, range: 0x0 to 0xF

When BMU detects abnormality or fault, it indicates abnormal component as shown below.

When multiple abnormalities are detected, the smallest detected value is provided.

		<b3> - <b0>: detail information							
		Communication failure	Power failure	Ground fault	Short circuit	Welding	Open	Circuit failure	Other failure
<b7> - <b4>: abnormal component		0x1	0x2	0x3	0x4	0x5	0x6	0x7	0xF
0x1	CMU	Communication Abnormality 2201	Power abnormality 3101	Power ground fault 3203	—	—	—	CMU abnormality (SNA)*2 3204	Automatic numbering abnormality 3103
0x2	Current sensor	Signal line abnormality 3207	Power abnormality 3206	—	—	—	—	Sensor body 3208/3210	—
0x3	Main circuit	—	—	—	—	—	SDC open*2 4301	—	SDC abnormality (main circuit voltage difference abnormality) 3209
0x4*1	Ground control relay	—	—	Ground fault 4108	Short circuit 4101	Welding 4201	—	—	—
0x5	Main contactor (P)	—	—	Ground fault 4105	Short circuit 4104	Welding 4203	—	—	—
0x6	Main contactor (N)	—	—	Ground fault 4107	Short circuit 4106	Welding 4202	—	—	—
0x7	Pre-charge contactor	—	—	Ground fault 4103	Short circuit 4102	—	—	—	—
0x8	BMU substrate	—	Reference voltage abnormality*2 3305	—	—	—	—	—	Parameter abnormality 3402
0x9	Non-volatile memory	—	—	—	—	—	—	—	Backup abnormality*2
0xF	Other components	—	Leak detection sensor power supply*2 3201	Current leak*2 3205	—	—	—	Current leak pre-check*2 3202	—

* The 4-digit number in each cell is the trace code provided on BMU Battery System Protection Item List in SPC-COM-E0036 Battery System Components for Social Infrastructure Applications Standard Specification.

*1: In the case of GND control relay use

*2: 0 is sent for 7.1.7.2 Status <b4> BMU abnormality.

7.1.7.24 BMU abnormality detail 2

CAN ID 0x50B+BMU No.×0x10

The bit information of every part where BMU has detected abnormality/failure is sent as 1.

The data is sent when BMU mode parameter 1 Operation mode bit1 is 1.

In the table below, M in the Byte column indicates the byte location from the top of the frame as defined for each type of the host system communication. M is 4 for Ethernet communication, and 2 for CAN communication.

Byte	Bit	Abnormality/failure details	BMU abnormality detail code (hex) Refer to 7.1.7.23
M	7	Current sensor abnormality (sensor body)	27
	6	Current sensor abnormality (power supply)	22
	5	Current sensor abnormality (signal line)	21
	4	CMU-UART communication abnormality	1F
	3	CMU abnormality	17
	2	CMU power supply abnormality	13
	1	CMU power supply shutdown abnormality	12
	0	CMU-CAN communication abnormality	11
M+1	7	Main contactor (P) drive circuit (short circuit)	54
	6	Main contactor (P) drive circuit (ground fault)	53
	5	GND control relay welding	45
	4	GND control relay drive circuit (short circuit)	44
	3	GND control relay drive circuit (ground fault)	43
	2	SDC failure / Fusing / Cable error between modules (open) / Contactor open failure	3F
	1	Reserved	-
	0	SDC open	36
M+2	7	Parallel connection configuration abnormality	8F
	6	AD reference voltage abnormality	82
	5	Pre-charge contactor drive circuit (short circuit)	74
	4	Pre-charge contactor drive circuit (ground fault)	73
	3	Main contactor (N) welding	65
	2	Main contactor (N) drive circuit (short circuit)	64
	1	Main contactor (N) drive circuit (ground fault)	63
	0	Main contactor (P) welding	55
M+3	4-7	Reserved	-
	3	Current leak sensor pre-check circuit abnormality	F7
	2	Cable error between modules (ground fault/current leak detection)	F3
	1	Current leak sensor power supply abnormality	F2
	0	Backup abnormality	9F
M+4 to M+6	0-7	Reserved	-

7.1.7.25 BMU Monitoring Information

CAN ID 0x25+BMU No.×0x20 (2nd byte)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0
<b0> Functional Safety Life Signal1 error	0:Normal/ 1:Abnormal							
<b1> Functional Safety Life Signal2 error	0:Normal/ 1:Abnormal							
<b2> Cell Voltage Deviation detection	0:Not detected/ 1:Detected							
<b3> reserved								
<b4> reserved								
<b5> reserved								
<b6> reserved								
<b7> reserved								

Functional safety life signal error information is valid only when using functional safety battery module.

7.1.7.26 Unit SOC [percentage]

CAN ID 0x25+BMU No.×0x20 (8th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x00 to 0x64 [0xFF: undetermined value (SNA)]

SOC is indicated by 1%/LSB (0%-100%).

Matching with (unit SOC / unit SOH) × 100. (truncated to a whole number)

7.1.7.27 CMU failure status (module 1 to 22)

CAN ID 0x26+BMU No.×0x20 (2nd to 4th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									-	-	b21	b20	b19	b18	b17	b16
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0

- 0x000000 to 0x3FFFFFF

<b0> CMU failure flag (module 1) 0: Normal/1: Abnormal

<b1> CMU failure flag (module 2) 0: Normal/1: Abnormal

<b2> CMU failure flag (module 3) 0: Normal/1: Abnormal

:

<b20> CMU failure flag (module 21) 0: Normal/1: Abnormal

<b21> CMU failure flag (module 22) 0: Normal/1: Abnormal

* If CMU communication abnormality (including SNA detection) occurs and causes the contactor to open, the failure flag is set to "1."

7.1.7.28 CMU failure status (module 23 to 28)

CAN ID 0x26+BMU No.×0x20 (5th byte)

Bit	7	6	5	4	3	2	1	0
	-	-	b5	b4	b3	b2	b1	b0

• 0x00 to 0x3F

<b0> CMU failure flag (module 23) 0: Normal/1: Abnormal

<b1> CMU failure flag (module 24) 0: Normal/1: Abnormal

<b2> CMU failure flag (module 25) 0: Normal/1: Abnormal

<b3> CMU failure flag (module 26) 0: Normal/1: Abnormal

<b4> CMU failure flag (module 27) 0: Normal/1: Abnormal

<b5> CMU failure flag (module 28) 0: Normal/1: Abnormal

* If CMU communication abnormality (including SNA detection) occurs and causes the contactor to open, the failure flag is set to “1.”

7.1.7.29 Module 1 cell failure/balance status

CAN ID 0x27+BMU No.×0x20 (2nd and 3rd bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	b15	-	-	-	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0

• 0x0000 to 0x0FFF, 0x8000 to 0x8FFF

<b0> Cell failure flag (cell 1) 0: Normal/1: Abnormal

<b1> Cell failure flag (cell 2) 0: Normal/1: Abnormal

<b2> Cell failure flag (cell 3) 0: Normal/1: Abnormal

:

<b10> Cell failure flag (cell 11) 0: Normal/1: Abnormal

<b11> Cell failure flag (cell 12) 0: Normal/1: Abnormal

<b15> Balance status 0: balance stop/1: balance discharge underway in some cell

* If overcharge/overdischarge (alert) occurs and causes the contactor to open, the failure flag is set to “1.”

(The same applies to cell failure/balance status for Modules 2 through 28.)

7.1.7.30 Branch parallel number

CAN ID 0x27+BMU No.×0x20 (8th byte)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	-	-	-	-	-	-	-	-	-	-	b5	b4	b3	b2	b1	b0

• 0x00 to 0x38 [0x01 means abnormality]

Branch number of the unit is provided within the range from 0 to 56.

0 means no branch in parallel connection.

7.1.7.31 Cell voltage 1 to 12

CAN ID 0x30+BMU No.×0x20 (2nd through 7th bytes)CAN ID 0x31+BMU No.×0x20 (2nd through 7th bytes)CAN ID 0x32+BMU No.×0x20 (2nd through 7th bytes)CAN ID 0x33+BMU No.×0x20 (2nd through 7th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x0000 to 0xFFFF [0xFFFF: SNA (undetermined or H/W failure)]

The cell voltage is indicated by 1mV/LSB (65534mV to 0mV).

(Note that the fixed value 0xAAAA is set for Cell voltage10 through 12 of 2p9s modules, or for Cell voltage12 of 2p11s modules, respectively.)

SNA is provided until CMU automatic numbering is completed or after BMU is stopped.

7.1.7.32 Data acquisition location module No.

CAN ID 0x30+BMU No.×0x20 (8th byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 (0x00: not specified)

The module No. used for acquiring voltage and temperature (6 points) is specified with module no. 1 through 56.

- 0x65 to 0x9C (Host system communication CAN/Ethernet)

The module No. used for acquiring voltage and temperature (13 points) is specified with module no. 1 through 56 + 100.

7.1.7.33 Cell temperature 1 through 6 (temperature 6 points)

CAN ID 0x36+BMU No.×0x20 (2nd through 7th bytes)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: Temperature abnormality or not determined (SNA)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C. Rounded off to the nearest whole number.

SNA is provided until CMU automatic numbering is completed or after BMU is stopped.

7.1.7.34 Cell temperature 1 through 7 (temperature 13 points)

CAN ID 0x37+BMU No.×0x20 (2nd through 8th bytes)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: Temperature abnormality or not determined (SNA)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C. Rounded off to the nearest whole number. The identical value is provided until the temperature acquired for the 13 points in BMU is updated.

SNA is provided until CMU automatic numbering is completed or after BMU is stopped.

7.1.7.35 Cell temperature 8 through 13 (temperature 13 points)

CAN ID 0x37+BMU No.×0x20 (2nd through 7th bytes)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x80 to 0x7F [0x7F: Temperature abnormality or not determined (SNA)]

The temperature is indicated by 1°C/LSB (2's complement). Usage range: -40 to 125°C. Rounded off to the nearest whole number.

The identical value is provided until the temperature acquired for the 13 points in BMU is updated.

7.1.7.36 CMU failure status (module 29 to 50)

CAN ID 0x500+BMU No.×0x10 (2nd to 4th bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
									-	-	b21	b20	b19	b18	b17	b16
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0

- 0x000000 to 0x3FFFFF

<b0> CMU failure flag (module 29) 0: Normal/1: Abnormal

<b1> CMU failure flag (module 30) 0: Normal/1: Abnormal

<b2> CMU failure flag (module 31) 0: Normal/1: Abnormal

:

<b20> CMU failure flag (module 49) 0: Normal/1: Abnormal

<b21> CMU failure flag (module 50) 0: Normal/1: Abnormal

* If CMU communication abnormality (including SNA detection) occurs and causes the contactor to open, the failure flag is set to "1."

7.1.7.37 CMU failure status (module 51 to 56)

CAN ID 0x500+BMU No.×0x10 (5th bytes)

Bit	7	6	5	4	3	2	1	0
	-	-	b5	b4	b3	b2	b1	b0

- 0x00 to 0x3F

<b0> CMU failure flag (module 51) 0: Normal/1: Abnormal

<b1> CMU failure flag (module 52) 0: Normal/1: Abnormal

<b2> CMU failure flag (module 53) 0: Normal/1: Abnormal

<b3> CMU failure flag (module 54) 0: Normal/1: Abnormal

<b4> CMU failure flag (module 55) 0: Normal/1: Abnormal

<b5> CMU failure flag (module 56) 0: Normal/1: Abnormal

* If CMU communication abnormality (including SNA detection) occurs and causes the contactor to open, the failure flag is set to "1."

7.1.7.38 Module 29 cell failure/balance status

CAN ID 0x501+BMU No.×0x10 (2nd and 3rd bytes)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	b15	-	-	-	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0

• 0x0000 to 0x0FFF, 0x8000 to 0x8FFF

<b0> Cell failure flag (cell 1) 0: Normal/1: Abnormal

<b1> Cell failure flag (cell 2) 0: Normal/1: Abnormal

<b2> Cell failure flag (cell 3) 0: Normal/1: Abnormal

:

<b10> Cell failure flag (cell 11) 0: Normal/1: Abnormal

<b11> Cell failure flag (cell 12) 0: Normal/1: Abnormal

<b15> Balance status 0: balance stop/1: Any cell under balance discharge

* If overcharge/overdischarge (alert) which causes the contactor to open occurs, the failure flag is set to "1".

(The same applies to cell failure/balance status for Modules 30 through 56.)

7.1.7.39 CMU-ID

CAN ID 0x50C+BMU No.×0x10 (2nd byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

• 0x01 to 0x38

CMU-ID in unit. Usage range: 1 to 56.

7.1.7.40 CMU Serial No.

CAN ID 0x50C+BMU No.×0x10 (3rd to 5th bytes)

Bit	7	6	5	4	3	2	1	0
Byte1	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶
Byte2	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸
Byte3	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

• 0x000000~0xFFFFFFFF

e.g. 0x001E61 = 7777

7.1.7.41 CMU Lot No. (the date of manufacture)

CAN ID 0x50C+BMU No.×0x10 (6th to 8th bytes)

Bit	7	6	5	4	3	2	1	0
Byte1	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶
Byte2	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸
Byte3	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

<byte1> Day : 0x01~0x1F (1 to 31)

<byte2> Month : 0x01~0x0C (1 to 12)

<byte3> Year : 0x00~0x63 (00 to 99 : Last two digits of 20xx)

e.g. 0x0F0B16 (Nov.15, 2022)

<Host system → BMU>**7.1.7.42 Status**CAN ID 0x300+BMU No.×0x10 (2nd byte) / 0x7E1 (broadcasting)

Bit	7	6	5	4	3	2	1	0
	b7	b6	b5	b4	b3	b2	b1	b0

<b0> Operation instruction 0: Stop/1: operation (contactor powering on permitted)

<b1> Power failure trip instruction 0: Normal/1: power failure stop instruction (trip)

<b2> Cell balance permit signal 0: Cell balance stop/1: cell balance permitted

<b3> Current leak detection instruction signal 0: Current leak detection stop/1: current leak detection instruction

<b4> to <b6> reserved Fixed at 0

<b7> Host system life signal (toggled at interval of 960 ms) 0/1 (inverse value is set at interval of 960 ms)

*1: The transmission of the instruction for operation stop (<b0>=0), power failure trip (<b1>=1) or current leak detection (<b3>=1) should be continued until the notification of contactor open is provided from BMU.

By default, BMU stops operation after *three* consecutive detections of the instruction, but can be made to stop with a *single* detection with a parameter change.

(For details, refer to the note in section 6.3.3 Normal stop procedure.)

*2: In case the toggle of the host system life signal remains unchanged for 2 seconds or more, BMU registers the abnormal log.

7.1.7.43 Main circuit voltageCAN ID 0x300+BMU No.×0x10 (3rd and 4th bytes) / 0x7E1 (broadcasting)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

• 0x0000 to 0xFFFF

The bus-bar voltage should be specified by 0.1V/LSB (6553.5V to 0V). The margin of error should be within 1%.

7.1.7.44 Host system instructed minimum cell voltageCAN ID 0x300+BMU No.×0x10 (5th and 6th bytes) / 0x7E1 (broadcasting)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

• 0x0000 to 0xFFFF

Host System Instructed Minimum Voltage (abbr. HSIMV) should be specified by 1mV/LSB (65535mV to 0mV). However, effective range equals to from “Balancer lower limit threshold” to “Balancer upper limit threshold” in BMU tuning parameter1.

Refer to “9.1 Cautions for controlling the battery unit” to control BMU properly in host system instructed cell-balancing.

7.1.7.45 Time data 1

CAN ID 0x301+BMU No.×0x10 (2nd to 5th bytes) / 0x7E2 (broadcasting)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ³¹	2 ³⁰	2 ²⁹	2 ²⁸	2 ²⁷	2 ²⁶	2 ²⁵	2 ²⁴	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x00000000 to 0xFFFFFFFF

The lower 32 bits of the elapsed seconds (64 bits) should be specified with start of "0" at 0:00:00 on January 1, 1970 (UTC).

7.1.7.46 Time data 2

CAN ID 0x302+BMU No.×0x10 (2nd to 5th bytes) / 0x7E3 (broadcasting)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	2 ³¹	2 ³⁰	2 ²⁹	2 ²⁸	2 ²⁷	2 ²⁶	2 ²⁵	2 ²⁴	2 ²³	2 ²²	2 ²¹	2 ²⁰	2 ¹⁹	2 ¹⁸	2 ¹⁷	2 ¹⁶
	2 ¹⁵	2 ¹⁴	2 ¹³	2 ¹²	2 ¹¹	2 ¹⁰	2 ⁹	2 ⁸	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x00000000 to 0x7FFFFFFF

The upper 32 bits of the elapsed seconds (64 bits) should be specified with start of "0" at 0:00:00 on January 1, 1970 (UTC).

The time data is to remain 0 until 06:28:15 on February 7, 2106.

The time data 1 and 2 should be sent immediately after the communication with BMU is established. After that, the data should be sent regularly at intervals of about 1 minute. For the timing, refer to [Figure 9](#).

7.1.7.47 Data acquisition location module No.

CAN ID 0x308+BMU No.×0x10 (2nd byte)

Bit	7	6	5	4	3	2	1	0
	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

- 0x01 to 0x38 (0x00: not specified)

The module No. used for acquiring the voltage and temperature (6 points) should be specified with 1 to 56. The data with the frame ID: 0x10 to 0x16 is returned once from BMU in the next cycle.

- 0x65 to 0x9C (host system communication CAN/Ethernet)

The module No. used for acquiring the voltage and temperature (13 points) should be specified with a value 1 to 56 added to 100. The data with the frame ID: 0x10 to 0x16 is returned once from BMU in the next cycle.

- 0x64 (host system communication Ethernet alone)

The independent transmission of the voltage and temperature (13 points) starts from Module 1, for eight modules at a time (continuous transmission). This should be specified after it is confirmed that the reception of the Ethernet frame from BMU has started.

When 0x00 is specified, the continuous transmission is stopped.

*The modules connected to BMU-2G sub connector (CN2-BMU) will be numbered as 29, 30, ..., 56 in order.

7.1.8 Communication sequence

BMU sends the data frame described in 7.1.6 to the host system at a constant interval. The communication sequence comes in 3 types depending on the BMU operation statuses. This section provides communication sequence diagrams to explain the procedures for BMU and the host system. Table 7 shows the list of parameters for the communication control.

- (1) Initialization status
- (2) Normal processing (regular processing) status
- (3) BMU abnormal status (e.g. abnormal module communication)

Table 7 CAN communication control parameter list

Detected by	Control parameter	Setting value
BMU	Frame transmission interval	In accordance with CAN2.0 spec. *
BMU	Frame transmission interval	2 types: 120 ms and 960 ms
Host system	Frame transmission interval	In accordance with CAN2.0 spec. *
Host system	Frame transmission interval	2 types: 120 ms and 960 ms
BMU	Module communication time out	660 ms
Host system	BMU communication time out	660 ms
BMU	Life signal abnormality	2 s exceeded
Host system	Life signal abnormality	2 s exceeded

* 22 μ s (3-bit inter-frame space + suspend transmission 8-bit, at the time of error active)
 6 μ s (3-bit inter-frame, at the time of error passive)

7.1.8.1 Sequence at initialization

Figure 9 shows the initialization sequence at the time of powering on (the CAN numbering procedure for BMU/CMU is omitted). BMU is activated when the host system inputs the power supply signal in the IO signal. The activated BMU completes the initialization and waits for CAN data ID:0x00 from the host system. To confirm that the communication between the host system and BMU is not disrupted, the host system should continue to send CAN data ID:0x00 at an interval of 120ms after BMU is powered on. BMU starts the transmission to the host system immediately after receiving CAN data ID:0x00, with cell voltages and temperatures sent as indeterminate values until the communication is established with the battery module. The host system should monitor the unit voltage value of ID: 0x03 and, after the voltage value is confirmed to be within the normal value range of the system, send the operation instruction of ID: 0x00. The charge/discharge current can be applied after the contactor in the battery unit is confirmed closed.

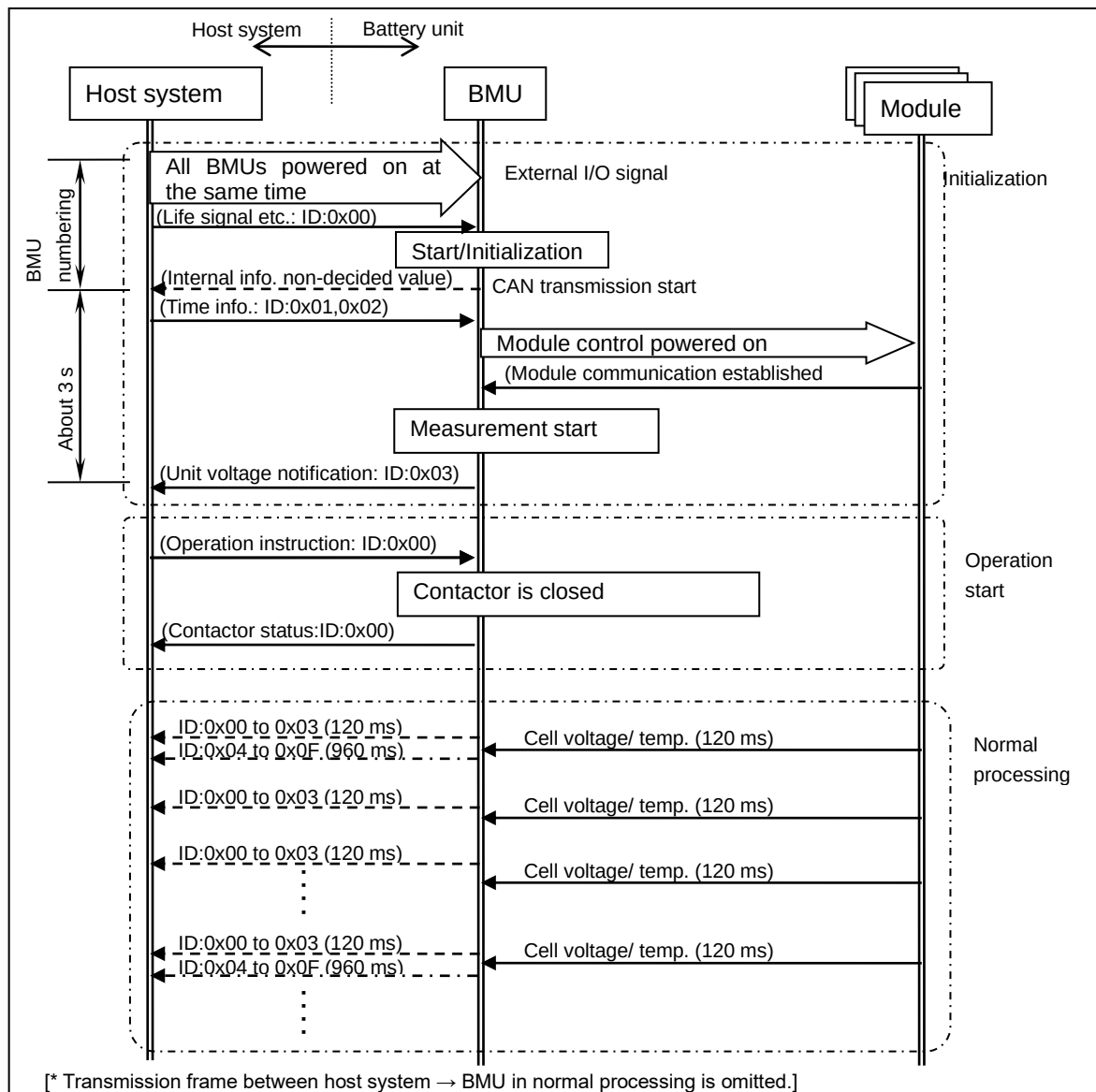


Figure 9 Communication sequence at initialization

【Note】 Figure 9 shows a case where the contactor included in the battery system components is used. In the case of using MCCB instead of the contactor, MCCB should be closed manually before the host system sends the operation instruction to BMU. Refer to 5.4 for the conditions for closing MCCB.

7.1.8.2 Sequence at normal operation status

Figure 10 shows the communication sequence in the normal processing status. BMU actively sends data to the host system at the data transmission intervals shown in 7.1.4. The data of the same transmission interval is continuously sent at the interval in accordance with CAN2.0 specification (i.e. inter-frame space). The data of 960 ms interval is sent together with the data of 120 ms interval at the 8th cycle of the data of 120 ms interval. Even if multiple BMUs are connected, each BMU sends the data in accordance with this sequence. The source of the data is identifiable by the CAN-ID (node ID).

The host system should send data to each BMU at the specified transmission interval similarly.

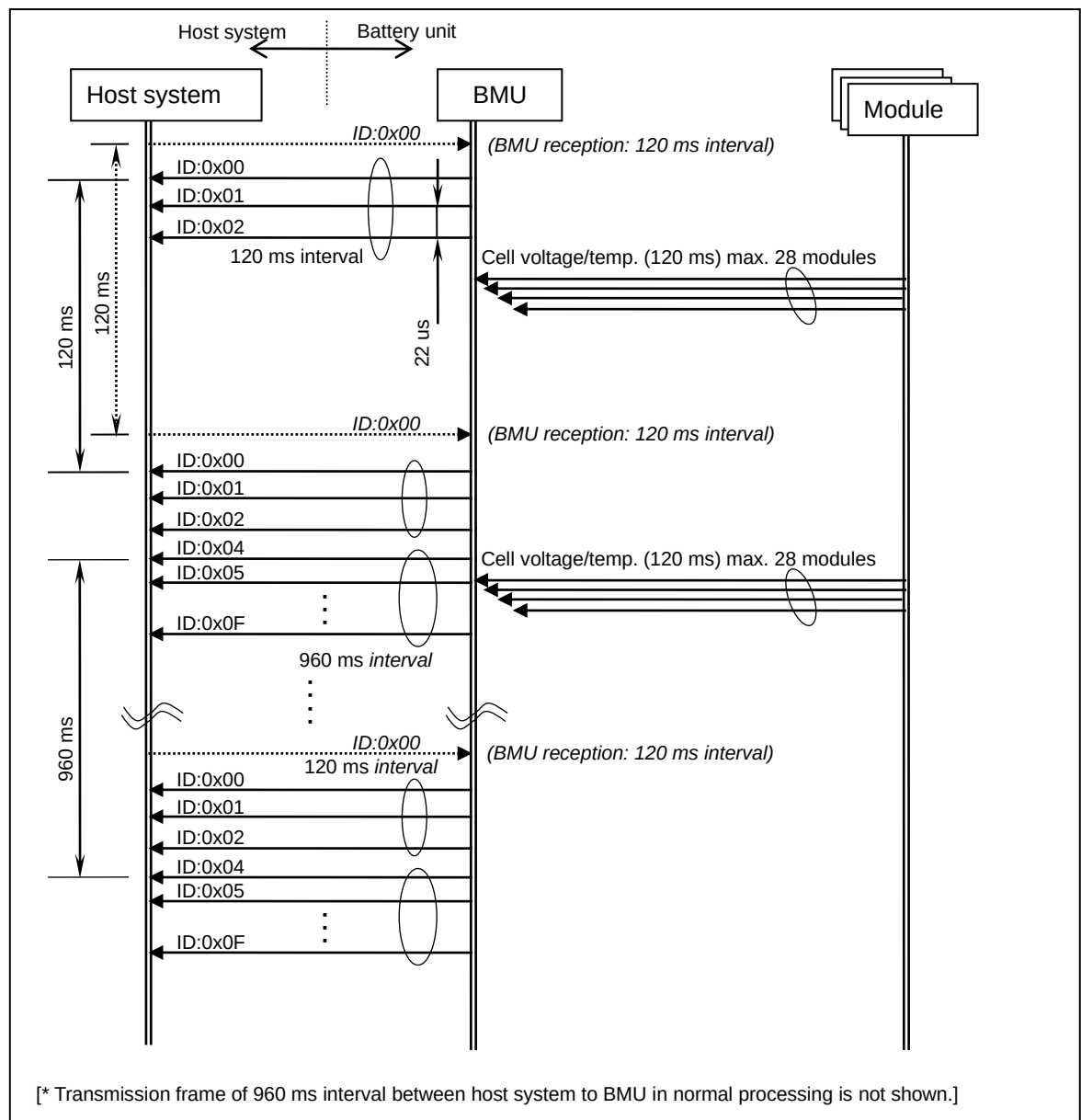


Figure 10 Communication sequence at normal processing status

7.1.8.3 Sequence at BMU abnormality

<At module communication abnormality>

If an abnormality occurs and continues in the communication between BMU and any of the battery modules for 660 ms or more, BMU judges it to be a module abnormality and opens the contactor, after which the contactor cannot be closed even if BMU receives the operation instruction from the host system. For restoration, it is necessary to remove the cause of the abnormality and restart BMU.

The communication between BMU and the host system continues even in the case of this abnormality, but the host system should not use the cell voltage and temperature values sent from BMU for the control, as the values are not valid.

Note that, besides the case of the communication abnormality with the battery module, the measures in this sequence are required if a battery unit abnormality is detected and the contactor is opened.

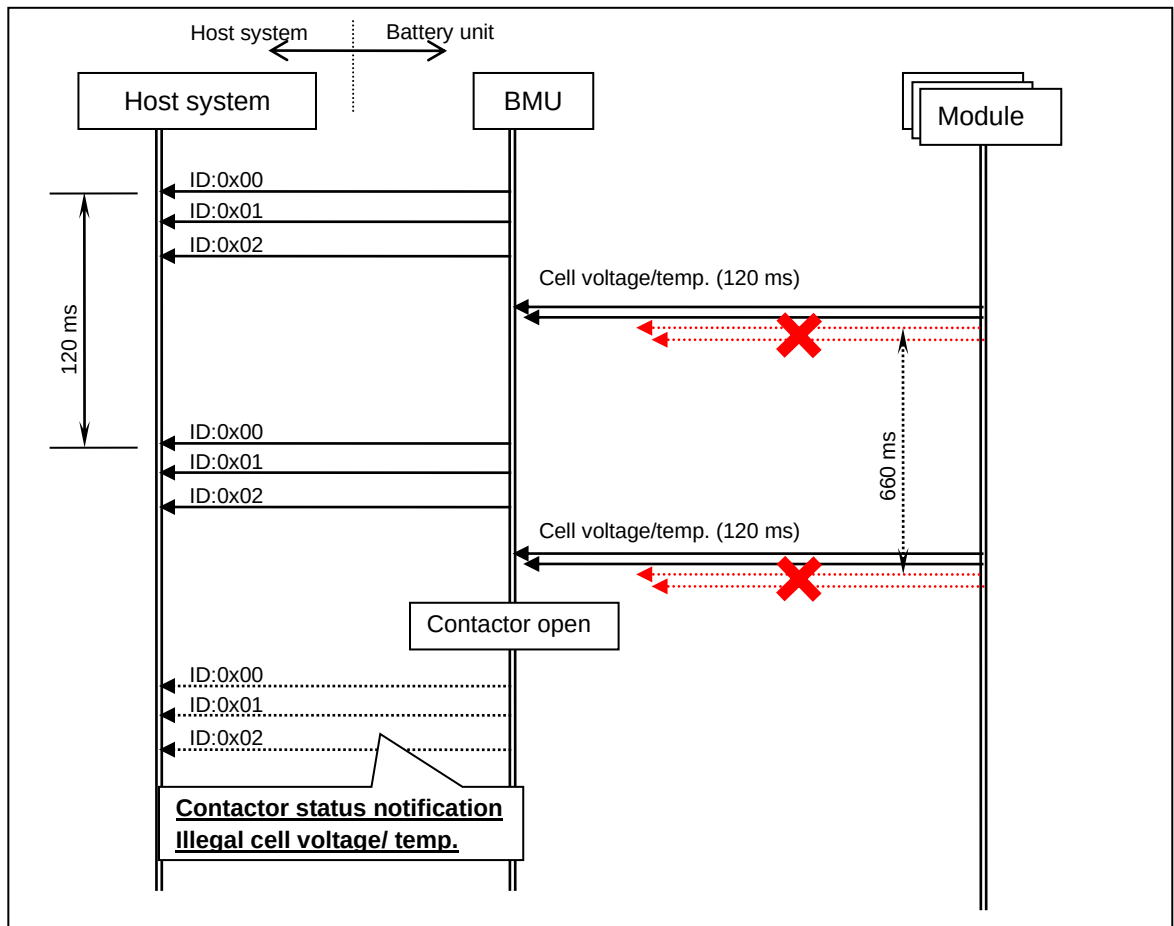


Figure 11 Communication sequence at module communication abnormality

Figure 11 shows the case where the contactor included in the battery system components is used. If MCCB is used instead, BMU opens MCCB, not the contactor.

<BMU CAN transmission abnormality>

If BMU detects an error in the transmission to the host system, it retries the transmission until BMU receives ACK from the CAN bus. For the specification of the CAN interface, refer to ISO11898 etc.

The abnormality caused by BMU transmission abnormality should be detected by the host system by way of the data frame reception timeout. Even if BMU detects this abnormality, it continues the normal operation.

<BMU CAN reception abnormality>

In the standard configuration, BMU does not detect any abnormality when receiving the data from the host system; i.e. BMU continues the normal operation.

Changes in the BMU settings can allow the data disruption from the host system for a set period of time to be detected as a BMU abnormality and open the contactor. For details on the BMU settings change, please contact our sales department.

<Life signal abnormality>

It is necessary for BMU and the host system to implement the function to detect an abnormality by mutually monitoring the life signals. BMU and the host system set the toggle signal which repeats ON/OFF at the 960 ms interval for the data (1 bit) shown in "Frame ID" by the equipment indicated in "Set" in the table below.

The equipment indicated in "Detect" should monitor the received life signal and, if the status continues for 2 seconds or more, take measures for communication abnormality.

If the host system detects this abnormality, measures for BMU failure (charge/discharge current stop) should be taken.

In the standard configuration, BMU only keeps records of the abnormality, but changes in the BMU settings can allow the contactor to be opened. For details on the BMU settings change, please contact our sales department.

	Name	Frame ID	Set	Detect
(1)	BMU life signal	0x00, Status bit 7	BMU	Host system
(2)	Host system life signal	0x00, Status bit 7	Host system	BMU

<Synchronization counter abnormality>

BMU and the host system need to make sure that continuous values emerge in the synchronization counters of the first byte in all data frames of each CAN-ID, with no values missed or duplicated. The frame should be discarded in case the value is not continuous with the value of the previously received counter. In this abnormality, no further action is required besides the discarding of the data.

The synchronization counter is judged to be continuous if the received counter is $n+1$ compared with the previously received counter n , wherein n ranges from 0 to 255 and 255 is followed by 0. (In case the previously received counter is 255, the counter is judged to be continuous if the received value is 0.) Note that the data in the interval following the judgment of discontinuity is regarded as continuous and thus is not discarded.

In the standard configuration, BMU continues the operation even when it detects a synchronization counter abnormality, but changes in the BMU settings can allow the contactor to be opened. For details on the BMU settings change, please contact our sales department.

7.2 Ethernet communication control

7.2.1 General specification

This section explains the Ethernet communication specification between BMU and the host system. BMU and the host system communicate via Ethernet as shown in [Table 8](#).

Table 8 General Specification of Ethernet communication between host system and BMU

Item	Description
Transmission method	(1) 100BASE-TX / 10BASE-T (IEEE802.3u) (2) Full duplex / Half duplex (3) Auto MDI/MDI-X [Note: Make sure the settings match with those on the host system side] * (1) and (2) are configurable by a parameter (default: auto negotiation) * (3) can be disabled by a parameter. (3) can only be enabled when used in combination with auto negotiation. Enabling (3) is prohibited if the transmission method is fixed. (default: enabled)
Connection method	Up to 22 BMUs by L2 switch (hub)
Transmission line	CAT5 or better cable, M12 connector, RJ45 connector
Isolation method	isolation transformer
MAC layer	Ethernet protocol (IEEE802.3 frames are not used)
Data length	min. 64 bytes, max. 1518 bytes (14 bytes of Ethernet header, 20 bytes of IP header, 8 bytes of UDP header, and 4 bytes of CRC)
Network layer	UDP/IP (periodic transmission, SNTP) TCP/IP (maintenance mode) ARP ICMP (ping response only)
Protocols on application layer	battery status notification, SNTP client, HTTP server (maintenance mode)

* The last three digits of the Model No on the label attached to BMU need to be equal to or greater than 102. "Model No: B2***102." In case the two digits are smaller than 102, the transmission method is fixed at the auto negotiation AutoMDI/MDI-X: ON.

7.2.2 IP address allocation

The IP address, allocated by way of the Class C private address as shown in Table 9, has the settings information of each individual BMU. The settings shown in the table are the default setting values, and can be changed by the parameters. Note that the actual IP address of BMU shown in No.1 through 22 is allocated automatically by adding the BMU No. to the base IP address.

In case the multicast address is designated for the IP address of the host system (destination), the MAC address is changed into the address for the multicast and transmitted. Reception multicast is not available.

In consideration of the use of a router or hub, the setting of the default gateway is possible, with the default value disabled. (Not shown in the table below.)

Table 9 IP address allocation

Instrument	BMU No.	Base IP address	Actual IP address*1	Subnet mask	Note
BMU1	1	192.168.1.100	192.168.1.101	255.255.255.0	
BMU2	2	192.168.1.100	192.168.1.102	255.255.255.0	
BMU3	3	192.168.1.100	192.168.1.103	255.255.255.0	
BMU4	4	192.168.1.100	192.168.1.104	255.255.255.0	
BMU5	5	192.168.1.100	192.168.1.105	255.255.255.0	
BMU6	6	192.168.1.100	192.168.1.106	255.255.255.0	
BMU7	7	192.168.1.100	192.168.1.107	255.255.255.0	
BMU8	8	192.168.1.100	192.168.1.108	255.255.255.0	
BMU9	9	192.168.1.100	192.168.1.109	255.255.255.0	
BMU10	10	192.168.1.100	192.168.1.110	255.255.255.0	
BMU11	11	192.168.1.100	192.168.1.111	255.255.255.0	
BMU12	12	192.168.1.100	192.168.1.112	255.255.255.0	
BMU13	13	192.168.1.100	192.168.1.113	255.255.255.0	
BMU14	14	192.168.1.100	192.168.1.114	255.255.255.0	
BMU15	15	192.168.1.100	192.168.1.115	255.255.255.0	
BMU16	16	192.168.1.100	192.168.1.116	255.255.255.0	
BMU17	17	192.168.1.100	192.168.1.117	255.255.255.0	
BMU18	18	192.168.1.100	192.168.1.118	255.255.255.0	
BMU19	19	192.168.1.100	192.168.1.119	255.255.255.0	
BMU20	20	192.168.1.100	192.168.1.120	255.255.255.0	
BMU21	21	192.168.1.100	192.168.1.121	255.255.255.0	
BMU22	22	192.168.1.100	192.168.1.122	255.255.255.0	
Host system (destination)	-	192.168.1.10	same as on the left	255.255.255.0	
Host system (NTP)	-	192.168.1.11	same as on the left	255.255.255.0	
BMU (broadcasting)	-	192.168.1.255	same as on the left	255.255.255.0	All BMUs target for transmission
Maintenance mode (BMU)	-	192.168.1.250	same as on the left	255.255.255.0	Common to all BMUs
Maintenance mode/ maintenance info. (PC/host system)	-	192.168.1.251	same as on the left	255.255.255.0	

*1 For BMU 1 through 22, the value of the actual IP address is the sum of the base IP address and the BMU No., because the BMU No. is allocated in accordance with the automatic numbering control (refer to chapter 7.3).

7.2.3 Port allocation

Table 10 shows the allocation of port numbers to communicate with BMU. For the periodic transmission, BMU allocates the four consecutive ports from 49152 (C000h), whereas the host system allocates the two consecutive ports from 49408 (C100h). For the maintenance mode, BMU allocates the port 53248 (D000h), whereas the tool (PC) allocates the port 53504 (D100h). (Eight ports in total.)

Table 10 port number allocation

type	content	source	destination port	destination	protocol	note
periodic transmission	battery status notification	BMU	<u>49152</u> , 49153 49154*, 49155	host system	UDP	See 7.2.4
		host system	<u>49408</u> , 49409	BMU	UDP	
	time calibration	BMU	123	host system	SNTP	used only if NTP server is enabled
		host system	-	BMU	SNTP	
maintenance mode	maintenance	BMU	<u>53248</u>	tool (PC)	KWP2000/UDP	service tool
		tool (PC)	<u>53504</u>	BMU	KWP2000/UDP	
	changing parameters	BMU	-	PC	HTTP	Web service
		PC	80	BMU	HTTP	

*To be used in BMU.

The **bold, underlined values** in the destination port column above are to be designated on the service tool. (Table 10 shows the default values.)

(Note) When changing the destination port numbers from the default values, make sure that all eight port numbers including the automatically assigned ones lie within the range from 1024 to 65526 with no duplication. Ethernet communication is not available with duplicate port numbers.

(Example) If the port number for the periodic transmission (BMU to host system) is to be changed from 49152 to 49408, the port number for the periodic transmission (host system to BMU) has to be set to a value other than 49408, 49409, 49410, 49411, 53248, or 53504 (to be set to 49400, for example).

7.2.4 Data format

Shown below are the contents of the periodic transmission when Ethernet is used. The data contents comply with the CAN data format shown in 7.1.7. For details on each data, refer to the CAN data format in 7.1.7. Note that BMU is available for a maximum of 22 parallel connections.

7.2.4.1 Data format (BMU to host system) frame 1

Data format (BMU to host system) transmission interval = 120ms

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49152									
0x20+BMU No. × 0x20	Sync. counter	Status	Final charge / discharge info.	Overcharge/ overdischarge/ overtemp. info.	Unit SOC		Unit SOH		
0x21+BMU No. × 0x20	reserved	Max. voltage value of cells in unit		Min. voltage value of cells in unit		Max. temp. value of cells in unit	Average temp. value of cells in unit	Min. temp. value of cells in unit	
0x22+BMU No. × 0x20	reserved	Max. cell voltage value module	Min. cell voltage value module	Max. cell temp. value module	Min cell temp. value module	don't care	don't care	don't care	
0x23+BMU No. × 0x20	reserved	Unit charge current		Unit discharge current		Unit voltage		BMU state supplementa l information	
(FCS : CRC-32)									

* An irregular value is entered in "don't care" fields. The data should be ignored.

7.2.4.2 Data format (BMU to host system) frame 2

Data format (BMU to host system) transmission interval = 960ms

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No. : 49153									
0x24+BMU No.×0x20	Sync. counter	Max. temp. value of cell in unit 2		Average temp. value of cells in unit 2		Min. temp. value of cell in unit 2		BMU abnormality detail	
0x25+BMU No.×0x20	reserved	BMU Monitoring Info. *2	reserved	reserved	reserved	don't care	don't care	Unit SOC [%]	
0x26+BMU No.×0x20	reserved	CMU failure status (module 1 to 22)			CMU failure status (module 23 to 28)	Module 28 Cell failure / balance status		reserved	
0x27+BMU No.×0x20	reserved	Module 1 Cell failure / balance status		Module 2 Cell failure / balance status		Module 3 Cell failure / balance status		parallel no. of branch	
0x28+BMU No.×0x20	reserved	Module 4 Cell failure / balance status		Module 5 Cell failure / balance status		Module 6 Cell failure / balance status		reserved	
0x29+BMU No.×0x20	reserved	Module 7 Cell failure / balance status		Module 8 Cell failure / balance status		Module 9 Cell failure / balance status		reserved	
0x2A+BMU No.×0x20	reserved	Module 10 Cell failure / balance status		Module 11 Cell failure / balance status		Module 12 Cell failure / balance status		reserved	
0x2B+BMU No.×0x20	reserved	Module 13 Cell failure / balance status		Module 14 Cell failure / balance status		Module 15 Cell failure / balance status		reserved	
0x2C+BMU No.×0x20	reserved	Module 16 Cell failure / balance status		Module 17 Cell failure / balance status		Module 18 Cell failure / balance status		reserved	
0x2D+BMU No.×0x20	reserved	Module 19 Cell failure / balance status		Module 20 Cell failure / balance status		Module 21 Cell failure / balance status		reserved	
0x2E+BMU No.×0x20	reserved	Module 22 Cell failure / balance status		Module 23 Cell failure / balance status		Module 24 Cell failure / balance status		reserved	
0x2F+BMU No.×0x20	reserved	Module 25 Cell failure / balance status		Module 26 Cell failure / balance status		Module 27 Cell failure / balance status		reserved	
0x500+BMU No.×0x10	reserved	CMU failure status (module 29 to 50)			CMU failure status (module 51 to 56)	Module 56 Cell failure / balance status		reserved	
0x501+BMU No.×0x10	reserved	Module 29 Cell failure / balance status		Module 30 Cell failure / balance status		Module 31 Cell failure / balance status		reserved	
0x502+BMU No.×0x10	reserved	Module 32 Cell failure / balance status		Module 33 Cell failure / balance status		Module 34 Cell failure / balance status		reserved	

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
0x503+BMU No.x0x10		reserved	Module 35 Cell failure / balance status		Module 36 Cell failure / balance status		Module 37 Cell failure / balance status		reserved
0x504+BMU No.x0x10		reserved	Module 38 Cell failure / balance status		Module 39 Cell failure / balance status		Module 40 Cell failure / balance status		reserved
0x505+BMU No.x0x10		reserved	Module 41 Cell failure / balance status		Module 42 Cell failure / balance status		Module 43 Cell failure / balance status		reserved
0x506+BMU No.x0x10		reserved	Module 44 Cell failure / balance status		Module 45 Cell failure / balance status		Module 46 Cell failure / balance status		reserved
0x507+BMU No.x0x10		reserved	Module 47 Cell failure / balance status		Module 48 Cell failure / balance status		Module 49 Cell failure / balance status		reserved
0x508+BMU No.x0x10		reserved	Module 50 Cell failure / balance status		Module 51 Cell failure / balance status		Module 52 Cell failure / balance status		reserved
0x509+BMU No.x0x10		reserved	Module 53 Cell failure / balance status		Module 54 Cell failure / balance status		Module 55 Cell failure / balance status		reserved
0x50B+BMU No.x0x10 ^{*1}		reserved	BMU abnormality detail 2						
(FCS : CRC-32)									

* An irregular value is entered in the "don't care" fields. The data should be ignored.

*1 The transmission can be permitted with the parameter setting if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 106, e.g. "Model No.: B2***106."

*2 BMU monitoring info. can be sent if the last three digits of the Model No. on the label attached to BMU are equal to or greater than 201, e.g. "Model No.: B2***201."

7.2.4.3 Data format (BMU to host system) frame 3

(1) Frame configuration for data acquisition (temperature 6 points)

- Data format (BMU to host system) transmission interval = non-periodic (1 response per request of data acquisition)

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49155									
0x30+BMU No.×0x20	Sync. counter	Cell voltage 1		Cell voltage 2		Cell voltage 3		Data acquisition location (Module No.)	
0x31+BMU No.×0x20	reserved	Cell voltage 4		Cell voltage 5		Cell voltage 6		reserved	
0x32+BMU No.×0x20	reserved	Cell voltage 7		Cell voltage 8		Cell voltage 9		reserved	
0x33+BMU No.×0x20	reserved	Cell voltage 10		Cell voltage 11		Cell voltage 12		reserved	
0x36+BMU No.×0x20	reserved	Cell temp. 1 (1°C/LSB)	Cell temp. 2 (1°C/LSB)	Cell temp. 3 (1°C/LSB)	Cell temp. 4 (1°C/LSB)	Cell temp. 5 (1°C/LSB)	Cell temp. 6 (1°C/LSB)	reserved	
(FCS : CRC-32)									

(2) Frame configuration for data acquisition (temperature 13 points)

- Data format (BMU to host system) transmission interval = non-periodic (1 response per request of data acquisition)

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49155									
0x30+BMU No.×0x20	Sync. counter	Cell voltage 1		Cell voltage 2		Cell voltage 3		Data acquisition location (Module No.)	
0x31+BMU No.×0x20	reserved	Cell voltage 4		Cell voltage 5		Cell voltage 6		reserved	
0x32+BMU No.×0x20	reserved	Cell voltage 7		Cell voltage 8		Cell voltage 9		reserved	
0x33+BMU No.×0x20	reserved	Cell voltage 10		Cell voltage 11		Cell voltage 12			
0x37+BMU No.×0x20	reserved	Cell temp. 1 (Temp. 13 points)	Cell temp. 2 (Temp. 13 points)	Cell temp. 3 (Temp. 13 points)	Cell temp. 4 (Temp. 13 points)	Cell temp. 5 (Temp. 13 points)	Cell temp. 6 (Temp. 13 points)	Cell temp. 7 (Temp. 13 points)	
0x38+BMU No.×0x20	reserved	Cell temp. 8 (Temp. 13 points)	Cell temp. 9 (Temp. 13 points)	Cell temp. 10 (Temp. 13 points)	Cell temp. 11 (Temp. 13 points)	Cell temp. 12 (Temp. 13 points)	Cell temp. 13 (Temp. 13 points)	reserved	
(FCS : CRC-32)									

(3) Frame configuration for data acquisition (cell voltage and temperature 13 points, continuous transmission)

- Data format (BMU to host system) transmission interval = 120 ms (transmission continued until the data acquisition stops)

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49155									
0x30+BMU No.×0x20	Sync. counter	Cell voltage 1		Cell voltage 2		Cell voltage 3		Data acquisition location (Module No.)	
0x31+BMU No.×0x20	reserved	Cell voltage 4		Cell voltage 5		Cell voltage 6		reserved	
0x32+BMU No.×0x20	reserved	Cell voltage 7		Cell voltage 8		Cell voltage 9		reserved	
0x33+BMU No.×0x20	reserved	Cell voltage 10		Cell voltage 11		Cell voltage 12		reserved	
0x37+BMU No.×0x20	reserved	Module A Cell temp. 1 (Temp. 13 points)	Module A Cell temp. 2 (Temp. 13 points)	Module A Cell temp. 3 (Temp. 13 points)	Module A Cell temp. 4 (Temp. 13 points)	Module A Cell temp. 5 (Temp. 13 points)	Module A Cell temp. 6 (Temp. 13 points)	Module A Cell temp. 7 (Temp. 13 points)	
0x38+BMU No.×0x20	reserved	Module A Cell temp. 8 (Temp. 13 points)	Module A Cell temp. 9 (Temp. 13 points)	Module A Cell temp. 10 (Temp. 13 points)	Module A Cell temp. 11 (Temp. 13 points)	Module A Cell temp. 12 (Temp. 13 points)	Module A Cell temp. 13 (Temp. 13 points)	reserved	
1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
: Voltage and temperature of Modules B,C, D, E, F, and G :									
0x30+BMU No.×0x20	Sync. counter	Cell voltage 1		Cell voltage 2		Cell voltage 3		Data acquisition location (Module No.)	
0x31+BMU No.×0x20	reserved	Cell voltage 4		Cell voltage 5		Cell voltage 6		reserved	
0x32+BMU No.×0x20	reserved	Cell voltage 7		Cell voltage 8		Cell voltage 9		reserved	
0x33+BMU No.×0x20	reserved	Cell voltage 10		Cell voltage 11		Cell voltage 12		reserved	
0x37+BMU No.×0x20	reserved	Module H Cell temp. 1 (Temp. 13 points)	Module H Cell temp. 2 (Temp. 13 points)	Module H Cell temp. 3 (Temp. 13 points)	Module H Cell temp. 4 (Temp. 13 points)	Module H Cell temp. 5 (Temp. 13 points)	Module H Cell temp. 6 (Temp. 13 points)	Module H Cell temp. 7 (Temp. 13 points)	
0x38+BMU No.×0x20	reserved	Module H Cell temp. 8 (Temp. 13 points)	Module H Cell temp. 9 (Temp. 13 points)	Module H Cell temp. 10 (Temp. 13 points)	Module H Cell temp. 11 (Temp. 13 points)	Module H Cell temp. 12 (Temp. 13 points)	Module H Cell temp. 13 (Temp. 13 points)	reserved	
(FCS : CRC-32)									

- (1) Once the module No. of the data acquisition location is provided as "100," BMU starts to send the voltage and temperature of the modules independently, starting from Module 1, for eight modules at a time (continuous transmission).

When the transmission is completed for the last module, the transmission is repeated from Module 1. If the number of the modules connected to BMU is not a multiple of eight, dummy data of Module No.0 comes after the last module. The cell voltage and temperature of Module No.0 are invalid data (where the value equivalent to the CAN ID is set for the first 2 bytes, the synchronous counter is set for the third byte, and 0 is set for the rest of the data).

Identical temperatures are sent for the temperature 13 points until they are updated via the temperature acquisition process of each module.

The CAN-ID of the first 2 bytes and the synchronous counter of the third byte are set for the entire eight modules and transmitted.

The continuous transmission is stopped when "0," "1 to 56," or "1 to 156" is provided for the data acquisition module No.

(Example) Shown below are the values of data acquisition location (Modules A through H) when 28 modules are connected.

Times of transmission	A	B	C	D	E	F	G	H
1st	101	102	103	104	105	106	107	108
2nd	109	110	111	112	113	114	115	116
3rd	117	118	119	120	121	122	123	124
4th	125	126	127	128	0	0	0	0
5th	101	102	103	104	105	106	107	108
6th	109	110	111	112	113	114	115	116
...	...	...	...	...	...	...	...	...

In the 4th transmission, the cell voltage and temperature for E, F, G, and H are invalid data.

- (2) The request for the continuous transmission (Module No.100) should be made after it is confirmed on the host system side that the unit voltage has been provided by BMU. On [Start 6] of BMU operation status and after, the transmission of the measured voltage and temperature becomes possible. After the BMU operation status transitions to [Stop] and CMU is shut down, the voltage and temperature of all connected modules are sent as SNA.

7.2.4.4 Data format (host system to BMU) frame 1

Data format (host system to BMU) transmission interval = 120ms

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49408									
0x300+BMU No.×0x10		Sync. counter	status	main circuit voltage		0x00	0x00	0x00	0x00
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	(FCS : CRC-32)	
(FCS : CRC-32)									

* The 1st and 2nd bytes should be CAN ID (0x300 + BMU No. × 0x10) of the CAN data format.

7.2.4.5 Data format (host system to BMU) frame 2

Data format (host system to BMU) transmission interval = 960ms

1 byte	2 byte	3 byte	4 byte	5 byte	6 byte	7 byte	8 byte	9 byte	10 byte
Ethernet header / IP header / UDP header (total 42 bytes) source / destination port No : 49409									
0x301+BMU No.×0x10		Sync. counter	Time (lower 32 bits)				0x00	0x00	0x00
0x302+BMU No.×0x10		0x00	Time (upper 32 bits)				0x00	0x00	0x00
0x308+BMU No.×0x10		0x00	Data acquisition location (Module No.)	0x00	0x00	0x00	0x00	0x00	0x00
(FCS : CRC-32)									

* The 1st, 2nd, 11th, 12th, 21st and 22nd bytes should be the CAN ID (0x301, 0x302, 0x308 + BMU No. × 0x10) of the CAN data format.

* Time (lower 32bits / upper 32bits) should be 0xFFFFFFFF, 0xFFFFFFFF respectively to specify only the data acquisition location. In this case, BMU doesn't update the Time.

* About data acquisition location (Module No.)

- Numbers 1 to 56 (CMU01 to 56) are for the demand made each time for the cell voltage and temperature 6 points of the specified module.
- Numbers 101 to 156 (CMU01 to 56) are for the demand made each time for the cell voltage and temperature 13 points of the specified module.
- Number 100 is for the demand for the start of the continuous transmission of the cell voltage and temperature 13 points of all modules. (This should be specified after the reception of the Ethernet frame from BMU has started.)
- Number 0 is for the demand for the stop of the continuous transmission of the cell voltage and temperature 13 points of all modules, or for the demand for the setting of the Time only.

7.2.5 Precautions

Although BMU is able to satisfy the Ethernet communication requirements as described in this document, BMU may operate abnormally, get hung up, or drop the transmitted data if a large quantity of frames are transmitted into the Ethernet interface and result in reception overload.

BMU with the Model No. (provided on the product label attached to BMU) whose last three digits are 104 or above is designed to temporarily stop the data reception upon detecting Ethernet reception overload, thereby avoiding abnormal operations or hang-ups. Note that, if the data reception is stopped, command from the host system will not reach BMU.

Be sure to follow the restrictions on the system configuration for Ethernet communication between the host system and BMU as shown below.

- *Make sure that the data transmission from the host system to BMU has the frequencies specified in this document.
- *Frequent use of broadcast communication should be avoided on the network where BMU is connected.
- *Prevent network loops from forming. The accompanied broadcast storm may cause Ethernet reception overload.
- *Use the system in a closed network environment that has no external connectivity, physical or logical.

7.3 Automatic numbering control

When controlling multiple battery units, the host system needs to execute the automatic numbering to determine the ID for BMU at the time of the start-up. This chapter explains the automatic numbering control.

7.3.1 Configuration

As shown in Figure 12, the host system and each BMU should be connected in the daisy-chain connection in accordance with ISO9141 (unidirectional communication). A maximum of 22 BMUs can be connected. The host system automatically assigns the ID from 1 up to 22 starting from the nearest BMU in the connection order.

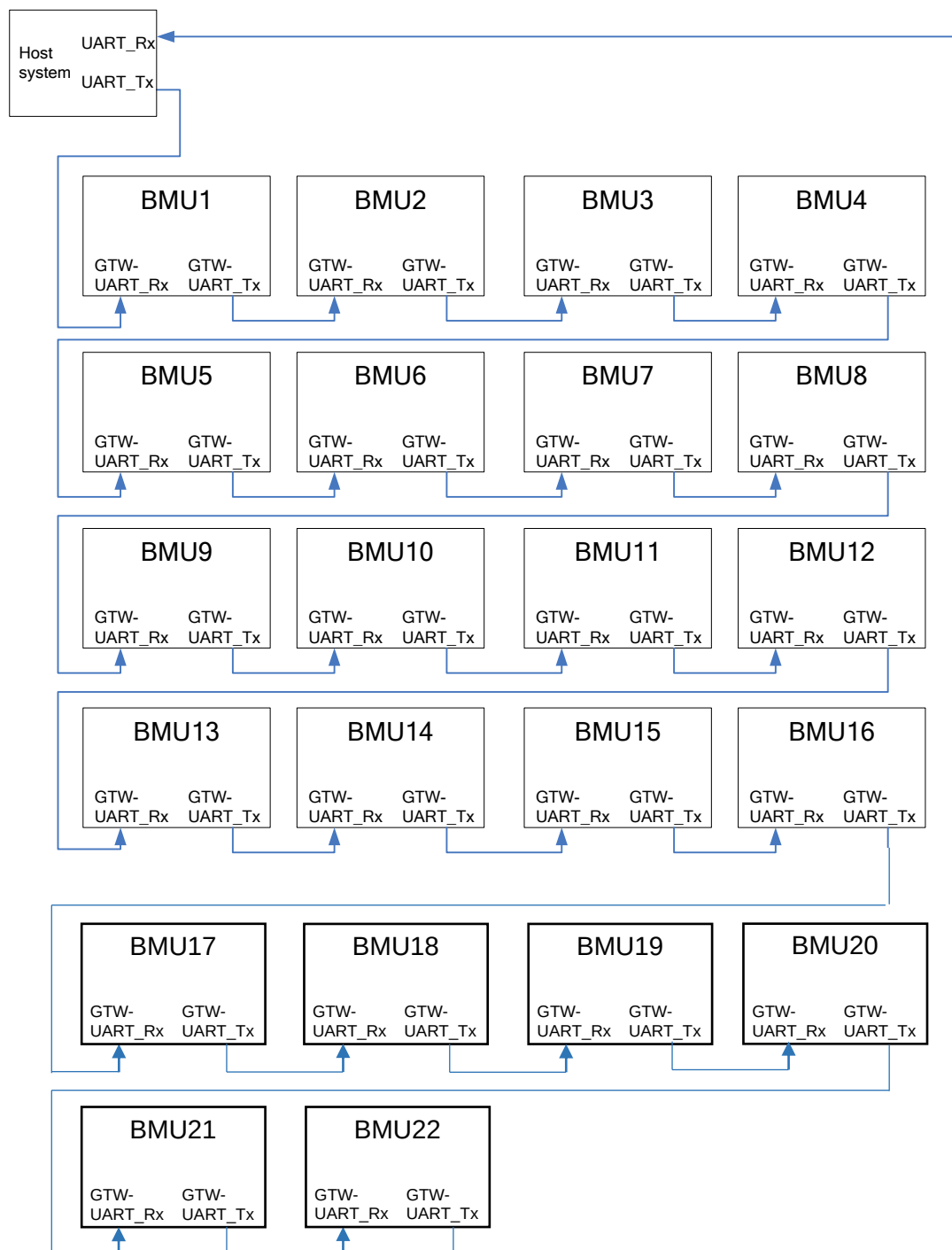


Figure 12 Connection configuration for BMU automatic numbering

7.3.2 Communication specification

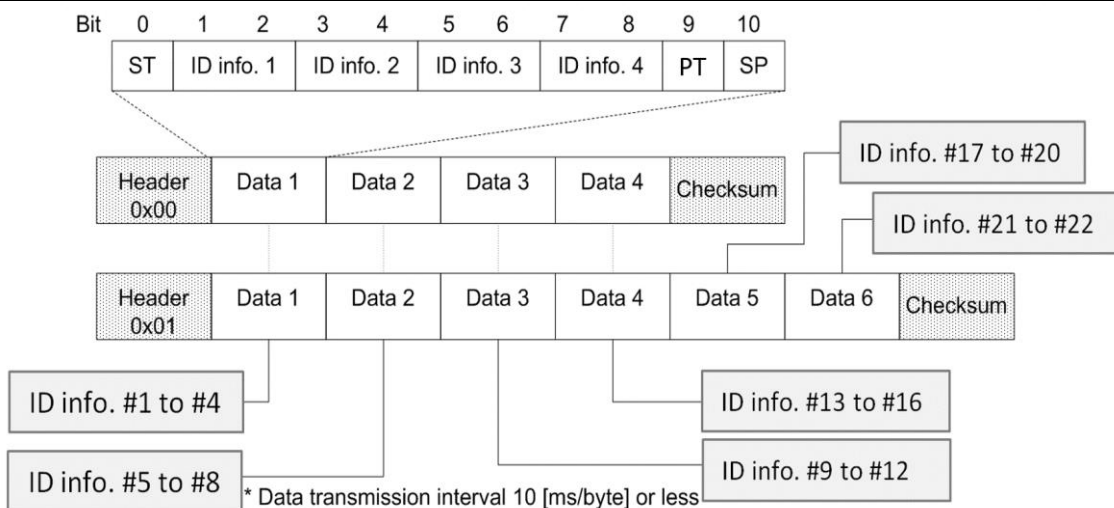
The table below shows the communication protocol specification and the frame structure.

Communication protocol	
Communication method	Unidirectional, start-stop system (compliant with UART)
Start bit (ST)	1 bit
Data bit	8 bits (LSB first)
Parity bit (PT)	1 bit, even parity
Stop bit (SP)	1 bit
Baud rate	1200bps $\pm 1\%$
I/F	In accordance with ISO9141 ^{Note)}
Error check items	Overrun error
	Framing error
	Checksum error
	Parity error

Note) The physical layer standard of the LIN-ISO9141 specification is complied with. RS-232C/485 cannot be connected.

Item	Specification
Frame	A series of data consisting of the header, 4 or 6 bytes numbering data, and checksum.
Header	Data byte used by the receive node to judge the kind of the frame. The header 00h indicates that the numbering data is 4 bytes, and 01h indicates that the numbering data is 6 bytes.
Numbering bits	2 bits assigned per BMU
Checksum	Lower 1 byte showing addition of 4 or 6 bytes numbering data

Data frame format



ID information	Specification
00	Default
01	Numbering completion
10	BMU ID change
11	Numbering error

7.3.3 Numbering control

The table shows the procedure of automatic numbering control for BMU.

Procedure	Control
1. BMU start	(1) Power on all BMUs. It is necessary to start them as simultaneously as possible. (2) BMU becomes ready to receive the automatic numbering data frame 1 second after the activation.
2. Numbering frame transmission	(1) Start transmission of the data frame. 00h should be set for all data; i.e. from "01" to "04". 00h should be set for the header and checksum as well. (2) It is necessary to start transmission immediately after 1 second from powering on BMU. If 5 seconds have elapsed, it is necessary to power off all BMUs and retry it. Note: If 5.5 seconds elapse after the activation, BMU automatically starts with internally-stored ID number. When configuring the battery system without the numbering control, be sure to assign ID to each BMU by the parameters in advance.
3. BMU processing	BMU receives the data frame from the prior BMU or the host system, sets the numbering status in the assigned numbering bits (2 bits), and sends it to the next BMU or the host system. Detailed control is as follows: (a) BMU assigns the ID information which first detects 00B to its own BMU ID as the ID information for its own BMU. (b) If the BMU ID is normally assigned, BMU sends its own ID information; i.e. 01B (new or without change) or 10B (ID update). (c) If BMU ID is not assigned normally, BMU sends its own ID information as 11B. (d) If the communication error or data frame abnormality is detected in the received data, all ID information is sent as 11B. - If other than 00B is detected in the ID information after its own ID information - If 00B is not detected in any ID information * * If all ID information is 01B or 10B, BMU sends a frame with 01h of header and 6 bytes of numbering data to next BMU or the host system.
4. Finish processing	(1) If the host system satisfies all conditions below, it judges the numbering as normal and completes the process. Read the values in the parentheses if the header of the reception frame is 01h. - All 6-byte (all 8-byte) data frames are normally received from the final-stage BMU. - ID information #1 to #16 (#22) in data 01 to 04 (06) is 01B, 10B, or 00B. - All bits are not 00B. In addition, 00B does not exist between 01B and 10B in the ID information #1 to #16 (#22). - Overrun error, framing error, checksum error, or parity error is not detected. (2) If any of the following occurs, it is necessary to take measures; e.g. retrieval, wiring check because of numbering failure: - The received data frame is not in 6 bytes (8 bytes). - All or any of ID information #1 to #16 (#22) is 11B or all of them are 00B. - 00B exists between 01B and 10B in the ID information #1 to #16 (#22). - Overrun error, framing error, checksum error, or parity error - Data frame cannot be received within 10 seconds after data frame transmission.

7.4 Reading BMU maintenance information

BMU-2G can send its maintenance information to the host system without switching to the maintenance mode. BMU sends a log as a response to the maintenance information request sent from the host system.

The communication is carried out with 8-byte wide frames for maintenance. The section 7.4.1 shows the method to communicate the maintenance frame over CAN or Ethernet, whereas the section 7.4.2 shows the data format and sequence of the maintenance frame.

In the standard configuration, the interface between the host system and BMU is CAN. Change of BMU settings is required for the use of Ethernet. For details on the BMU settings change, please contact our sales department.

7.4.1 Communication interface

7.4.1.1 CAN communication

The contents of the data field of CAN are the same as those of the maintenance frame.

• data format (host system → BMU)

CAN ID	1byte	2byte	3byte	4byte	5byte	6byte	7byte	8byte
0x75F+ BMU No.	maintenance request frame							

• data format (BMU → host system)

CAN ID	1byte	2byte	3byte	4byte	5byte	6byte	7byte	8byte
0x77F+ BMU No.	maintenance response frame							

7.4.1.2 Ethernet communication

The IP address for the transmission/reception is the same as that for the periodic transmission. Refer to Table 9 for details.

The port numbers (default value) are provided in Table 11 below. Refer to 7.2.3 for details.

Table 11 Port number for reading logs

Source	destination port	destination	protocol
BMU	53248	host system	UDP
host system	53504	BMU	UDP

The tables below show the data format of UDP payload, with the bytes arranged in the big-endian order.

• data format (host system → BMU)

1byte	2byte	3byte	4byte	5byte	6byte	7byte	8byte	9byte	10byte
0x75F+BMU No.		maintenance request frame							

• data format (BMU → host system)

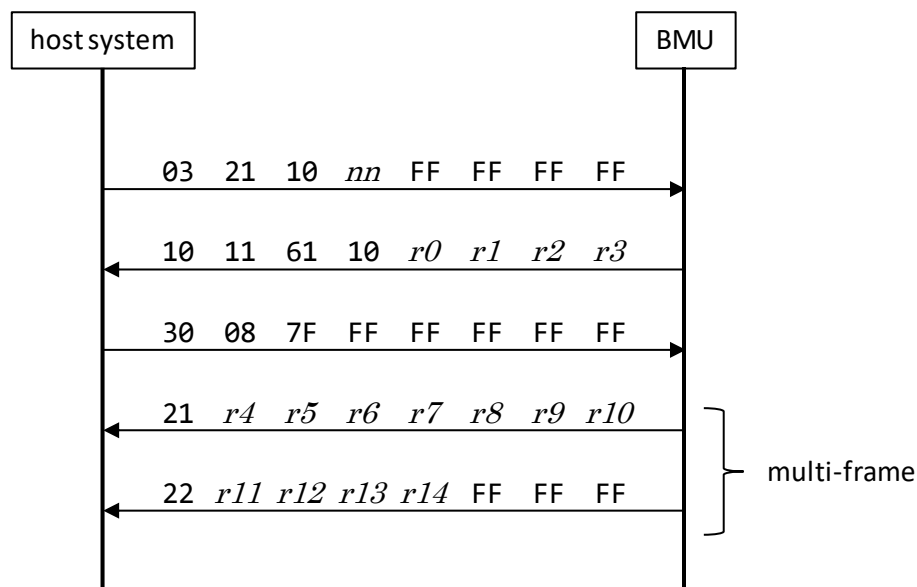
1byte	2byte	3byte	4byte	5byte	6byte	7byte	8byte	9byte	10byte
0x77F+BMU No.		maintenance response frame							
0x77F+BMU No. (optional; only in case of multi-frame)		maintenance response frame (optional; only in case of multi-frame)							
⋮		⋮							

“Multi-frame” is packed in one UDP frame. Multi-frames are defined as consecutive maintenance frames.

7.4.2 Communication sequence

The sequence diagrams below show the maintenance request/response frames between the host system and BMU.

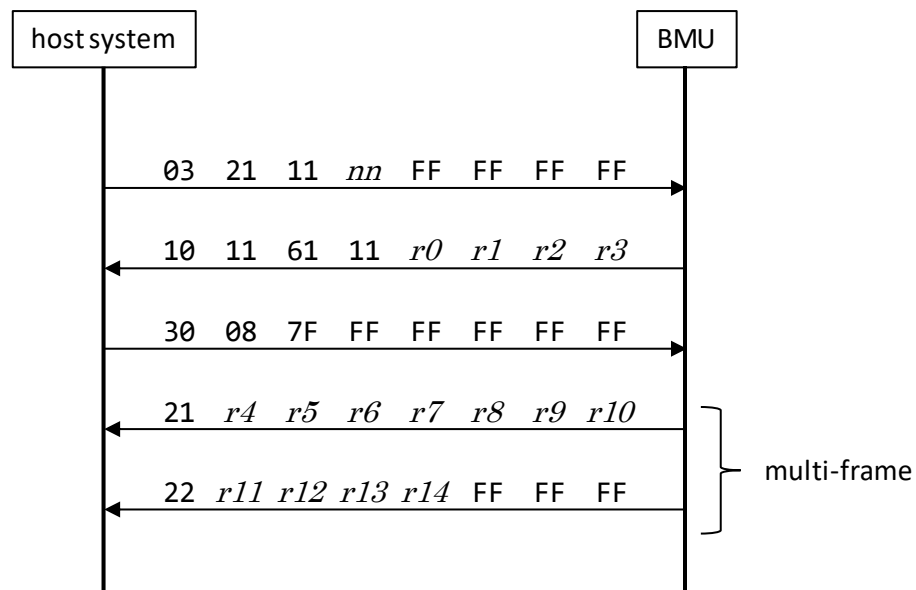
7.4.2.1 Acquiring normal trace



<i>nn</i>	Trace No. (0x00 (newest) to 0xAF (oldest))
<i>r0</i>	Designated Trace No.
<i>r1 .. r4</i>	Time of occurrence (UNIX time)
<i>r5 .. r6</i>	Trace code
<i>r7</i>	(reserved)
<i>8</i>	Number of occurrences
<i>r9 .. r14</i>	Historical data

One record is acquired in one sequence. *nn* is the number of the record to be acquired. The length of the record is 15 bytes. Repeat the sequence 176 times and read all records.

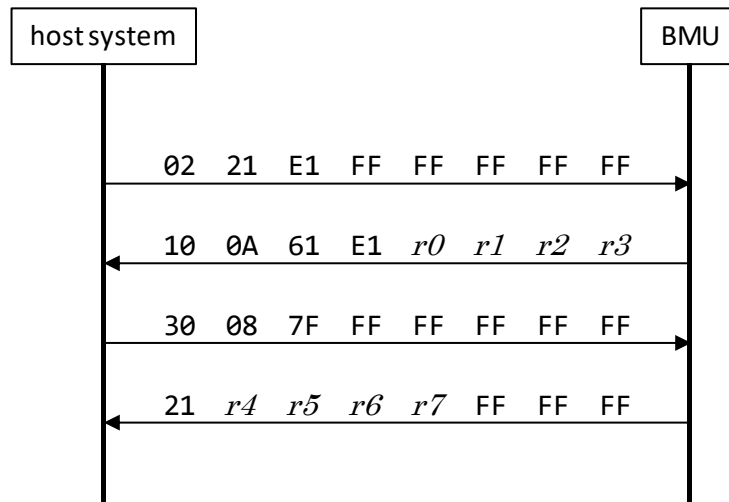
7.4.2.2 Acquiring serious fault trace



<i>nn</i>	Trace No. (0x00 (newest) to 0x3F (oldest))
<i>r0</i>	Designated Trace No.
<i>r1 .. r4</i>	Time of occurrence (UNIX time)
<i>r5 .. r6</i>	Trace code
<i>r7</i>	(reserved)
<i>r8</i>	Number of occurrences
<i>r9 .. r14</i>	Historical data

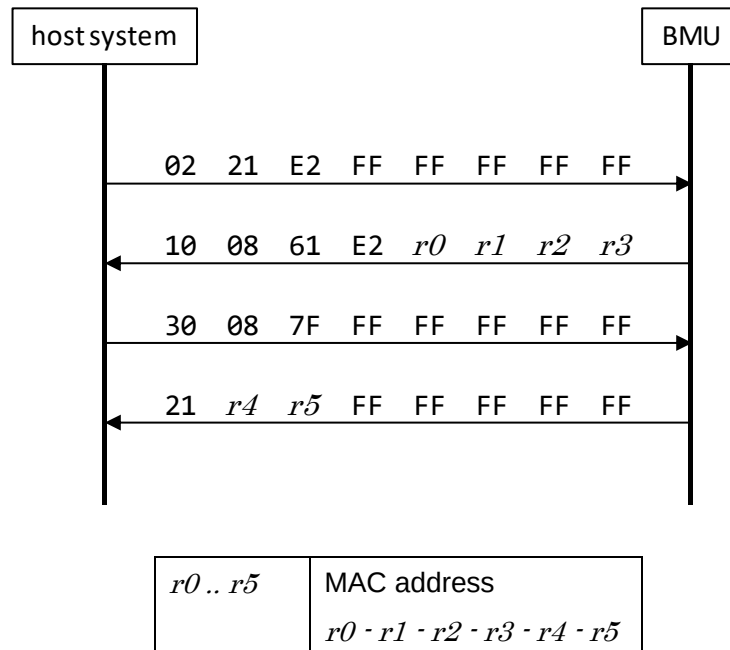
One record is acquired by one sequence. *nn* is the number of the record to be acquired. The length of record is 15 bytes. Repeat the sequence 64 times and read all records.

7.4.2.3 Acquiring production date and serial number

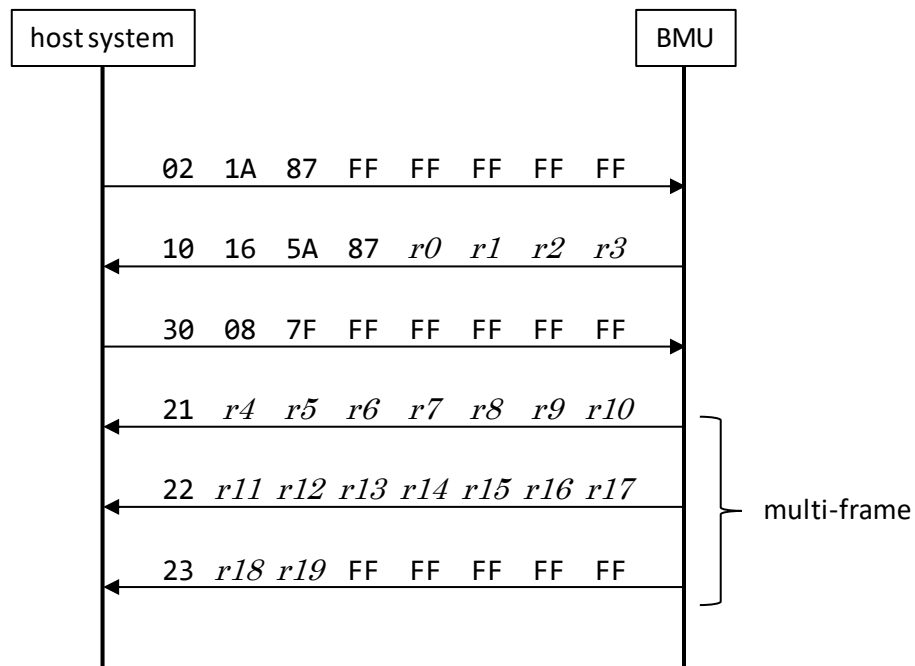


<i>r0</i>	production day, the tens digit (0 to 3)
<i>r1</i>	production day, the ones digit (0 to 9)
<i>r2</i>	production month (1 to 12)
<i>r3</i>	production year (0 to 99)
<i>r4</i>	0x00
<i>r5</i>	serial number (upper 8 bits)
<i>r6</i>	serial number (middle 8 bits)
<i>r7</i>	serial number (lower 8 bits)

7.4.2.4 Acquiring MAC address



7.4.2.5 Acquiring version



<i>r0</i>	0xFF
<i>r1</i>	0xFF
<i>r2</i>	0xFF
<i>r3</i>	0xFF
<i>r4</i>	0xFF
<i>r5</i>	Hardware Version (Major Version)
<i>r6</i>	Hardware Version (Minor Version)
<i>r7</i>	Software Version (Major Version)
<i>r8</i>	Software Version (Middle Version)
<i>r9</i>	Software Version (Minor Version)
<i>r10 .. r19</i>	Parts Number (ASCII code) BMU-2G is 'SSSIB3MA□□' (□ is blank)

7.4.3 Precaution

- This function is compatible with BMU-2G alone.
- This function is available only when BMU is numbered, and is unavailable with BMU ID=0.
- This function does not work in the maintenance mode.
- The following are error cases that require the restart of the sequence from the beginning.
 - The maintenance response frame received from BMU is different from the value shown in the sequence diagram.
 - No response comes from BMU for one second or more.

8 Abnormality Detection/Diagnosis

The battery unit requires the abnormality detection processing and recovery processing described in this chapter. Mandatory processing is indicated as [Mandatory] in the tables below. The abnormality detection processing and recovery processing should be implemented properly in the host system.

8.1 Failure judgment conditions

(1) Abnormality judgment table

Item	Abnormality judgment				
	Diagnosis state	Abnormality detection condition	Judgment value	Recovery action at abnormality detection	Remarks
communication abnormality	At start, at normal condition	[Mandatory] Error detection in communication with BMU	Continuous detection of BMU communication timeout (660 ms or more)	BMU is powered on up to 3 times. If 3 times is exceeded, the corresponding unit is judged as failure.	
		[Optional] Detection of abnormality in BMU life signal	Without BMU life signal for 2 seconds or more	Ditto	Ditto
Automatic numbering abnormality	At start	[Mandatory] (not necessary for series-only battery connection) UART communication abnormality	UART communication abnormality (see 7.3.3)	For UART communication retrial, BMU is powered on and communication is started up to 3 times. If 3 times is exceeded, the corresponding unit is judged as failure.	
Fuse abnormality	At start, at normal condition	[Optional] Detection of blown fuse between the BMU power supply and BMU	(Depending on implementation)	-	

(2) Diagnosis judgment table

Item	Abnormality judgment				
	Diagnosis state	Abnormality detection condition	Judgment value	Recovery action at abnormality detection	Remarks
Main contactor (H) driven output	At start, at normal condition	[Mandatory] (not necessary for series-only battery connection) Detection of abnormal value in the unit discharge current received from BMU The unit discharge current exceeds the rated current of the fuse in the service disconnect.	(Depending on fuse rating) When using FMW-GAA 0003P, 120A or more is detected once or more	Decrease PCS input/output current [mandatory]. If the abnormality is detected after stopping PCS input/output, stop the detected BMU (optional).	If stop is not done, the fuse may be blown. BMU does not open the contactor with the overcurrent .

8.2 Failure restoration conditions

(1) Abnormality restoration judgment table

Item	Restoration judgment			
	Failure judgment permission condition	Judgment time	Normality restoration processing	Remarks
communication abnormality	Restoration of communication with BMU	Successful if communication with BMU continues restoration time (660 ms) or more.	Continuous processing	
	Restoration of heart beat signal	-	Continuous processing	
Automatic numbering abnormality	Successful UART communication	See 7.3.3	Continuous processing	
Fuse abnormality	Detection of normal fuse	(Depending on implementation)	Continuous processing	

(2) Diagnosis restoration judgment table

Item	Restoration judgment			
	Failure judgment permission condition	Judgment time	Normality restoration processing	Remarks
Main contactor (H) driven output	Restoration by inspection of failure point	-	-	

9 Restrictions/Cautions

9.1 Cautions for controlling the battery unit

Care should be taken for the following with regard to controlling the battery unit:

- BMU abnormality

If BMU detects an abnormality; e.g. overcharge or failure, the contactor should be opened, and then the module should be powered off and the processing stopped. Upon receiving BMU abnormality detail from BMU, the host system should find out the location and factor of the failure and take actions as appropriate. To restart, it is necessary to remove the cause of the failure and power off BMU.

- Abnormality between BMU-CMU

If a communication abnormality occurs between BMU and CMU, it takes a maximum of approximately one minute after the starting of BMU to notify the host system of the abnormality. Care should be taken for the failure judgment.

- Notification of SOC

When the module starts up with the cell temperature of 56°C or above, or -31°C or below, BMU transmits SNA (0xFFFF) because SOC calculation is unavailable. The ambient temperature should be controlled in order for the cell temperature to be between -30°C and 55°C. BMU transmits the correct values 120 seconds after the cell temperature reaches the range between -30°C and 55°C.

- Final charge, final discharge, nearly final charge, and nearly final discharge

If any of the flags of final charge, final discharge, nearly final charge, and nearly final discharge are reported by BMU, restrict the charge/discharge current in accordance with the table below. Wrong actions may lead to the detection of overcharge or overdischarge, and result in the contactor being opened.

	Ambient temperature	
	Below 15°C	15°C or above
Final charge	Stop charge within 1 second.	Stop charge within 1 second.
Nearly final charge	Restrict charge current to 1It or less.	Restrict charge current to 3It or less.
Nearly final discharge	Restrict discharge current to 1It or less.	Restrict discharge current to 3It or less.
Final discharge	Stop discharge within 1 second.	Stop discharge within 1 second.

Note also that, in case the units are connected in parallel, an imbalance in the current may be caused by the difference in capacity or internal resistance between the units near the final charge/discharge, resulting in a significant amount of current that exceeds the rated current of the fuse. If the units are connected in two or three parallels, it is recommended that, at the nearly final charge/discharge, the charge/discharge current should be restricted to $3 \times (N-1)/N [It]$ or less (N: number of connections in parallel), or to the smaller of the values shown in the table above.

- Prevention of overdischarge

When the host system receives the notification of the final discharge from BMU, the discharging should be stopped and either of the actions 1) or 2) below should be taken to prevent overdischarge.

- 1) Charge the battery immediately within about 1 hour.
- 2) In case it is impossible to charge the battery, stop supplying the power to CMU in either of the ways below.

[1] Stop 12 V DC power supply to BMU, or

[2] Turn BMU power supply signal off

Also, after the BMU operation is stopped or when CMU power supply shutdown abnormality is received, the BMU power supply should be immediately turned off to prevent overdischarge.

To prevent the battery performance degradation, do not leave the battery in the final discharge state even if the powering to CMU is stopped. The battery should be charged within about one week to prevent battery failure due to overdischarge, since self-discharging continues to gradually consume the energy even with the power supply to CMU turned off.

- Implementation on the host system when MCCB is used

When BMU detects an abnormality, BMU opens MCCB by driving the MCCB control port.

Accordingly, if the power supply for BMU is lost, MCCB cannot be opened. The implementation on the host system side should ensure that MCCB should be opened in case the host system cannot supply the power for BMU.

- Implementation on the host system instructed cell balancing control

Normally, the BMU can recognize the minimum cell voltage in the BMU and perform cell balancing control simply by sending "Cell balance permit signal:1 (permitted)" from the host system to BMU.

Besides, BMU can be specified the minimum cell voltage(as HSMV) from the host system (*).

* The last three digits of the Model No on the label attached to BMU need to be equal to or greater than 110. "Model No: B2***110."

If this control is not necessary, set the HSMV to 0mV.

When specifying HSMV, the host system should closely observe "Min. voltage value of cells in unit" from BMU, and control not to output unintended cell voltage.

9.2 Differences from BMU-1G

Table 12 shows the differences between BMU-1G and BMU-2G. BMU-2G can change timings to the BMU-1G equivalence mode. Change of BMU settings is required for the change to the BMU-1G equivalence mode⁴. For details on the BMU settings change, please contact our sales department.

Table 12 Differences between BMU-1G and BMU-2G

Specifications	BMU-1G	BMU-2G	
		Standard specification	BMU-1G equivalence mode
At startup, time from receiving operation instruction to start of closing contactors	10ms – 130ms	440ms – 560ms	80ms – 200ms
Time from start of closing contactors to start operation (with GND control relay)	approx. 1,320ms	approx. 2,160ms	approx. 1,320ms
Time from start of closing contactors to start operation (without GND relay)	- (BMU-1G requires GND control relay)	approx. 1,440ms	approx. 600ms
Time from opening main contactor (P) by operation stop instruction or power failure trip instruction to opening of contactor (P)	less than 120ms (one-time stop judgement)	385ms – 505ms (three-time stop judgement) 145ms – 265ms (one-time stop judgement *1)	385ms – 505ms (three-time stop judgement) 145ms – 265ms (one-time stop judgement *1)
Time from detection of protection/failure to opening of contactor (P)	less than 120ms	25ms – 145ms	25ms – 145ms
Time from opening contactor (P) to opening contactor (N)	240ms	960ms	240ms
Time from opening contactor (N) to opening GND control relay	120ms	approx. 300ms	120ms
Start timing of sending CAN	after finish of BMU numbering and receiving a CAN from host system	shortly after power-on	
Unit voltage notification when units are connected in parallel	When in progress of startup CMU, parallel connection is not considered and sum of cell voltage is provided as unit voltage. Accurate unit voltage is sent after CMU are started.	BMU outputs correct unit voltage at all times.	
Failure detection and protection when waiting for operation instruction	BMU doesn't execute failure detection or protection when waiting operation instruction	BMU executes all of failure detection or protection when waiting operation instruction	
Time control	Initial value is 0 (1970/1/1 0:00:00 UTC). It is re-initialized to 0 at every reboot or power-off.	Initial value is 56FD3B70 (2016/4/1 0:00:00 JST). BMU saves the value before reboot or power-off, and restores it at startup.	
Transition to maintenance mode during operation	Operation continued (contactor closed)	Transition to stop (contactor open) Maintenance (operation) recorded on log	

*1 One-time stop judgment can be set with the parameter change. To use this function, the last three digits of the Model No. on the label attached to BMU need to be equal to or greater than 110, e.g. "Model No.: B2***110."

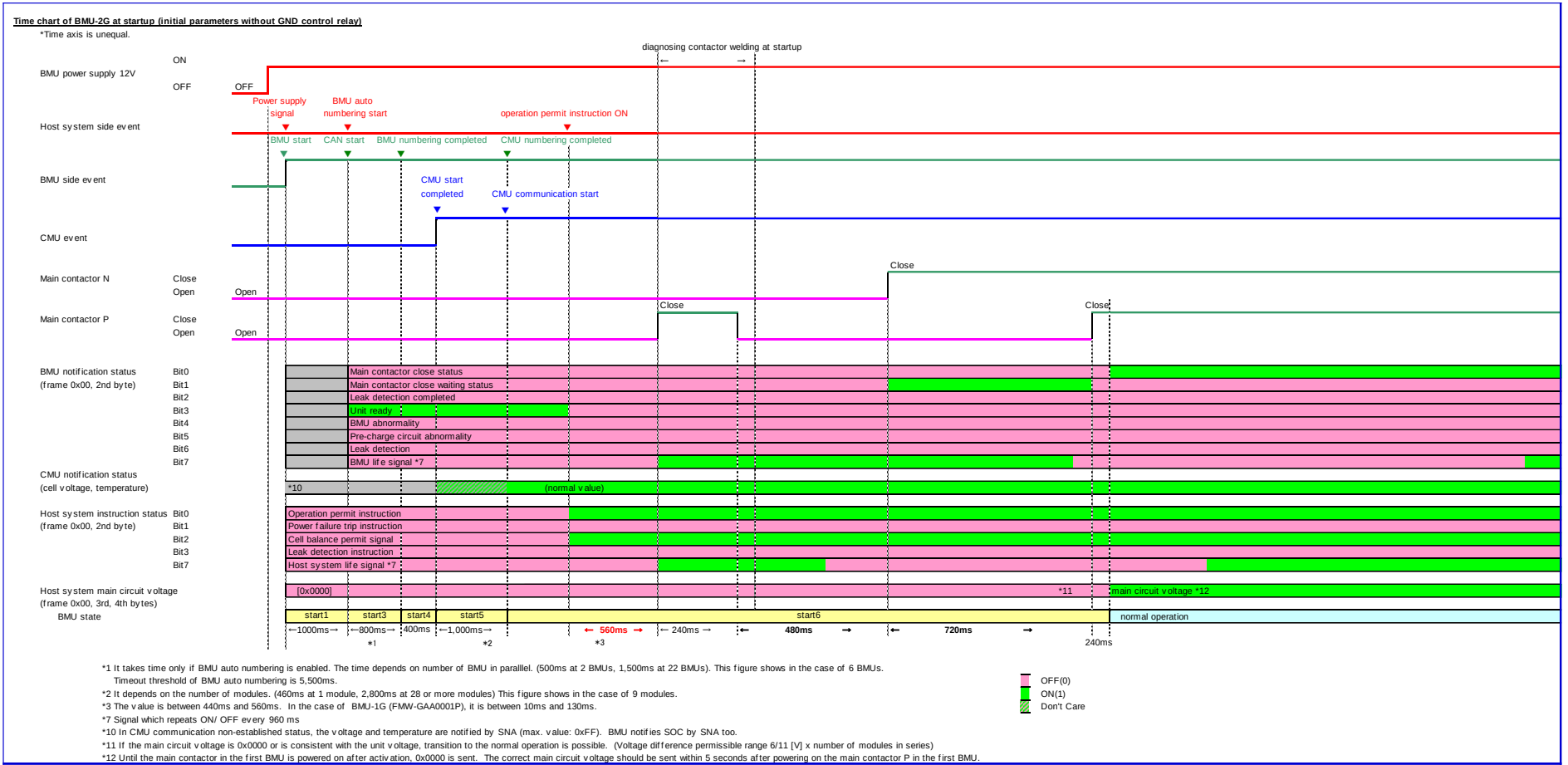
⁴ To turn on the BMU-1G equivalence mode, the last three digits of the Model No on the label attached to BMU needs to be equal to or greater than 102. "Model No: B2***102"

10 [Appendix] Timing Chart

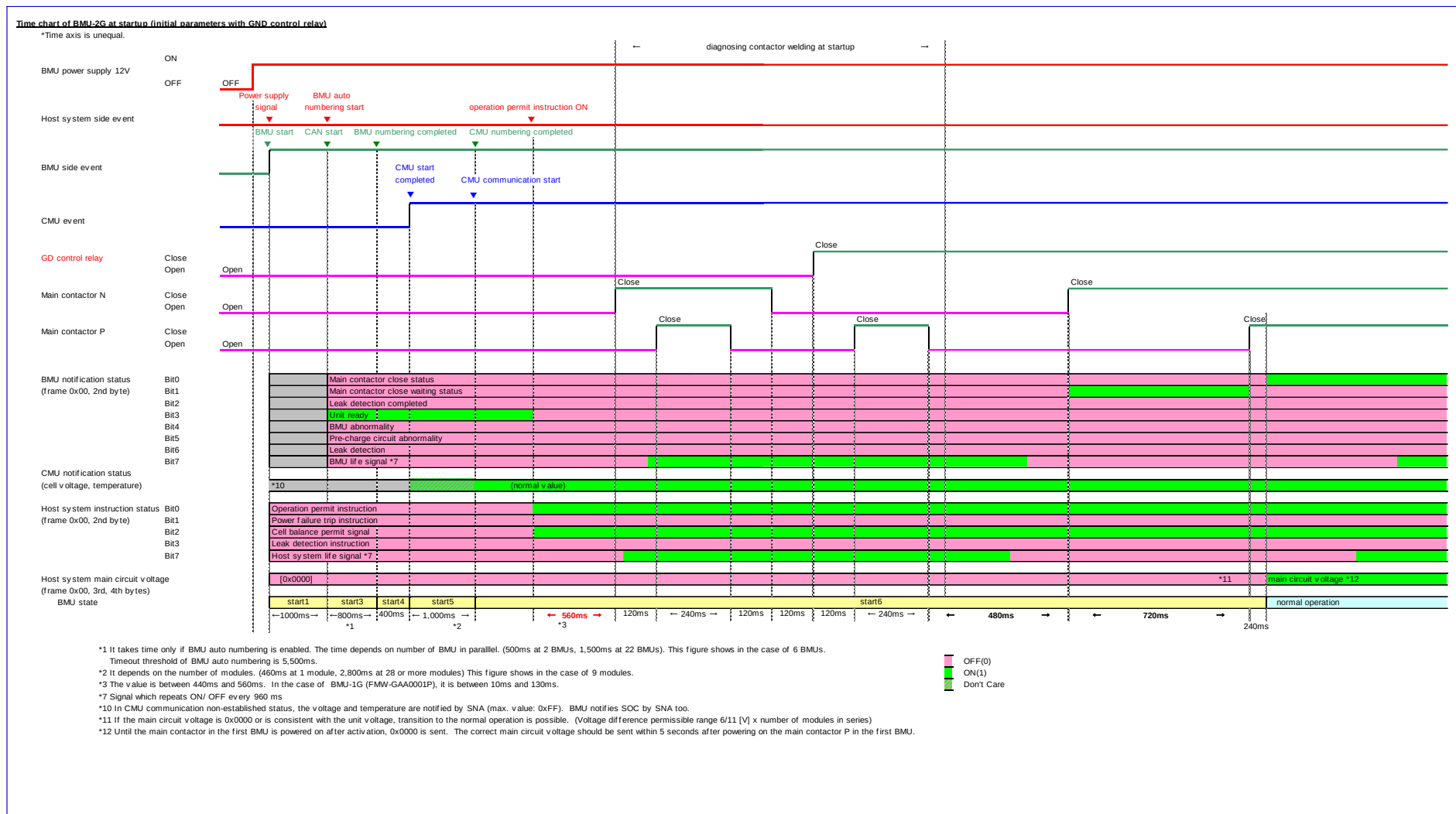
Following are the charts related to the BMU control timing, where (3) and (5) show the operations in the BMU-1G equivalence mode intended for the BMU replacement in a system originally configured with BMU-1G. Change of BMU settings is required for the change to the BMU-1G equivalence mode. For details on the BMU settings change, please contact our sales department.

(1)	BMU start (initial parameters)
(2)	BMU start (initial parameters + GND control relay)
(3)	BMU start (BMU-1G equivalence mode, GND control relay)
(4)	BMU stop (initial parameters + GND control relay)
(5)	BMU stop (BMU-1G equivalence mode, GND control relay)

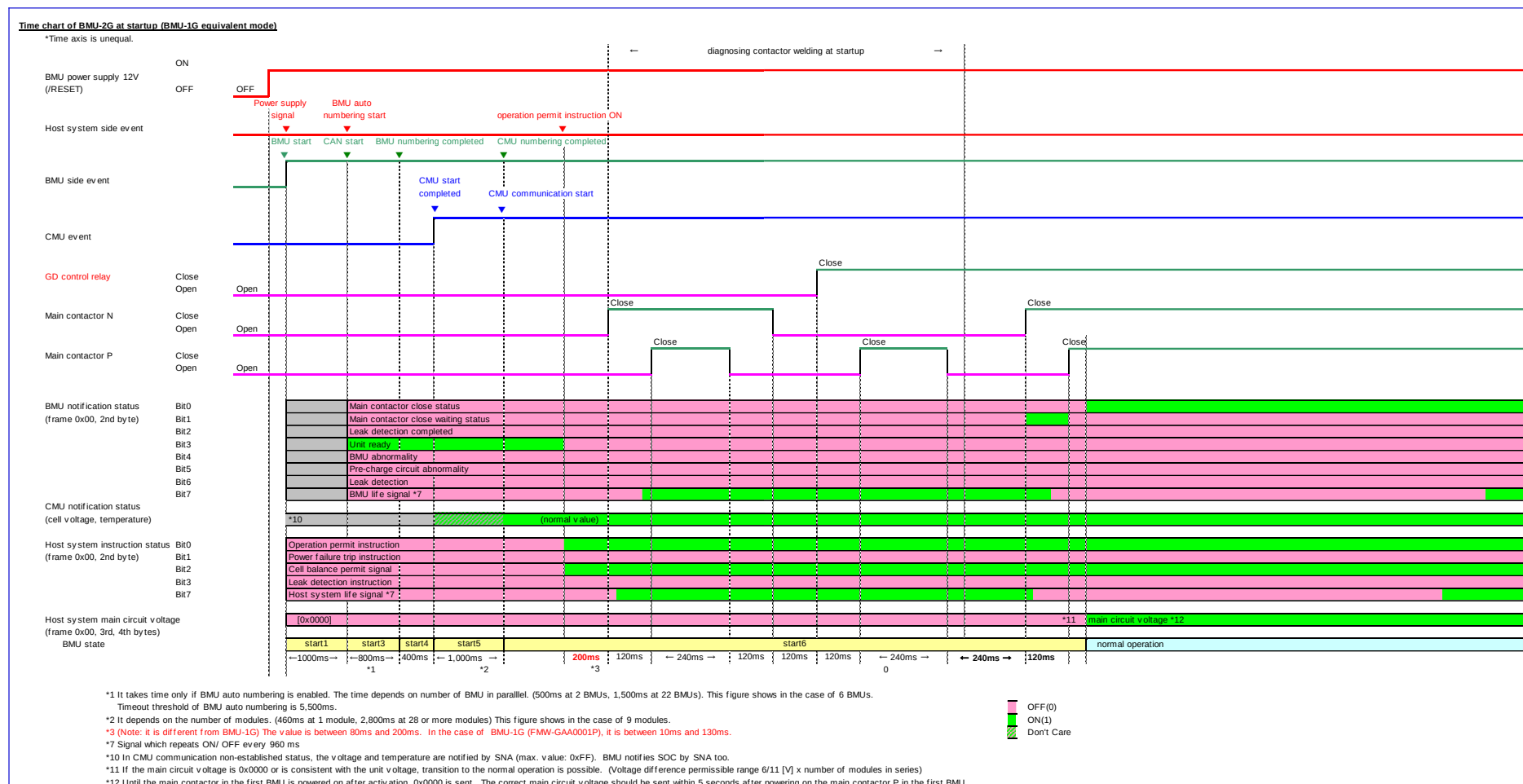
(1) BMU start (initial parameters)



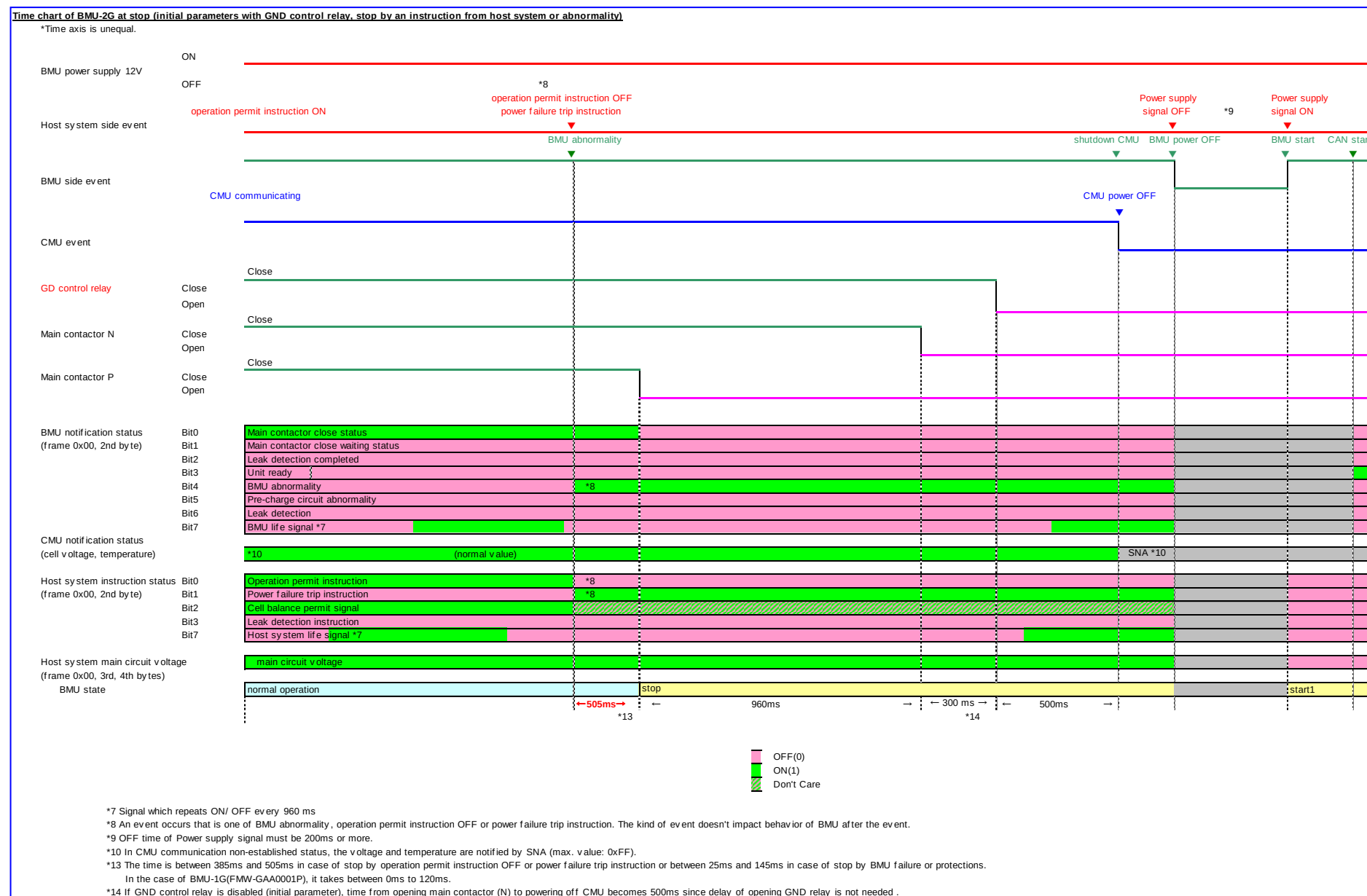
(2) BMU start (initial parameters + GND control relay)



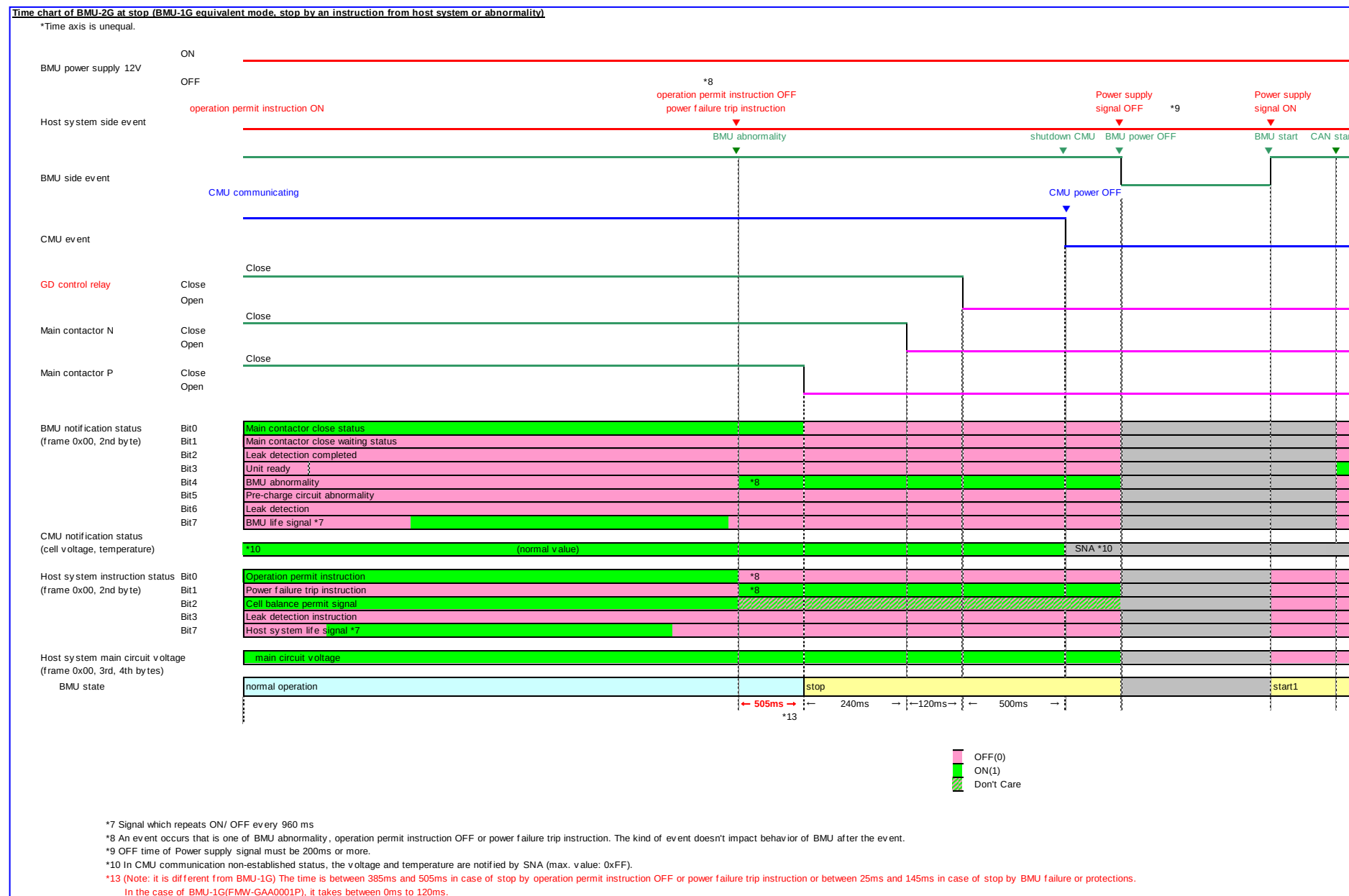
(3) BMU start (BMU-1G equivalence mode, GND control relay)



(4) BMU stop (initial parameters + GND control relay)



(5) BMU stop (BMU-1G equivalence mode, GND control relay)



11 Contact information

For inquiries regarding this document, please contact us at the contact information below.

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